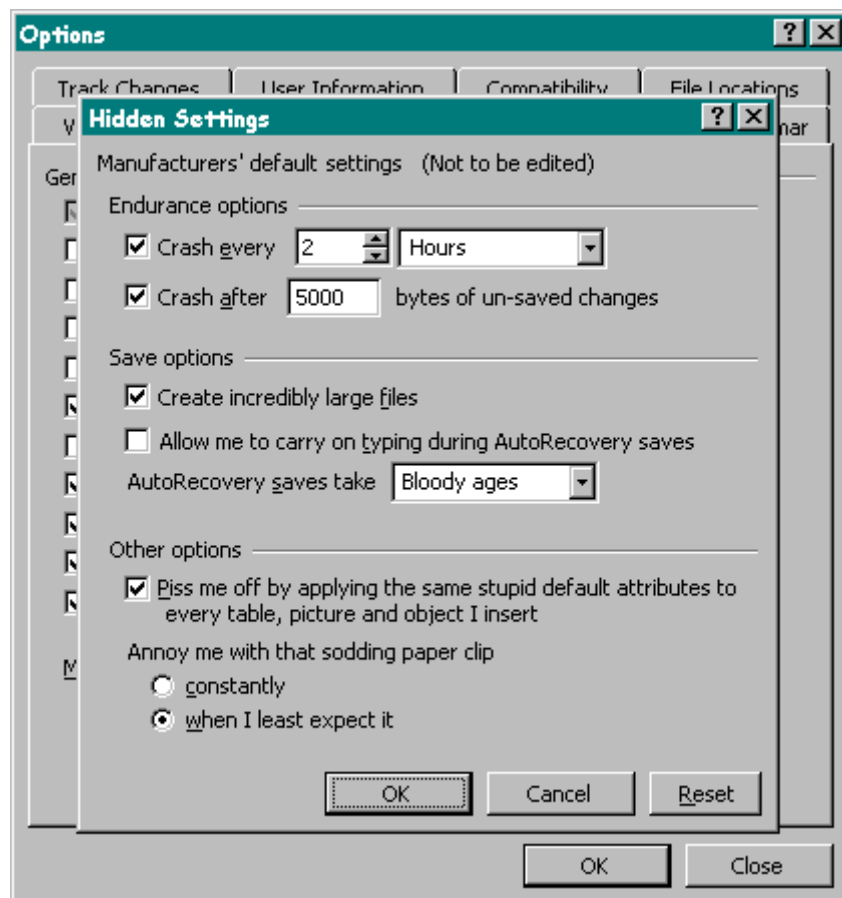


PATN

TECHNICAL

REFERENCE

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PATN

OUTLINE

The 'Front-End'

The DOS version of **PATN** can be run either with or without a '*front-end*'; a simple menu-style navigation program that maybe useful for beginners to get accustomed to **PATN**. The major options provided by the menu are

- pre-processing
- analysis
- post-processing/evaluation

Sub-menus are provided to guide you down the hierarchy to the point where individual modules are listed for selection. For example

- pre-processing/data display/HIST

For most operating systems, this front-end program makes minor use of computer resources. In the case of the standard DOS-version, **PATN** is somewhat limited by the resources available. It may be that if there are TSR (Terminate and Stay Resident) programs or drivers), insufficient memory will be available for the front-end and the running **PATN** module if a 'largish' data set is being worked on. If memory problems occur, try running the module without the front-end (stand-alone mode). Once you know your way around **PATN**, it will probably be more convenient and efficient to run the particular module, simply type:

- HIST

Function

The function of the front-end is to provide a classified list of major options (modules) available and activate the selected **PATN** modules.

Typing a ? results in a single screen detailing all the available **PATN** modules. They are listed with a brief description in the order of use; *pre-processing*, *analysis* and *post-processing*.

Typing a module name followed by a ? results in a page of help about the selected module. The purpose, basic algorithm and major options are displayed.

If the logging option in PRAM is set to 1 or 2, **PATN** will echo (log) information to the file *PATN.LOG*. This happens on the activation of the selected module; the front-end is not involved.

Lastly, **PATN** runs the selected module. In the case of DCOR (the DECORANA program pre-processor), it will also run DCA (DECORANA) and POST, the post processor program. When *not* using the **PATN** front-end, you may have to type:

DCOR
DCA
POST

to get a DECORANA ordination (or Reciprocal Averaging) results in a **PATN** format.

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OPTIONS

Help

Help on major options is available by using a ?. For example, to find out more about data input and output at the PATN prompt, enter-

PATN<: DATN?

On DOS systems, enter a '?' and follow the menu to the point of interest, the information about the module will then be displayed.

Operating system commands

Using \$ as the first character in any command to PATN will re-direct that command to the operating system (, VMS, UNIX, PRIMOS...). A similar feature is not available on the DOS menu system.

Escaping to the operating system may be useful when there is a need for operations like copy, rename, delete, print and type while in **PATN**. If logging is on, these commands will be echoed to *PATN.LOG* for later replay in batch.

Comments

An ! anywhere in a **PATN** command is ignored. This feature may be used to add useful comments to *PATN.LOG*. **PATN** uses ! to add it own commands on each command or parameter in this file, starting at position 43. You may however, use additional commands that begin with ! to provide additional commentary on the analysis. For example-

```
PATN<: !
PATN<: ! Now I am going to analyse the environmental data
PATN<: ! using the file supplied by Malcolm 'X'
PATN<: !
PATN<: ! First, get the ASCII file into PATN: set-up parameters
PATN<: !
PATN<: PRAM
PATN<: ....
PATN<: ....
PATN<: !
PATN<: ! Now for the data....
PATN<: !
PATN<: DATN
PATN<: ....
PATN<: ....
PATN<: !
PATN<: ! Now to get decent labels from X.LAB
PATN<: !
PATN<: LABN
PATN<: ....
PATN<: ....
```

If logging is on, these comments, along with those added by PATN, form a useful record of the details of what may be a complicated analysis covering many operations. The date and time of activation of any of the major options is also recorded by **PATN**. For further details about the logging facility, see the chapter on PRAM below or the section on logging in the **PATN USERS GUIDE**.

DEFINING DATA PARAMETERS: PRAM

OUTLINE

PRAM is used to list and alter the parameters of the current data file and a number of **PATN** environmental **parameters** as maintained in an *unformatted form* (non-ASCII) in the file *PATN.PRM*. These parameters provide **PATN** with -

1. a TITLE (80 characters)
2. a currently active **PATN** internal data file (43 characters max.),
3. the numbers of rows in the data (objects),
4. the number of columns in the data attributes,
5. the number of groups of objects,
6. the number of groups of columns,
7. the missing data value and
8. the level of logging.

Alteration of Parameters

Unless the data generation module RAND is used, these parameters are assumed to have been initialised by you through PRAM. Parameters 1 through 4 are mandatory while 5-8 will usually default to reasonable values. Once created, parameters may be modified by PRAM or by implication from some other module. For example, the PRAM command can be used to alter the value to be used for missing data while the GDEF module (generate a set of group definitions) may be used to alter the current number of row groups. Once the **PATN** parameters have been defined, either the user (directly) or other commands (indirectly) may update them.

The file *PATN.PRM* cannot be edited. The parameters it contains must be modified using PRAM.

Default Values

Pressing the <RETURN> key in response to any prompts by PRAM will result in the **defaults** being accepted. Some parameters, such as the number of rows and columns have no meaningful defaults and must be supplied.

OPTIONS

Option 1 TITLE

TITLES may be up to 80 characters in length, and should be used to describe the type of data as well as the nature of the analysis. It is wise to use this feature to assist in recording data status and analysis details; looking back at results after a few years, anything helps!

When the logging parameter is greater than ZERO (see below), the title is recorded on the log file (PATN.LOG) as each module is selected.

To assist updating the title, a SUB-STRING may be substituted, starting from any character position in the current string. The character (#) in the first character position is used to signify such a sub-string. IMMEDIATELY following the (#), a one or two, 2 digit number indicates the sub-string START POSITION. For example, given the current title -

UNIFORM RANDOM DATASET : TESTING ALOC CLASSIFICATION
--

If the first three words are to be retained, adding subsequent comments commencing in column 26, the following would suffice:

#26 NOW TESTING FUSE METHODOLOGY

This would lead to a new title:

UNIFORM RANDOM DATASET : NOW TESTING FUSE METHODOLOGY

NOTE: **PATN** scans the incoming text string locating the last non-blank character. It then INSERTS this string segment into the 80 character title string at the indicated position. It will NOT blank the rest of the title. For example, in the example above, if there was text beyond position 53 (26+27), it would be RETAINED. This permits substitution in the centre of the title without affecting other characters. Blanking the rest of the TITLE is achieved by adding blanks, then any non-blank character must be used to end the string. If you place this character beyond position 80, it will be truncated off the title string and won't appear. If a (#) is not detected as the first character option 1 will produce a COMPLETELY NEW title.

Option 2 Database file NAME

This is the name of the file that contains data in **PATN** *internal format*. This file has to be created using DATN on an ASCII file with any file name that is supported by the operating system. The name must be legal in the sense of being comprehensible by the operating system. It therefore should contain only printable characters and no blanks. The maximum file name, including directory prefix on **ANY** operating system is currently 43 characters. In DOS, the file name is limited to 13 characters. On other operating systems such as UNIX, the limit is far greater.

The structure of a legal name on a VMS system could be:

SYSTEM::DEVICE:[DIRECTORY]ROOTNAME.EXTENSION

where DEVICE:: is the computer where the file is to be found, DEVICE: usually refers to a disk and so on. An example file name on this system could be:

DUA1:[LEE.PATN.TEST]TESTFILE.DAT

refers to the default computer, a disk device DUA1:, a directory [LEE.PATN.TEST] and a file name TESTFILE with an extension .DAT. If no DATA FILE NAME has been defined, the default supplied by the package is <UNDEFINED>.

Note that UNIX, unlike DOS or VMS is case sensitive on commands and file names.

The default data file is an internal file. It is usually a rectangular object (row) by attribute (column) matrix (a *-.dat* file). In some cases, primary data may be matrix of association values (*-.aso* file).

Option 3 Number of ROWS

This is the number of ROWS or OBJECTS in the data matrix or the order of the association matrix. It will be this number of records that **PATN** will use to read and write during the analysis. This value is defined by the data and remains unchanged, unless MASK (masking), SAMP (sampling) or other modules are used to alter the structure of the data.

Option 4 Number of COLUMNS

This is the number of ATTRIBUTES defined in the data matrix. If an association matrix is the raw data, then this number will be ignored. This parameter is used to determine the length of the LOGICAL records **PATN** will read and write. This value is defined by the data and remains unchanged, unless it is altered by modules such as MASK (masking) or SAMP (sampling).

Option 5 Number of ROW GROUPS

Normally this value is undefined (0) until a set of row groups is created by commands such as GDEF (group definitions), ALOC (non-hierarchical clustering), MDIV, PDIV (hierarchical divisive clustering) etc.

Normally **PATN** will update and maintain this value as groups are defined in the analysis. There are however, reasons why it may be altered manually. If externally defined groups are to be analysed by **PATN**, this value must be set.

This value refers to the *groups* as defined by the ROWS of the current data matrix. This may correspond to objects or attributes depending on which one forms the rows of the matrix. Transposition (DATN) reverses rows and columns of the data matrix.

Option 6 Number of COLUMN GROUPS

Similar to option 6 except it applies to the *columns* of the data matrix. This may correspond to groups of attributes (usually the columns) or objects, depending on which are forming the columns of the current data matrix.

Option 7 MISSING data VALUE

This is the value which **PATN** will recognise as corresponding to a *missing value* in the data file or association matrix.

The DEFAULT value for this parameter is -9999. *Do not use zero unless you really have no non-missing zero values in your data.*

Data scanning commands such as HIST, SCAN and SCAT will take into account missing values in attempting to list characteristics of the data set. ASO will skip a comparison of two rows on a single attribute if either has a missing value. This should have minimal impact on overall association as long as there are not too many missing values in each comparison. Missing values are detected in the following commands and the following actions are taken:

ALOC	Skip missing attributes
ASO	" " "
SCAN	" " "
PCC	" " rows
GSTA	Use only non-missing data for summaries
HIST	Tabulate using non-missing data and list number of missing values zx
SCAT	Skip plotting point with missing x or y cord, use M for missing z
TRND	Don't transform missing values
TWAY	Annotate missing values with 'M'

Option 8 LEVEL of LOGGING

The extent of listing user-actions to the log file PATN.LOG depends on the selection of a single parameter. *Logging* refers to the echoing of information entered by the user to the special file PATN.LOG. There are 3 types of information that can be recorded by PATN -

1. Echo of **PATN** modules as activated by the user-
 - a. name of module (Eg. PRAM),
 - b. date and time of activation and
 - c. current title.
2. Echo of all input -
 - a. all **PATN** modules and
 - b. options selected to the terminal and log file.
3. Each option selected will have appended, a comment detailing the nature/purpose of the option. This annotation begins with an exclamation (!) mark at column 43 and continues to column 80 or as necessary.

This parameter controls the *type* and *extent* of logging. The acceptable values and corresponding functions (above) of the logging parameter are:

Logging parameter	Logging action
0	none
1	(1)
2	(1) + (2) + (3)

Use of '0' adds nothing to the log file PATN.LOG. It requires no overheads and is therefore fast; use it when familiar with the package.

'1' is useful when getting used to PATN. It provides an outline of activities during sessions. When encountering problems or trying to remember what was done some years ago, a copy of the log file is indispensable.

'2' commits PATN to log all modules used *and* all subsequent *options* (all keystrokes). The format of the logging is such that it may be edited and re-submitted to PATN as an input command file. For example, the log file could be copied to another file, edited for a different data set and submitted to PATN as a set of batched commands. Level 2 logging is recommended for new users. If problems arise, a comprehensive record of activities is then available. A test data set may be quickly generated using RAND and used to familiarise yourself with PATN. Once satisfied, copy and edit PATN.LOG to conform to a real data set and submit it to PATN for execution.

The use of stored commands to run PATN is different, depending on the operating system in use. With VMS, the structure of the command is-

PATN INPUT.DAT

where INPUT.DAT contains a set of commands as found in the PATN log file with level 2 logging.

For DOS, re-direction may be used, but at the module level. For example, the command-

PATN <INPUT.DAT

will *NOT* work because as each PATN module is activated, the pointer in the re-directed input file is reset to the *first record*. A PATN stand-alone utility called LOG2B will split PATN.LOG into separate re-directed input files to each separate PATN module. LOG2B produces a batch file called PATNBAT.BAT which contains something like this-

```
DATN <DATN.INP
ASO  <ASO.INP
FUSE <FUSE.INP
DEND <DEND.INP
GDEF <GDEF.INP
.....etc.
```

There is a chapter on LOG2B at the end of this manual.

Option 9 Saving Environmental Parameters

This option *saves* the contents of the parameter file PATN.PRM into the file -.prm where (-) refers to the *root* of the data file name as defined in option 2. For example, given a data file name FRED.dat, option 9 would result in the file FRED.prm being created, having the same format as PATN.PRM (unformatted).

Option 10 Restoring Environmental Parameters

This option requests the name of the previously *saved* parameter file, and once supplied restores the file *PATN.PRM* from this file. The original *PATN.PRM* is **OVERWRITTEN** and the saved parameter file used for restoration remains as it was before this option was invoked. The *root* name of the file only need be supplied, **PATN** adds the *.PRM*.

Option 11 Stopping PRAM

Selecting this option completes the PRAM command, saving any changes that have been made to the parameter file *PATN.PRM*

DATA INPUT-OUTPUT:

DATN

OUTLINE

DATN is the module for data input and output. It **must** be used after PRAM to read objects by attribute data into **PATN's** internal binary (unformatted) format. It is in this binary (machine readable) format that PATN assumes that internal data files are stored. Data labels, parameters and association measures in a binary form because-

1. FORTRAN formats are only required for data input/output to or from **PATN** (not within) and
2. Unformatted input and output is more than twice as fast as formatted.

Data Structures

DATN **must** be used to read rectangular (objects by attributes) data into **PATN** and may be used to re-format data between a variety of different formats and structures. These options should permit sufficient flexibility for data exchange with a variety of other programs as well as data entry into **PATN** using an editor.

A **PATN** data set is stored in a *complete* form. All values in the data matrix must eventually be specified. In the case of the compressed formats used by DATN, uninitialized data is assumed to have a value of zero. All other data must have a numeric value that represents an actual measurement or missing data (see PRAM).

Formats

DATN can use any of the following formats-

1. FORTRAN floating point (F) format fixed length (integers can be read using F-formats),
2. FORTRAN free format (values separated by spaces or commas or a combination of both).
3. DBASE ASCII format (ASCII-type output from DBASE II or III with row labels on the end of each record),
4. Cornel's CEP format (a compressed-type format as used by programs such as DECORANA and TWINSpan: only the non-zero values are nominated).
5. 'Relational' format (non-zero values at row and column indices: as stored in relational databases).
6. The quick format from the PRIMER package (data with no labels).

NOTE: the maximum number of characters that may be used to define the data format, including the left and right parentheses is 80. This should not be a limitation because the format for a single record is all that is required. PATN will re-use the format for each record of data. Formats can usually be simplified. For example, (1x,F3.1,F4.1) can be replaced by (2F4.1)

Most data is read and stored in *floating point* (F) format (CEP formats are an exception). Data that has no decimal places may be read and written effectively with F-type formats. For details regarding FORTRAN formatting as used by PATN, see the chapter in the **PATN Users Guide** on FORTRAN formatting conventions.

When reading data into **PATN**, LABN will update the parameter file *PATN.PRM* and create a copy of this file in *-.prm* where '-' refers to the root of the output file.

Updating

It is dangerous to have the same file name for input and output files. Always change the name of the output file to reflect the actions being performed. Never overwrite an old file unless you really are very certain of the outcome.

In some situations **PATN** will note some alteration of parameters and ask if they are to be *updated*. What this means is whether or not *PATN.PRM* is to be altered to reflect the changes. Either way, the *-.prm* file *will* be altered. In this situation, a new data file is usually created and you are provided with the option of carrying on further analysis with this new file or remaining with the old. The choice is yours.

If the new file overwrites the old, you have no choice about updating. You must update otherwise the parameters and data (including row and column labels) will not match!

DATN provides a simple program for interactive data entry into **PATN**. A better way of entering data is by use of a database or spreadsheet program on a PC (for example, Lotus, Dbase, Excel, Paradox). These are better suited for data entry and storage than is PATN (it's forte is analysis).

For data input and output, DATN will usually need to know-

1. the *name* of the data file and possibly
2. the FORTRAN *format* of the data.

Depending on the option selected, DATN may not read or generate row and column labels. The exceptions are CEP and archive formats. LABN is used for label input and output when data does not contain labels or where DATN cannot effectively read them.

It is **important** to realise that DATN will assume that PRAM has been used to nominate the *size* and *name* of any data set to be read. In the case of input, DATN will request the name of the file. For output options, current parameters are assumed.

REFERENCES

- Belbin, L., Faith, D.P. and Minchin, P.R. (1984). Some algorithms contained in the Numerical Taxonomy Package NTP. CSIRO Division of Water and Land Resources Technical Memorandum: 84/23.

OPTIONS

Option 1 ASCII file --> PATN

This option reads a ASCII files using either FORTRAN F-style (floating point: see PATN Users Guide) *fixed format* or *free format*. It assumes that all data values are specified with floating point (have a real or implied decimal point) or missing values. DATN uses the FORTRAN format you nominate to determine the format/type of file to be read. **PATN** will echo when it has read every 'x' objects where x is a function of the total number of objects. The function is-

If $N < 100$ $x=1$ (every object)
 if $100 < N < 1000$ $x=10$ (every 10 objects)
 if $1000 < N < 10000$ $x=100$ (every 100 objects) and
 if $10000 < N$ $x=1000$ (every 1000 objects).

Fixed Format

This style of format means that all the data is perfectly lined up in the same columns. Each variable takes up exactly the same number of characters for each record in the file. Reading this style of data is usually done with some type of FORTRAN 'f-style' format. For example if records were as follows

1.012	2.023	5.765	4.543	5.432
9.123	2.222	6.987	0.012	1.001
4.546	6.789	10.432	1.555	5.111

then the format would be something like '(5f7.3)' or 5 floating point values each of 7 character positions and 3 decimal places.

With this style of format, it is assumed that the column labels are in a separate file and that row labels must be skipped over using FORTRAN 'x'-style formatting. If for example, labels were in the first 8 character positions on the previous example, the format may be something like '(8x,5f7.3)'. In this case, LABN should subsequently be used to read in the labels from the data file using format '(a)' which implies that the row labels are in the first 8 characters positions of each record.

Free Format

If an '*' is placed anywhere in the format, data is read using what is termed *free format*. In this situation, commas or spaces are used to delimit labels and data values. Labels are assumed to be surrounded by quotation marks (eg: "label") while data values are not. One comma and/or any number of spaces can be used to separate values.

Free-format data is assumed to have column labels as the first record of the ASCII file while row labels are assumed to be the first field on each record. If they are the last field on each record, then use '-dba' as the format (see below). An example of a free-formatted ASCII version of the example above would look like this-

"RowLab",	"Ht",	"Width",	"Length",	"Depth",	"Breadth"
"spec-1",	1.012,	2.023,	5.765,	4.543	5.432
"spec-2",	9.123	, 2.222,	6.987,	0.012,	1.001
"spec-3",	4.546,	6.789,	10.432,	1.555,	5.111

Note that there is a 'dummy label' for the row labels on the first record. This label must be included for consistency. It is not stored in **PATN**. Note also that the fields do not have to line up, that the commas may be anywhere between the fields and that blanks can be placed anywhere with no effect (except inside the quotes in labels).

FORTTRAN free format represents an effective way to read a data file that has been created using a standard (ASCII) editor or an *unformatted* save form a word-processor (if a document is formatted inside a word processor, it will be in binary rather than ASCII format). With this method, the only specification in the FORTRAN format is '*', and the data does not have to be lined up in fixed columns. The only problem occurs when a number is left out. With free format, blanks do not represent zeros, so that the reading of data records gets out of sync. Be aware of this if read errors occur (the end-of-file will be detected if one or more numbers are missing).

Free-format is ideal for reading data that has been created from a spreadsheet or database program using *-.csv (comma-separated-values)*, but remember that the first record must contain a set of m+1 column labels with quotes.

'DBA' ASCII format

This format is identical to free-format, except that *ROW LABELS* are added at the end of the record (in quotes). As with free format, commas may be used to separate values and multiple records may also be used. For example-

```
1.1 2.2 3.3 1.4 9.5 "Fred"
2.1 7.4 5.8 2.9 5.2 "George"
```

would be in DBA ASCII format. To nominate this format to DATN, supply the letters-

```
DBA or dba
```

in the FORTRAN format field.

TAB delimited ASCII (eg. EXCEL) format

DATN is informed that the ASCII file is TAB-delimited when a 't' or 'T' (standing for tabs) *as the first character of the input format statement*. Microsoft EXCEL can export data files using TAB characters between label and data values. DATN is capable of reading labels and data values from this type of file.

Labels should be the first 8 characters on each record and be surrounded by quotes. All field in the file are assumed to be separated by TAB characters. DATN parses each record in the data file, locates the position of the TABS and reads and stored the labels and data values.

An example of such a file (^ is used to show the location of TAB characters in the file)-

```
"RowLab"^^"Ht"^^"Width"^^"Length"^^"Depth"^^"Breadth"
"spec-1"^^1.012^2.023^5.765^4.543^5.432
"spec-2"^^9.123^2.222^6.987^0.012^1.001
"spec-3"^^4.546^6.789^10.432^1.555^5.111
```

Option 2 PATN --> ASCII file

This option is the reverse of the previous one. It takes a standard **PATN** file as nominated by the parameters and writes an ASCII equivalent using the FORTRAN format you nominate. Output can be either *fixed* or *free* format. Labels may be optionally exported with the data in either fixed or free format.

This option is convenient for checking what data values PATN is using; for checking original data input or after transformations or standardisation. The option also provides a method to interface with other software such as statistical packages.

Free-format *output* using FORTRAN has complications. FORTRAN examines the type of data and allocates a number of significant digits by default. This generally means that you cannot leave it up to FORTRAN to output data values. **PATN** gets around this problem by simply getting you the user to nominate the output format exactly. For example, if you wish a comma between data values, then it is up to you to nominate them in the output format specification. For example, the format '(4(f5.1,','),f5.1)' directs FORTRAN to write a comma between the five data values. The reason a '4' is used in the repetition (see the Users Guide) is that we don't want a comma after the last data value.

If labels are to be output *using free-format* an '*' must be nominated in the FORTRAN format specification **and** the format of the label and data values for each data record must be nominated. The '*' is relevant only for the output of the column labels to the first record of the free-format file. It is best placed as the first character of the FORTRAN format (although it can occur anywhere in the format specification) for simplicity. An '*' in the format of an ASCII file has no meaning (as you **must** insert ','s between the data values yourself using the FORTRAN format), and will be ignored.

All **data values** must be output using an 'F' or floating point FORTRAN format specification while row-labels must be output using an 'A'-style format. Read the Users Guide chapter on FORTRAN formatting for an outline of conventions and tricks.

Labels can be exported with the data in both free and fixed formats. With fixed-format export, the column labels are listed, one per line (A8) on the first M (=number of attributes) lines of the output file. Row labels are placed into the first position on each logical record. The user must supply the format of one complete logical record, for example -

```
(A,2X,10F6.1,/12F7.2,/50(F2.0,TL1)).
```

If labels are requested with free format (*), the column labels are written the first record using quote delimiters and comma separators instead of one label per record (A8). The remainder of the format is accepted as is and used for the remainder of the data, including row labels. For example, an export format-

```
*('"'',a,'" ',10(' ',f6.1)).
```

will export column labels as the first record and format row labels as the first field of the second record with the 10 data values formatted using 'f6.1' with commas between the data values. This style is most useful when exporting data for use by spreadsheets such as LOTUS (tm) or EXCEL(tm) and is particularly suited to the statistical package Splus(tm).

If no labels are required and the data is to look like .csv format (comma separated values), then use a format like this-

```
(9(f5.1,','),f5.1)
```

when you don't want a comma after the last data value. **NOTE:** DATN archive format can also output data with the parameter headers and label footer in free format (see options 6 and 7 below).

Option 3 CEP --> PATN

The Cornell Ecology Program or CEP format is used by programs such as DECORANA (Detrended Correspondence Analysis) and TWINSpan (Two-Way Indicator Species). The data structure is designed to only nominate the non-zero data values.

A sub-option is used to select between one of the two forms available. The first option applies when the non-zero values are assumed to be '1's (presence/absence data). The second applies when the non-zero values can take any positive floating-point (decimal value).

CEP Presence/Absence Format

For the presence/absence (0/1) form of this format, the **location** or column associated with the 1's or presence values are coded for each object. The specification for the format is -

RECORD 1: An 80 character title

RECORD 2: a 65 character FORTRAN format followed by a right justified integer (I5) referring to the maximum number of non-zero integer values on any data record.

RECORD 3+: An INTEGER value on all records nominates the *object sequence number* (not label) to be associated with the following values. Given this, records can be entered in any order but the record for each object should be contiguous. Subsequent values on each record represent the sequence number of the non-zero attributes.

LAST OBJECT: the end of the data is signified by the object number being zero. A blank line will do.

LABELS: directly following the blank line are one or more records nominating the column labels and then, on a new record, the row labels. The format must be FORTRAN (10A8) or 8 characters per label and 10 labels per record. The order for the column and row labels is assumed to match the object sequence numbers.

An example of this style of data for 3 rows and 11 columns is -

```

This is an example title (max 80 characters)
(I2,1X,5I2)
1  1 3 5 7 9
1 11
2  2 4 6 8
3  1 2 3 4 5
3  6
(blank line)
Col-1    Col-2    Col-3.....Col-m
Row-1    Row-2    Row-3.....Row-n

```

this would give rise to the **PATN** file -

```

10101010101
01010101000
11111100000

```

If no labels are found at the bottom of the CEP file, an error message will be displayed, but the data file is still created. In this case, labels can be read separately or created using LABN.

CEP Continuous Format

The non-presence/absence or continuous version of this format is the same as the presence/absence style, EXCEPT that data values are not assumed to be ONE (presence) and are therefore entered AFTER the INTEGER values nominating the location of the non-zero value. The maximum number of non-zero values in this case, represents the number of PAIRS.

Using the previous example, the continuous format could appear something like this:

```

This is an example title (max 80 characters)
(I2,1X,5(I2,F3.1))
1  1 .2 3 .1 5 .5 7 .1 9 .8
1 11 .1
2  2 .2 4 .4 6 .6 8 .8
3  1 .1 2 .2 3 .7 4 .6 5 .6
3  6 .4
(blank line)
Col-1    Col-2    Col-3.....Col-m
Row-1    Row-2    Row-3.....Row-n

```

This would lead to the following data being created by **PATN**-

```

.2 .0 .1 .0 .5 .0 .1 .0 .8 .0 .1
.0 .2 .0 .4 .0 .6 .0 .8 .0 .0 .0
.1 .2 .7 .6 .6 .4 .0 .0 .0 .0 .0

```

Option 4 PATN --> CEP Format

For this option, the data can be output to CEP format in either the presence/absence or continuous format. A sub-option is used to request which of the two styles is required. The MAXIMUM number of attributes (presence's or pairs of identifier-amount) per record is also requested.

This option is useful if data are sparse (there is a significant proportion of zeros) and are to be archived. In such cases, a significant amount of disk storage may be able to be saved by using the CEP-style as an output format from **PATN**. This option may be used in tandem with OPTION 3 for archiving and retrieval as an alternative to OPTIONS 6 and 7.

Details of the two formats are outlined in OPTION 3 above. To get an idea of the style, you may simply view the data as output from this option and compare it with data as output from OPTION 2. Both use ASCII but have different styles.

Option 5 Mixed Attribute --> PATN Format

Throughout PATN, attribute types are referred to as being one of -

1. NOMINAL
2. ORDINAL
3. INTERVAL or
4. RATIO

The **PATN** Users Guide provides a detailed definition of each of these formats. Generally speaking, **PATN** assumes that attributes are interval or ratio. This option can re-code any mixed set of attributes containing *nominal* attributes to a set of *ratio* attributes.

Nominal->Ratio Re-Coding

Nominal attributes are those where, usually an integer code refers to different states of the attribute, and where these states are nominal in the sense that any one state is not greater or less than any other state. For example, if a colour coding:

```
1 = yellow
2 = magenta
3 = cyan
```

was used, then this option would detect 3 different levels of the attribute 'colour' and code 3 new presence/absence (1/0) attributes labelled:

```
yellow
magenta and
cyan.
```

As an example, input nominal coding using the above colour on 4 samples could be:

```
colour
-----
1
3
2
1
```

while the output data file would correspond to:

yellow	magenta	cyan
-----	-----	-----
1	0	0
0	0	1
0	1	0
1	0	0

By definition, a presence on one of the new attributes excludes any other '1' state on other new attributes. New column labels, one for each state of each nominal attribute will be requested from the terminal.

Option 6 PATN --> Archive File

Options 6 and 7 have been designed for the archiving and retrieval of **PATN** data files. Some **PATN** functionality is based on having parameters, data and labels in separate files. While this is convenient when using the package, these options provide a means of merging all related files into one making it more appropriate for archiving. No data compression is used and the files, as with all other **PATN** files, are in a readable ASCII format.

Archiving requests the name of a file to be used as the archive file (*-.arc*) and then appends:

1. The PARAMETER FILE (*PATN.PRM*) file,
2. The DATA (*-.dat*) or ASO (*-.aso*) file
3. The *-.rlb* and *-.clb* files and
4. Any additional text material you may care to append to the end of the file. Eg
 - a. references
 - b. notes
 - c. formulae
 - d. methods used
 - e. copy of PATN.LOG

These files are concatenated without any intervening blank records into the archive file. Additional (optional) text material is placed at the end of the file. No interpretation is made of any comments etc. If the original data is an *-.aso* format file; a source (readable) file with the given FMT (parameter) format and in a lower triangle (minus diagonal) form.

Free format output can be achieved by using an '*' at the beginning of the format specification. If DATN detects this, the parameters and labels are output with quote delimiters and comma separators. Comma separators between the data values must be specified in the output FORTRAN format by you. For example, a format of the style -

```
(9(f5.1,', '),f5.1)
```

should be used (see option 2 above). The use of '9' here and the tailing 'f5.1' means that there will not be a comma after the last data value on each record.

After an analysis, it may be useful to *append* the *-.gdf*, *-.ssh* or *-.pcc* files to the archive file for future reference. This can be done either with the DOS copy command eg-

```
copy fred.arc+fred.gdf new.arc
```

Format

The format of the archive file is fixed for the parameter section and the labels. The format of the data is user-defined and is nominated in the parameter section.

At the head of the file are the parameters. The format is-

Record 1:	the Title (1X,A80)
Record 2:	the name of the data set (1x,A43)
Record 3:	FORTTRAN format of the data (1X,A80)
Record 4:	number of objects
	number of attributes
	number of object groups
	number of attribute groups
	logging-level (0=none, 1=intermediate, 2=full)
	missing data value (default: -9999) (1X,5I6,E13.4)
Records 5+:	the data in FORTRAN f-style format)
Records k-2+:	the row labels (10A8)
Records k-1+:	the column labels (10A8)
Records k+:	any other information you wish

An example of segments of a typical archive file follows-

```
Fablelands reduced set of 336 long-lats
f.dat
(12f7.1)
336      12      3      1      2  -0.9999E+04
6.6      9.2      0.4      2.0      19.6      -3.7      23.4      3.1      12.7      12.9      0.3 1509.0
6.8      9.7      0.4      2.0      20.0      -4.1      24.0      3.4      13.0      13.1      0.4 1413.0
6.5      9.8      0.4      2.0      19.7      -4.5      24.2      3.2      12.7      12.9      0.1 1374.0
.....(more of the same)
11.0      11.5      0.5      2.0      22.6      -0.7      23.4      16.1      6.2      16.2      5.5 830.0
11.8      11.7      0.5      2.0      23.3      -0.2      23.5      16.9      7.1      17.0      6.3 823.0
13.3      11.4      0.5      1.0      24.1      1.3      22.8      18.2      8.8      18.2      8.1 840.0
13.8      11.1      0.5      1.0      24.2      1.9      22.3      18.5      9.4      18.5      8.7 827.0
Obj00001Obj00002Obj00003Obj00004Obj00005Obj00006Obj00007Obj00008Obj00009Obj00010
Obj00011Obj00012Obj00013Obj00014Obj00015Obj00016Obj00017Obj00018Obj00019Obj00020
.....(more of the same)
Obj00311Obj00312Obj00313Obj00314Obj00315Obj00316Obj00317Obj00318Obj00319Obj00320
Obj00321Obj00322Obj00323Obj00324Obj00325Obj00326Obj00327Obj00328Obj00329Obj00330
Obj00331Obj00332Obj00333Obj00334Obj00335Obj00336
T.Annua T.DRang T.Iso  T.CoFv  T.Max   T.Min   T.ARang T.Wet.Q  T.Dry.Q  T.Hot.Q
T.Cold. P.Annua
```

Option 7 Archive file --> PATN

This is the retrieve component of the archive/retrieve pair. DATN requests the name of the input archive file and separates this file into the appropriate **PATN** files:

1. Parameter file *PATN.PRM* containing parameters as at archival time
2. Data or association file (*-.dat* or *-.aso*)
3. Row label file (*-.rlb*)
4. Column label file (*-.clb*)

These file are then immediately available to **PATN** in a suitable form (*PATN.PRM* is updated immediately).

Free-format archive files may be input through DATN. This is the reverse of free-format output of option 6 above. DATN detects free-format archive file by the presence of a quote character (") in the first character position of the first record of the file. If a quote is detected, DATN will use free format to read the archive file.

Option 8 Transpose Data and Label Files

This option is provided to exchange the rows and columns of a data file and to swap the row and column label files. Transposition is where the first row becomes the first column, the second row the second column, etc.

The input data file and format is as nominated by the parameter file and the output file name and format are requested. As with all options producing legal **PATN** files, the user is given the option for updating the parameters in the parameter file.

DATN transposition is required for any classification, ordination or other analysis of the columns of data. Some care is required to name files appropriately when commands such as TWAY are used. If TWAY is to be used, perform the analysis *first* on the items that will form the columns of the two-way table. This will save time because, after the first analysis and transposition, the data will be in the form that it is to be listed in the two-way table.

Option 9 Interactive Data Entry

This option permits interactive entry of a data matrix directly into **PATN** format. DATN expects that the number of rows and columns in your data have been defined using PRAM (this amount of memory is set aside to store your data). Optionally, you may enter minimum and maximum data values for each of the columns of data to be entered. If supplied, **PATN** will use these values to range check the data values as you enter them.

Data is to be entered either by row or by column. If row entry is selected, all data values for row 1 would be entered, one after the other, from column 1 to column M where M is the number of columns as defined in *PATN.PRM*. The second row would then be entered.

This option must be used for a complete data matrix; data entry cannot be interrupted, and then re-commenced after exiting from DATN. You can however enter a sub-matrix and subsequently append or merge the separate files. This option will not create ROW and COLUMN labels, so use LABN for this.

Option 10 Interactive Data Editing

This options provides a simple means of editing values in the data matrix currently defined by *PATN.PRM*. Given that data is stored in an unformatted form, this represents a simpler method than exporting, editing and re-importing the data.

The ROW and COLUMN sequence numbers are requested and the current value is shown on the terminal. A new value may be entered or, a <CR> will retain the current value. This is a simple method of displaying selected data values.

DATN will continue to request row and column numbers until a '0' ROW sequence number is entered. At this point, the output file name is requested and written.

Option 11 'Relational' format -->PATN

This data structure is termed "relational" because it is often the style associated with the storage of sparse (many zero entries) in relational databases. This format, like CEP-style, only nominates the data entries that are *non-zero* in a data matrix (flat-file). This style of format looks like this:

1	2	1
1	5	2
2	1	4
3	9	1
3	8	2
.	.	.
.	.	.
.	.	.
109	73	1

Two or three column versions of this format can be used. The first column in the table above refers to the *row-index* or the object-number, the second column to the *column-index* or attribute-number while the third column is the *value*. In the example, above, a '1' would be placed in a (zeroed) matrix at position (coordinate if you like) row 1, column 2. Similarly, a '2' would be placed at row 1, column 5, a '4' placed in row 2, column 1 and so on down to a '1' placed in row 109, column 73. These two numbers must be able to be read and interpreted as whole numbers (integers using FORTRAN 'I' format)

If the third column is absent, a '1' is assumed (presence/absence data). **NOTE:** in this instance free-format cannot be used to read the data. All entries in the matrix that are not specifically nominated in the input file are coded as '0.000' (zero). The values in this third column, if present, are read as FORTRAN floating-point numbers ('F'-style format). **If using fixed-style format, the format specifier of the third column must be stated, even if the record only contains the two index values.** Simply tack an ',f1.0' on the end of the format if this is the case. This will enable blanks to be read as zeros, which PATN will convert into '1's. Typical formats would therefore look like:

```
(2i5,f8.3)
```

or

```
(i5,i4,f1.0)
```

or

```
*
```

if using free-format.

As usual, PRAM must be run to establish the actual number of rows and columns (N,M) to be anticipated in the resulting data matrix (flat-file) that PATN will store. This is **not** the number of rows and columns in the input file. In the example above, it may be that the resulting file will be 109 rows by 73 columns if these are the corresponding row and column-index **maxima**.

The data not have to be in any particular order, for example, you could nominate row 50 column 20 before row 1, column 10. In the situation where a row-column pair have already been used, PATN will write a warning to the screen but will none the less, use the last value as the value to be placed at that location.

Option 12 PATN -> 'Relational Format'

This is the reverse of option 10 (Relational to PATN format). DATN accepts as input, a standard PATN data file and exports it into a form that containing three columns.

The first column is comprised of integers nominating the row of the data values in the PATN data file. The second column contains integers referring to the column of the data value. The third column contains the actual data values.

For example a value of say 3.5 in the third row and seventh column of the PATN data file would be represented in the relational file as '3 7 3.5'. Each non-zero value in the input PATN data file is therefore represented by one record of the export relational-format file.

Row and column labels are not exported from PATN's label files using relational format. In this instance, use LABN to export the label files in ASCII format. I suggest a format specifier in this case should be '(A)'.

Option 13 'PRIMER' -> PATN

PRIMER is a similar package to PATN. PRIMER's quick format is similar to the CORNELL (CEP) format in that only non-zero values are nominated. In other words, these formats are designed to handle what is termed 'sparse' data. That is, the situation where the data matrix has a relatively large number of zero values. This is often the case with 'ecological' data where species usually only occur at a subset of possible sites.

Parameters must be specified using PRAM before this option can be used. The number of rows and columns must match the maximum values that are to be expected in the imported PRIMER file. For example, the largest site identifier (number) must be less than or equal to the number of rows in the parameters. If the maximum site identifier is less than the number of rows parameter, zeros values will be used to fill the additional sites. Likewise, the number of columns nominated by parameters must be greater than or equal to the maximum species identifier. Unused species identifiers will likewise be filled with zero values.

PRIMER's quick format **does not include** any row or column labels. These must be handled through LABN in PATN.

The format is based on specifying the number of species at each site and then nominating the species identifiers and the values. All site and species identifiers must be sequential integer values.

The format can be considered as a repetition of three components. The first component, on one record in free format (integer values separates by at least one space and or a comma) states the site number and the number of non-zero species at that site. The second component, on the following records, nominates, as pairs of integers, the species identifier and the abundance. For some strange reason, the next record must contain an asterisk (*) in the first position of the record. This is the third component.

The three segments are repeated for each of the N sites in the data set. An example of the format would look like this-

```

1 4
2 10 15 5 25 4 31 1
*
2 2
7 9 11 3
*
3 1
9 21
*
...

```

The first 'site' contains four species, identified as '2', '15', '25' and '31' with abundances 10, 5, 4 and 1 respectively. The '*' is used as a record separator. This pattern is repeated for site '2' and site '3'.

PATN will by default, expect that the name of the import PRIMER file will have the extension *-.qck*. This is the standard extension produced by PRIMER for the 'quick' format.

Option 14 PATN -> 'PRIMER'

This option is the reverse of option 13. PATN will assume that the current default parameters are identifying a file that you wish to be exported to PRIMER's quick format.

All of PRIMER's format specifiers are of the type INTEGER. **This means that all decimal values will be truncated on export to PRIMER format.** If decimal places are significant, try using TRND to multiply by 10 or 100.

PATN will export data in exactly the same format as specified on import from PRIMER (see option 13). PATN will ask for an output file name and a format. Integer formats only should be used.

Option 15 EXIT

This option exits DATN without doing anything. It is included for consistency with other modules to see the options without committing yourself to data input or output.

LABEL INPUT-OUTPUT:

LABN

OUTLINE

LABN deals with the input and output of labels to and from **PATN**. It is **required** to get a set of labels into the package the first time, and may be used to create, output, tabulate or match label files.

Most **PATN** options require a set of labels up to 8 characters in length for each row and column of the current data file. The **PATN** label files are -

-.rlb (row labels) and
-.clb (column labels)

where '-' is the ROOT name in *PATN.PRM*. For example, a data file-

fred.dat

should have the associated files-

fred.rlb
fred.clb and
fred.prm (see PRAM).

The label files are *unformatted*. They cannot be printed or edited and must be created via the LABN module.

Labels are used by **PATN** to annotate all output and should be chosen in a meaningful fashion. The labels provided to **PATN** will also be used to reflect operations on a data file such as:

1. Masking (MASK),
2. Merging (MERG)
3. Sorting (SAMP)
4. Transposition (DATN)
5. Sampling (SAMP).

The LABN module has the same list of options for both row and column labels. A complete set of labels contains 'N' labels in *-.rlb* plus 'M' labels in *-.clb*.

OPTIONS

Option 1 Auto-Generation of Labels

This option is used when the labels are not important; where sequence numbers will suffice. This might occur when 'playing' with **PATN** with a test set of data or when foreign data is imported for a quick analysis. This option requests a three character *base* leaving up to 5 digits for the sequence numbers that **PATN** will automatically generate. For example, if *SAM* is the base and there are five rows, the labels that would be generated using LABN option 1 would be:

```
SAM00001
SAM00002
SAM00003
SAM00004
SAM00005
```

By default, *ROW* is used as the base for row label generation and *COL* for column labels.

Option 2 Input or Edit Labels from Terminal

This option is useful when the number of labels to be entered is not great. If more than about 30 labels are to be entered for the first time, it is probably more efficient to use an editor or word processor to enable concurrent editing of labels as they are entered (LABN will not allow corrections once the <CR> key is pressed) and periodic saving of labels if entry is spread over a number of sessions.

If a set of labels exists, they will be used as defaults when LABN prompts for any alterations. Selection of which labels to enter or edit is done through an integer value -

1. a positive *integer* selects, via the sequence number, a single label.
2. ZERO requests LABN to prompt for alterations to ALL labels in sequence order (from 1 to M) while
3. A negative sequence number stops entry and editing.

If labels exist, pressing <CR> will retain the existing label while any other entry will overwrite the existing label. If no labels exist, the default will be 8 blanks.

Option 3 Read labels from an ASCII file into PATN

This option is suitable when more than about 30 labels are to be entered into **PATN** for the first time. It is easier to use an editor and create the labels at leisure than to attempt to enter labels as a response to prompting by PRAM. Using the editor, typographical errors can be corrected on the spot, while incorrect input to a running program is not readily retrieved and corrected. In PRAM, typing errors can only be corrected by re-running the module after all labels have been entered.

The *name* of the file containing the ASCII labels and the associated FORTRAN A-type format is required. This combination allows for labels to be read from just about any file and format combination. For example, labels may be read as the first or last field from a fixed format data file.

Free Format: if an '*' is given as the format of the labels, LABN will assume that the labels are in the form:

"label11"	"label12"	"label13"	"label14"	"label15"	"label16"	"label17"
"label18"	"label19"	..."labelM"				

Labels should be enclosed in single or double quotes. There can be any number of spaces between labels and any number of records can be used. LABN will read until M labels are obtained.

As a check on the input format, LABN examines the contents of each of the labels read and if any contain the default 8 blank characters, a warning message is given.

Option 4 Write PATN labels to an ASCII file

This is the reverse of the previous option. The row or column label file is written out to a user-specified file in ASCII using a supplied FORTRAN format. For example, a format of (A8) will write one label per record, using N records or rows or 'M' records for columns.

The default extensions used for the ASCII files are *-.row* for row labels and *-.col* for column labels.

Option 5 Tabulate the Labels to an ASCII file

This option tabulates labels against sequence numbers. The files used for output are *-.trl* (rows) or *-.tcl* (columns) using a two column format.

A sub-option asks if the labels are to be sorted using the standard ASCII collating sequence (Yes/No). If labels are not to be sorted they are listed in sequence number order. If sorting is to be performed, LABN requests the first and last character position to be used for the sorting process and performs an ascending sort.

Option 6 Indices of Sub-set of the Default Labels

This option has been designed to determine matches between two sets of row and/or column labels. This procedure is useful when masking a secondary set of data so as to match a primary set. In this case, a series of masking/re-ordering options may have been performed on the secondary set and it is now desired to align the primary and secondary set for further analyses (eg. GSTA)

A common application for this option arises when dealing with an *extrinsic* data set. This refers to a secondary data set that has the same set of objects as the primary set used in the analysis, but with a different set of attributes. For example, in biological applications, sites are often classified on the basis of species (intrinsic attributes). A second set of data contain a set of environmental attributes (extrinsics) such as rainfall, temperature, geology, altitude etc. In this situation, the species set is often masked to eliminate sites that have all or no species, or the data set is sampled by sub-setting. The analysis results in a group definition file and an ordination and the impact of various species is determined through GSTA and PCC.

What may be required in this instance is to determine if there is any pattern of relationships between the site groups based on biotic information and the environmental attributes (extrinsics). To do this however, the extrinsic data set must be made to match the species data, at least in terms of the number and order of sites in this example. This LABN option does this. The extrinsic set must be the current data set as defined by the parameters (*PATN.PRM*). In the example, LABN would then read in the row (site) labels of the species data set and match them to the site labels of the environmental data. This would create a set of indices that could be used to MASK and re-order the rows of the environmental data so as to match the species data.

The group definition file *-.gdf* would then apply to the extrinsic as well as intrinsic data set. GSTA could then be used on the extrinsics. In addition, PCC could also be used to correlate extrinsics to ordination results.

Two files are produced, both ASCII. A *-.trl* or *-.tcl* file will be created if mismatches occur. This means that the sub-set label didn't match any of the existing labels. One line is written detailing which labels didn't match.

The second file contains, in user defined FORTRAN (I) format, the index (sequence number or position) of the sub-set label in the default set.

Another example this time using columns. MASK may have been used to select (a direct MASK) a sub-set of rows of the data matrix for analysis. Indirect masking subsequently eliminated a number of columns. If an identical analysis on another sub-set of rows with the same columns is required, this LABN option could be used to construct the data set.

First, set the parameters to refer to the original (complete) data matrix and select this option for the column labels in LABN. Provide the name of the file containing the column labels referring to the first sub-set data and the result will be a file containing index numbers that can be used for the columns in MASK for further sub-setting of the rows of the original matrix.

Option 7 Skip Row or Column Phase

All of the above options can relate either to row labels, to column labels or to both. In many instances, input or tabulation of labels may only apply to either row or column labels. This option can be used to skip the unwanted phase (row or column).

ASSOCIATION

INPUT-OUTPUT:

ASON

OUTLINE

ASON is designed to input and output files containing a set of *associations* in various formats and structures. It is the counterpart to PRAM, LABN and DATN for association measures.

Structure

PATN has an internal form and structure for storage of an association matrix. This is a row-wise, lower symmetric matrix of off-diagonal (usually) dissimilarity values in binary (unformatted) format. An example of the *structure* (unformatted files are not readable) of an order 4 matrix is -

```
0.1000
0.2000 0.3000
0.4000 0.5000 0.6000
```

where:

```
0.1000 refers to the dissimilarity between Objects 2 & 1,
0.2000 " " " " " " " 3
1,
0.3000 " " " " " " " 3
2,
0.4000 " " " " " " " 4
1,
0.5000 " " " " " " " 4
2,
0.6000 " " " " " " " 4
3.
```

Using this style of format, it is the *position* or order in the file that determines what pair of objects a particular value refers to. There are no associated labels, except as stored by **PATN** in the associated label file (*-.rlb*). There is however an annotation option.

The various structures that can be used for this form may be useful in the context of some commands in **PATN** or they may represent forms that are more 'digestible' to other programs and packages. The formats included in ASON are-

1. Lower symmetric ('standard' **PATN** form)
2. Annotated lower symmetric
3. Upper symmetric
4. Full (square) symmetric
5. Single Row Vector (as data format)
6. Specified individual comparisons

Each of the options below performs a translation of one style of format to another. In some cases, ASON may need to be run more than once. For example, to transform an upper symmetric form to a symmetric form. In this case, the upper to lower option and then lower to symmetric.

Symmetry

The symmetrical nature of the matrix arises because the relationship between a pair of objects is symmetrical; the value of dissimilarity between Object 4 and 2 is the same as the value between 2 and 4. The diagonal is absent in this format because it is usually assumed to contain zeros, ie. the 'distance' between an object and itself is usually assumed to be zero. In situations where this is not the case, a separate file (*-dia*) is used.

It is customary to use the term *order* to refer to the size of the association matrix. An order 10 matrix has $10 * (10-1) / 2 = 45$ values.

Input/Output

All **PATN** modules, other than ASON assume a lower symmetric matrix (the default). If an association matrix was generated outside **PATN**, ASON **must** be used to read it and to put it into this standard form.

ASON will always request an output file NAME and, depending on the option, a FORTRAN format. The standard **PATN** name for the default/internal association file is

-.aso.

As with most data in **PATN**, the format is applied to *floating point variables* and consequently needs to be one of *E*, *F* or *G* type.

REFERENCES

- Faith, D.P., Minchin, P.R. and Belbin, L. (1987). Compositional dissimilarity as a robust measure of ecological distance. *Vegetatio*, 69: 57-68.
- Belbin, L. (1991). Semi-strong hybrid scaling, a new ordination algorithm. *J. Vegetation Science*, 2: 491-496.

OPTIONS

Option 1 Lower --> Annotated Lower

This option provides an output file containing the lower symmetric matrix with marginal row labels. The output is in a multi-page tabular form where individual dissimilarity values for any pair of objects can be quickly located. The structure of the symmetric matrix for orders larger than the width of the page (seven values) may be reconstructed by cutting across the pages on the table breaks and placing each section to the right of the previous one.

An example of the output format for the example above is:

	Object-1		Object-2	
Object-2	0.1000			Object-3
Object-3	0.2000	0.3000		
Object-4	0.4000	0.5000	0.6000	

Option 2 PATN --> Upper Symmetric ASCII file

This options takes as input, the standard, unformatted lower symmetric *-.aso* matrix and *reflects* it about the diagonal to output it in a formatted upper symmetric form. The output file name and FORTRAN format is requested.

The effect of the transformation, given as input-

0.1000			
0.2000	0.3000		
0.4000	0.5000	0.6000	

would be-

0.1000	0.2000	0.4000
	0.3000	0.5000
		0.6000

Option 3 ASCII upper --> PATN

This is simply the reverse of option 2 above. When a non-standard form is input, ASON requests:

1. The input file NAME and
2. The data format and
3. The output file name.

Option 4 PATN --> ASCII Symmetric (Square)

With this option, ASON will reflect the contents of the standard **PATN** lower symmetric matrix around the diagonal so as to produce a symmetric square matrix. A value to be placed in all diagonal positions is requested. For example, if the input matrix was:

0.1000		
0.2000	0.3000	
0.4000	0.5000	0.6000

then the output matrix (using 0.0 for the diagonal values) would appear as:

0.0000	0.1000	0.2000	0.40000
0.1000	0.0000	0.3000	0.50000
0.2000	0.3000	0.0000	0.60000
0.4000	0.5000	0.6000	0.00000

This option may be of use to external multidimensional scaling packages such as ALSCAL or KYST.

Option 5 Symmetric --> PATN

This is the reverse of option 4 above. A symmetric matrix is read, the diagonal is dropped and the lower triangle portion is output in *standard PATN* form.

Option 6 PATN --> Vector

This option writes the standard **PATN** lower symmetric matrix in a linear or vector form with a user defined FORTRAN format. For example, if the input file was:

0.1000			
0.2000	0.3000		
0.4000	0.5000	0.6000	

and the output format requested was:

(1X,16F8.4)

then the output file would appear as:

0.1000	0.2000	0.3000	0.4000	0.5000	0.6000
--------	--------	--------	--------	--------	--------

This option may be useful when comparing the association values for a number of different association measures. In effect, this option transforms an association-type matrix into a data-type matrix containing one row (object) and $N*(N-1)/2$ attributes. An alternative to this would be to use the subtraction option of the **PATN** command TRNA.

Option 7 Vector --> PATN

This is the reverse of option 6 above. An input file name and format are requested and the **order** of the matrix is assumed to be as detailed by the parameter file *PATN.PRM*. The output file is in standard **PATN** form (lower symmetric unformatted).

Option 8 PATN --> Lower Symmetric ASCII

This option provides a readable version of the standard *-.aso* matrix. A FORTRAN format is requested. Any 'E', 'F' or 'G' format may be used. For example the format (1x,3f8.4) produces-

0.1000		
0.2000	0.3000	
0.4000	0.5000	0.6000

while (1X,10F4.1) gives-

```

0.1
0.2 0.3
0.4 0.5 0.6

```

Option 9 Lower Symmetric ASCII --> PATN

This is again, the reverse of the previous option. A formatted lower symmetric matrix, stored row-wise in an ASCII file is read into a standard unformatted **PATN** form. The name and format of the incoming matrix is requested.

Option 10 PATN --> Masked PATN

This option is the counterpart to MASK for association files. While MASK accepts details of rows and columns of the data matrix that are to be retained, a list of objects that are to be **deleted** from the output matrix are requested.

For each object eliminated from the matrix, one row and one column are *deleted* from the lower symmetric matrix. For example, consider the matrix-

```

0.1000
0.2000 0.3000
0.4000 0.5000 0.6000

```

Deleting object 3 would result in a matrix-

```

0.1000
0.4000 0.6000

```

A new label file is created. Both the input and output matrices are in standard **PATN** (unformatted) form.

Option 11 Pairs --> PATN

This option is designed for entry of a limited number of association measures as 'raw data'. This form will be convenient if some or many values are missing. ASON will assume that any missing pair-wise association value is set to the user defined **missing value**. The order (size) of the matrix is assumed from the parameter file *PATN.PRM*.

The input data file name is requested. The format is assumed to be FORTRAN free. The number of pair-wise entries is determined by the number of records in the file. The structure of the input data is:

```

I1  J1  D1
I2  J2  D2
I3  J2  D3
.   .   .
.   .   .
.   .   .

```

where I1 and I2 refer to the sequence numbers of the objects and D1 is the value of dissimilarity between them

For example, the input file:

```

2  1  .1
3  1  .2
3  2  .3
4  1  .4
4  2  .5

```

with the missing value '-9.', would result in:

0.1000	
0.2000	0.3000
0.4000	0.5000 -9.0000

Option 12 Histogram of Associations

This option, while fundamental, has significant implications for (Pattern) analysis techniques that use the association measures (most do!). A histogram is created from the association values in default *-.aso* file. The shape of the distribution and the values involved provide insights into the structure of the data and the nature of the association measure.

If the distribution was smooth and unimodal, one would assume that no distinct (natural?) groups are present in the set of data. A bimodal distribution may, on the other hand, signal two groups. In many cases, the shape of the distribution will reflect the limitations of the measure of association used. It is noted (Faith, Minchin & Belbin, 1987 and Belbin, 1991), that once objects fail to share non-zero values on attributes, even the best association measures *underestimate* the dissimilarity. When using Bray & Curtis or Kulczynski measures (see ASO), a dip in the curve at 0.9 and subsequent heaping up of values at 1.0 is often noted (see SSH). This is due to underestimation of values at around values greater than 0.8-0.9.

This limitation can be addressed in a number of ways. TRNA has a number of options to re-estimate the larger association values. The best option there is by the *shortest path* algorithm. This however, may result in over-estimating some of the larger values. This may however be preferable to the situation where there are a significant number of tied association values of 1.0 (see TRNA). If ordination is to be performed on the association measures, SSH can be used to re-estimate the larger values and attempt to find an optimal configuration of the objects. Lastly, the flexible UPGMA algorithm in FUSE may be used, selecting a slightly negative β value (-0.1) to dilate the multivariate space and thereby, re-estimate the larger values (see FUSE).

Option 13 Edit Association Values

This is a similar option to the editing option in DATN except that the object (row) pair is nominated and the associated value of association is listed for editing. A new value may be entered and the <RETURN> key pressed, or the old (current) value may be retained simply by pressing the <RETURN> key without altering the displayed value.

Option 14 Export Association Values & Group Codes

This option has been designed to enable the export of association and group values to statistical packages where they may be further analysed.

This option has been prompted by a number of requests from users who want to perform some sophisticated analysis that further the ANOSIM approach (see ASIM). Basically, ASIM examines the between and within group variances based on association values. The raw data for ASIM are the pair-wise association values and an identification if the pair are in the same or different groups.

This option requires two files as input, a standard PATN association file (*-.aso*) and a file nominating groups. The latter could either be a standard group-definition file (*-.gdf*) or a group affinity vector (*-.gav*). The former is the PATN standard and is in the form of groups by members and the latter is in the form of members by groups. The association file is *binary* (not directly readable) while the grouping file is ASCII (readable).

The format of the data exported from the option is ASCII (readable) and contains four columns and one record for each pair of values. The first column contains the sequence number (identifier) of one of the pair of sites. The second column nominates the second object of the pair. The third column contains either a '0', stating that the pair of objects are in different groups or a '1' stating that the pair are in the same group. The fourth and last column is the association value between those pair of sites.

An example of the format follows-

```
2  1  0  0.7564
3  1  1  0.9876
3  2  0  0.1327
4  1  1  0.2376
4  2  0  0.3471
4  3  0  0.0192
...
...
```

This shows that objects 3 and 1 are in the same group. Likewise objects 4 and 1 are in the same group, the other pairs as shown in the example are not in the same group.

The file is written in the order of a lower triangular matrix. There will therefore be $N*(N-1)/2$ records in the output file, one for each pair of objects nominated by the current parameters in PATN.PRM.

Option 15 EXIT

This option exits ASON without doing anything. It is included in case you want to see what options are available without committing yourself to data input or output.

GENERATING DATA:

RAND

OUTLINE

The RAND module generates DATA matrices of N rows and M columns where N and M are user defined. Any sequential group of 1 or more rows and columns may be generated by any of the following RANDOM VARIATES:

1. Uniform
2. Uniform integer
3. Log-random
4. Complimented log-random
5. Poisson
6. Exponential
7. Normal
8. Log-Normal
9. Gamma

Purpose

RAND is useful for the fast GENERATION of data, either for getting acquainted with PATN, or for testing and comparison of various **PATN** algorithms or options. RAND generates:

1. A **PATN** data file,
2. *PATN.PRM* and *-.prm*
3. Label files for rows and columns

providing an efficient means of trying out the various options in **PATN** without the concern for maintaining the integrity of your real data. Options may be selected and re-selected with different parameters with the test data until you are familiar with how **PATN** is used. At this stage, all files resulting from experimentation can be deleted by a command such as:

DELETE *random**.* (assuming you have no other “random” files!)

If default file names are used in RAND and maintained throughout the analysis, all files generated will begin with the name *random* and have different extensions, depending on the options chosen.

Because cluster distributions with a variety of specifications can be built up with RAND, the testing of specific options and sub-options and the comparison of the applications of different algorithms in **PATN** is facilitated. For example, a set of clusters may be defined and the data applied to FUSE and to PDIV. Comparison of the results should provide insights into the advantages and disadvantages of the two methods.

Data Blocks

The data matrix may be 'built-up' by the addition of one or a range of contiguous rows and columns which together define a *block*. This block of data may be generated using any one of the random variates and may be at any position in the overall data matrix, for example, a normal distribution with mean of 10.0 and standard deviation of 5.0. Any number of blocks may be generated or regenerated over the top of pre-existing data. The largest defined row and column specifies the size of the *enveloping matrix* that is output.

Consider an example with 4 blocks:

1. ROW 1 to ROW 10, COLUMN 1 to COLUMN 3 with UNIFORM
2. ROW 1 to ROW 10, COLUMN 4 to COLUMN 6 with NORMAL
3. ROW 1 to ROW 10, COLUMN 7 only with POISSON distribution
4. ROW 11 to ROW 12, COLUMN 1 to COLUMN 7 with LOG NORMAL

Parameters & Labels

The resulting matrix is 12 rows by 7 columns. Block specification terminates on entry of a **zero** for the first row to be generated. Values in the matrix within the enveloping matrix that have *not been defined* have value of zero. The row and column label files contain the labels -

```
"ROW 1" to "ROW N" and
"COL 1" to "COL M"
```

The master parameter file *PATN.PRM* is automatically updated.

Presence/Absence Data

Presence/absence (0/1) data may be generated by a number of different methods. Probably the simplest specification is:

1. the *uniform* distribution
2. *minimum* of 0.0
3. *maximum* depending on sparsity ratio (non-zero/total values)

```
range 0.0 - 1.49 (sparsity ratio = 0.5)
range 0.0 - 0.7 (sparsity ratio = 0.3)
```

4. format of (F2.0) or (F2.0,TL1).

Random Variates & Seeds

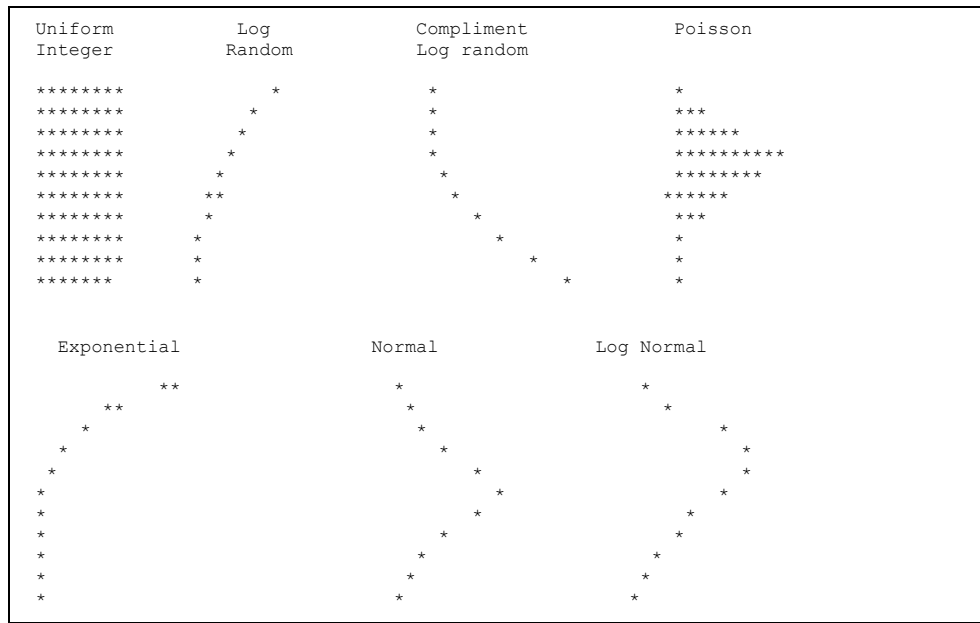
The method used for generating **random numbers** is based on the multiplicative congruent method. The basic relationship is:

$$X_{n+1} = a * X_n \pmod{m}$$

where *a* and *m* are non-negative numbers. The method appears reasonably robust and the major advantage seems to be that the same numbers are generated on any computer using 4 byte floating point storage.

When generating random numbers for any of the distributions, RAND as other options in **PATN** requests a *seed* value. This is the value that *initialises* the sequence of random numbers in the range 0.0 to 1.0 as generated by **PATN**'s internal random number generator. Just about any value will do as the seed. The significance of the seed is that the same *sequence* of random variates will be generated by the use of the same seed value (regardless of the computer). This is useful for re-generating particular sets or sequences of data.

The general SHAPES of the distributions are as follows:



The γ distribution is particularly useful, in that various shapes can be generated between the exponential to right skewed normal by varying only the mean and variance.

Precision

RAND stores the random variates in four-byte floating-point words. When the data generation has been completed, it is output in binary form to the selected data file. Under some circumstances, full precision is not desired. An option is provided to reduce the number of significant decimal places. Integers may be created by using 0.

REFERENCES

- Box, G.E.P. and Muller, M.E. (1958). A note on the generation of random normal variates. *Annals of Math. Stat.* **29**, 610-611.
- Marsaglia, G. and Bray, T.A. (1964). A convenient method for generating normal variables. *SIAM review* 6(3), 260-264.
- Phillips, D.C (1971). Generation of random gamma (γ) variates from the two-parameter gamma (γ). *AIIE transactions* 3(3).
- Shannon, R.E. (1975). *Systems Simulation*. (Prentice Hall).

Belbin, L., Faith, D.P. and Minchin, P.R. (1984). Some algorithms contained in the Numerical Taxonomy Package NTP. CSIRO Division of Water and Land Resources Technical Memorandum 84/23.

OPTIONS

Option 1 Uniform

This is the simplest of the distributions available. The technique used is a simple linear transformation of the output from PATN's random number generator which produces values in the range 0.0 to 1.0. The option will request the minimum and the maximum and generate floating point values, uniformly distributed between the minima and maxima. The formula used is:

$$\text{Value} = \text{minimum} + \text{range} * \text{random value}$$

Option 2 Uniform Integer

This option is produced as the nearest integer of the values produced by option 1.

$$\text{Value} = \text{nearest integer} (\text{minimum} + \text{range} * \text{random value})$$

Option 3 Log Random

The log-random function is generated using the uniform distribution as follows:

$$\text{Value} = E (\log(\text{min}) * \log(\text{range}) * \text{random value})$$

Option 4 Complimented Log Random

The complimented log-random function is generated using the formula:

$$\text{Value} = \text{max} - e (\text{random value} * \log(\text{max} + 1 - \text{min})) + 1.0$$

Option 5 Poisson

The Poisson distribution is based on a set of independent Bernoulli trials (yes/no) each having a small probability of the event occurring. The distribution is used for the number of events in a period of time, for example car/plane crashes and other disasters. Like the uniform integer distribution, the Poisson takes discrete integer values with a mean equal to the variance. The Poisson distribution is derived from the exponential on the basis that if the time period between events is exponentially distributed, then the number of arrivals per unit time is Poisson distributed.

Option 6 Exponential

If the probability of an event in a time unit is small and such events are independent, then the time interval between events is exponentially distributed. Length of telephone conversations, time between component failures, people arriving at a queue, time between disasters, are all examples of exponential distributions. The distribution is based on the formula:

$$\text{value} = -\text{mean} * \log (\text{random value})$$

Option 7 Normal

The normal or Gaussian distribution is probably one of the most common distributions where the measurement of some attribute tends toward the mean value, dropping off in probability symmetrically on each side of the mean. The method used to generate normal variates is based on Box and Muller with a modification proposed by Marsaglia and Bray.

Option 8 Log-Normal

Values of this distribution are derived directly by using the normal distribution as outlined above.

Option 9 Gamma

The Gamma distribution is underrated because it can simulate a family of unimodal curves ranging from log-type skewed, through to normal using different means and variances. The method used to generate Gamma variates is due to Phillips. It is versatile in allowing non-integer means. The parameters most often used in referring to the Gamma distribution are

$$\begin{aligned}\alpha &= (\text{mean} * \text{mean}) / \text{variance} \\ \beta &= \text{mean} / \text{variance}\end{aligned}$$

SAMPLING DATA:

SAMP

OUTLINE

SAMP provides a number of options for extracting a sample of rows or columns from the current **PATN** data file. This function may be useful for at least the following situations:

1. The basic data file is larger than the capacity of one of the **PATN** options.
2. When a quick profile of a large data set is required.
3. When a set of seed points is required for ALOC.
4. When it is suspected or known that rows may contain duplicate data.
5. When a data file, after various pre-processing operations, may contain duplicated columns. This often occurs when columns are species and they have been given slightly different names (labels).
6. When a random re-ordering of rows is required.
7. When a group or groups of data defined from any prior (internal or external) analysis is to be analysed.
8. When performing any type of Monte-Carlo type sampling.
9. When it is desired to take a proportional sample of all defined groups.

SAMP may create the following new files for each run:

1. The sample data set and OPTIONALLY:
2. New row label file (*-.rlb*),
3. New column label file (*-.clb*),
4. *PATN.PRM* and a
5. saved parameter file (*-.prm*)

The sampling schemes currently available are:

1. Regular
2. Random
3. A pre-defined group of Rows
4. Unique Rows
5. Unique columns
6. Monte-Carlo
7. Sort
8. Proportion of all groups

The first two options are straight-forward sampling options that you would use to **subset** data. The third option is best used when a more detailed analysis of a particular classification group is required. Option 8 is similar except that a fixed proportion of objects in each defined group is sampled.

The options for unique rows and columns are superficially similar but have radically different approaches for different requirements. The unique rows option is for eliminating duplicated **data** while the unique columns option eliminates columns with duplicated **labels** (regardless of data).

The **Monte-Carlo** sampling is a recent addition. This type of statistical modelling appears to be increasingly useful in pattern analysis and cladistics. This option simplifies within column randomisation.

The **sorting** option has been added to replace the module SORT. A simple re-ordering of rows of the data matrix based upon values in a single column is performed. Multi-key sorting is not supported in SAMP.

If the parameters are not updated, SAMP assumes either that the sample file is for use as seed values to ALOC and no **labels** are required, or that the sample file is not required immediately. SAMP will ask if labels are to be created and if so, in addition to the modified label files, a new parameter file (*-.prm*) is created to match the output file. The *-.prm* file can be copied to *PATN.PRM* if the modified file is required in **PATN** at some later time. If no updating and labels are requested, then the output sample file should NOT have the same *ROOT* file name as the input data file.

REFERENCES

- Belbin, L., Faith, D.P. and Minchin, P.R. (1984). Some algorithms contained in the Numerical Taxonomy Package NTP. CSIRO Division of Water and Land Resources Technical Memorandum 84/23.

OPTIONS

Option 1 Regular Sample

The **regular** sampling scheme requests the sequence number of the first row and a value for k where k represents sampling every kth row after the first. ie.

$$\text{Next row} = \text{current row} + \text{increment}$$

If the sequence of the data set in some way corresponds to a stratification of attribute values, the regular scheme provides a meaningful sub-set of the data. For example, to take one fifth of the rows as samples, use a start row of one and an increment of 5.

With this method, the exact number of rows in the sample is not immediately apparent and so SAMP will inform you of the number.

Option 2 Random Sample

Random sampling selects k rows from the data set in a **random** order. You select 'k' and a seed value and SAMP does the rest. Using the same seed value will result in the same number and sequence of sample rows. The random option may be used to randomise the complete data set by selecting N (total number of rows) as the same size.

The algorithm generates N random values between 0-1. These values are then sorted into ascending order and the first K indices of the sorted list select the rows to extract from the data.

Option 3 Group Sample

Any **group** of rows, as defined by either

1. a standard group definition file *-.gdf* or
2. a group affinity vector

may be extracted to the sample file (along with associated parameters and labels). This option may be useful when further analysis is required to stratify or subdivide a sub-group of objects.

This option operates on any data file and the associated label files. SAMP requests the sequence number of the group representing the objects (rows) to be extracted from the input file to the sample file.

Option 4 Sample Unique Rows

Sampling **unique** rows of data should be used when duplicated data is suspected. The probability of this occurring increases with presence-absence data sets with an increasing number of rows and decreasing number of columns.

Using this option, SAMP writes the **first** row to the sample file and then checks all **subsequent** rows looking for a match. Any matches found are noted and eliminated from subsequent searches. The next row that remains is then selected and the procedure repeated until all rows have either been selected or eliminated.

The scanning process corresponds to the examining of the indices of an upper triangular matrix ie:

```

Row 1      is compared with 2  --> N
Row 2+ "    "                  "    3+ --> N
Row 3+ "    "                  "    4+ --> N
.    .    .    .    .    . and so on

```

In the following data rows 2 and 4 are duplicates, row 2 will remain-

```

0000100000010000
0100010001000001
0010010100001000
0100010001000001

```

Option 5 Sample Unique Columns

This option compares column **labels** for duplicates, not data. A number of options are provided to allow for deletion or merging of the data in the duplicated columns. Once a duplicated column label is found, the options are-

1. Do *nothing*, ie allow both columns to be retained in the data set. Possibly rename either or both of the labels using LABN.
2. Perform an *addition* operation with the two columns of data. If the data is quantitative (not presence/ absence), and it is decided that the two columns contain correct data, then a logical thing to do is to ADD the two columns together. In this, or the next operation the result of the operation will always be placed in the column with the lower sequence number, for example -

Column 4 = Column 4 + Column 15.

3. Perform an ORing operation between the two columns. This is usually the desired action if the data in both columns is correct and is of the presence/ absence type. In this situation, if either of the two columns contain a '1' (presence) value, then the resulting column will also contain a '1', otherwise it will contain a '0' (absence).
4. Deleting the leftmost (lowest sequence number) column.
5. Deleting the rightmost (highest sequence number) column.

Option 6 Monte-Carlo re-ordering within Columns

Monte-Carlo studies of various results from **PATN** may be performed using this option. The method uses *randomisation* of the *order* of values within one or more columns of data and subsequent analysis to test if the result produced could have been produced by chance alone.

SAMP permits any or all of the M columns to be permuted (randomised) in a single run. Given a single random number seed, a set of N (one per row) uniform random variates between zero and one are generated. These are then sorted and the indices from the sort are used for re-ordering the values within the selected columns.

Each column is treated *independently*. For example, if a single column is selected, the output data file will be identical to the input file except for randomisation within the selected column. Data *values* are not altered.

A single random number *seed* value is requested regardless of the number of columns for re-ordering. To reproduce the same data after randomisation, both the same seed and the number of selected columns will have to be the same.

To select the columns for random re-ordering, a MASK-style of input is used. This facilitates the selection of a subset, or all columns without the need for much typing. As with all MASK-type input, the selected column sequence numbers can be read from a file. The output row and column labels (if created), will be identical to the input.

Option 7 Extracting a SORTED sample

The SORT module in PATN has been replaced by a sorting option in SAMP. Selecting this option results in a simple re-ordering of the rows based upon an ascending sort of the values in a single column. Multiple keys are not supported. The row labels are re-ordered commensurate with the data.

Option 8 Proportion of defined GROUPS

This option is similar to options 2 and 3 except the sample is achieved by proportionally sampling each of the defined groups. A new data set is produced which contains a number of objects that is, (as close as feasible) to the selected proportion of the input data set.

The proportion requested must be between zero (no objects sampled) and one (all objects sampled). For example, consider an input data set containing 100 rows, 50 columns having 10 groups defined by a -.gdf file. A 0.5 proportion would result in a new data set of approximately 50 rows by 50 columns.

Each group is proportionally sampled except for single member groups. One object will be selected from such groups. To achieve around the correct overall proportion, the sample number for all but single member groups is rounded down. For example a 50% proportion of group containing nine objects will result in a sample of four objects.

The process used to select the sample uses a random number generator with a user-supplied seed value. This method is used rather than say selecting the first 'k' samples. The reason is that the order of objects in the -.gdf file depends on how the groups were generated. If the classification was produced from FUSE, the ordering of objects is as seen on the dendrogram. The ordering is not random but determined by the sequence number of the objects within the branches. If ALOC was used, the ordering within each group is by sequence number. Other classification methods such as PDIV are different again. Hence, a completely (?) random selection is made (without replacement).

HISTOGRAMS & UNIVARIATE STATS: HIST

OUTLINE

HIST provides *univariate* statistics and *histograms* of any or all rows or columns of the current data file. The user defines the number of bars to be used in the histogram. The statistical information listed includes-

1. Number of values
2. Minimum
3. First quartile
4. Median
5. Mean
6. Third quartile
7. Maximum
8. Inter-quartile range (third quartile-first quartile)
9. Average deviation
10. Standard Deviation
11. Variance
12. Range
13. Sum of values
14. Number of values > 0
15. Skewness
16. Kurtosis

The *mean* is calculated as the arithmetic average of all the values in the row or column. The *median* corresponds to the middle value if the values in the row or column are first sorted. The *standard deviation* is taken as the standardised deviation from the mean; the formula used is that for the population, ie, the divisor is (N-1).

The *inter-quartile range* is the value of the third quartile minus the first quartile where the quartile is defined in an analogous way to the median. The *first quartile* is that value where, one quarter of the values are below it, similarly, the *third quartile* is a value that has one quarter of the values above it.

The *skewness* is calculated as a deviation from the normal distribution as the mean minus the mode divided by the standard deviation. *Kurtosis* is an indication of the 'flatness' (platykurtic) or sharpness (leptokurtic) of the distribution curve. It is calculated from the quartiles as

$$\text{Kurtosis} = (M_4 / (M_2 * M_2)) - 3$$

This results in a value of zero for the normal distribution (mesokurtic), a negative value for platykurtic curves (flat) and a positive value for leptokurtic curves (peaky). HIST and TRND may be used in tandem to transform and monitor the transformation of the selected rows or columns.

REFERENCES

Belbin, L., Faith, D.P. and Minchin, P.R. (1984). Some algorithms contained in the Numerical Taxonomy Package NTP. CSIRO Division of Water and Land Resources Technical Memorandum 84/23.

OPTIONS

Selecting ROWS or COLUMNS

HIST optionally produces a histogram of one or more rows or columns of the data matrix. The rows or columns to be operated on are selected using the standard MASK-style of input. You first select whether rows or columns are to be used. Then, you have the option of

1. Using the LIST-style format to read the row or column sequence numbers from the terminal or
2. a file, or
3. automatically select all rows or columns within the current PATN data set to be used.

A tabulation of row and column labels against sequence numbers can be produced using LABN (option 5).

SCANNING

PRESENCE / ABSENCE DATA:

SCAN

OUTLINE

SCAN is designed as the counterpart of HIST for the **description** and summarisation of sets of data composed of presence/absence data. SCAN produces an output file (*-.scn*) sorted by the first values in the list below. It is a useful method for checking data and for reference during subsequent analysis.

A separate list for each object and attribute of the data matrix contains-

1. The number of NON-ZERO values,
2. the number of MISSING VALUES,
3. the MINIMUM value,
4. the MAXIMUM value,
5. the SUM and
6. the AVERAGE value.

Within the tables objects and attributes are sorted independently by SUM. At the end of the two tables, SCAN also lists -

1. sparsity ratio
2. D-ratio
3. Whittaker's (1980) β diversity (columns = species)
4. Routledge's (1977) β diversity (columns = species)
5. total missing values in the data,
6. the overall minimum data value and
7. the overall maximum data value.

These are intended to provide overall features of the data set. The formulae for these summary measures are -

```

Sparsity ratio = number of non-zero value / # rows * # columns
D ratio = number of D's (0-0 matches) / maximum possible D's
Whittaker Diversity (W) = s/a-1

where      s = total number of species (columns) in the data
           a = average number of species per sample

Routledge Diversity (R) =  $\log_e T - |1/T \cdot \sum E_j| \cdot \log_e E_j$ 
                         $- |1/T \cdot \sum A_i| \cdot \log_e A_i$ 

where T    =  $\sum E_j$ 
       $E_j$    = number of sites with species j
       $A_i$    = richness of site i
      log   = log to base e
  
```

The file manipulation command MERG has the ability of *merging* the row and column frequencies as an extra column and row respectively to the data matrix or any other matrix containing extrinsic attributes for the same set of rows. To achieve this, MERG expects that SCAN has been run on a data set of the same number of rows that are defined by the current data file.

REFERENCES

- Belbin, L., Faith, D.P. and Minchin, P.R. (1984). Some algorithms contained in the Numerical Taxonomy Package NTP. CSIRO Division of Water and Land Resources Technical Memorandum 84/23.
- Routledge, R.D. (1977). On Whittaker's component of diversity. Ecology, 58: 1120-7.
- Whittaker, R.H. (1960). Vegetation of the Siskyou Mountains, Oregon and California. Ecological Monographs 30:279-338.

OPTIONS

none

SCATTER PLOTS:

SCAT

OUTLINE

SCAT is the bi-variate equivalent to HIST. It produces 2-dimensional (x-y) scatter plots with an option for plotting a z-value at the x-y coordinates. The plots, along with optional tabulations and regressions are produced for a standard line-printer. SCAT provides a variety of options for:

1. the input file structure
2. Scale type
3. Annotation type
4. Output device
5. Tabulation of x-y-z's
6. Row and/or column selection.

The input file can either be a file containing a data matrix or an ordination file where the columns represent scores from one of the ordination techniques. In addition, rows or columns of the data matrix may be plotted.

The scale types can be either dependent or independent scaling. Dependent scaling is used where the values used for the x-axis and y-axis are commensurate. This is usually the case with attributes measured on the same scale or with the loading of rows on ordination vectors. Independent scaling is the alternative; where the values for the x and y axis are not commensurate and not measured on the same scale. An example of this would be a scatter plot of number of road accidents against year.

SCAT provides four different methods for the annotation of the values on the plot at the x-y intersections:

1. Number of values at position the position
2. Five level light -> dark density scale of (1)
3. The Sequence Numbers of the rows or columns plotted
4. Z-values on an INTERVAL scale.

Option 1 adds up the number of points (Rows or Columns) that have been scaled into a character position (the resolution limit) of the scatter plot. Values greater than zero are as a single integer in the range 1->9 with values greater than 9 being plotted as '*'. Option (2) corresponds to a transformation of Option (1) into a five level grey scale as generated by the characters:

blank	-	+	*	#
-------	---	---	---	---

The most common annotation option is the third. This lists the sequence numbers (1->N) corresponding to the rows or columns plotted at the x-y coordinate. With options 3 and 4, SCAT annotates on a "first come - first plotted" basis; the first case to be written at a particular position on the plot takes precedence over all subsequent candidates for that position. When this occurs, the points unable to be plotted are flagged in the tabulated output (if produced) as "NOT PLOTTED".

For annotation option 4, SCAT linearly interpolates the values of the z-attribute into the user selected number of classes using the following symbol table:

1=1	2=2	3=3	4=4	5=5	6=6	7=7	8=8	9=9
A=10	B=11	C=12	D=13	E=14	F=15	G=16	H=17	I=18
J=19	K=20	L=21	M=22	N=23	O=24	P=25	Q=26	R=27
S=28	T=29	U=30	V=31	W=32	X=33	Y=34	Z=35	*=>35

If the row or column to be used as a z-value is coded in a MERISTIC scale (positive INTEGER values between 0 and infinity) then selecting ZERO class intervals for option 4 will force SCAT to plot the ACTUAL values rather than bothering to interpolate.

SCAT can output the plots either to:

1. A video terminal capable of handling 132 column mode and
2. A standard line printer.

Because the scale factors for the two devices are different, one or the other must be pre-selected. The video output is scaled to 21 rows and 101 columns while the line-printer plot is scaled to 61 lines and 101 columns (11 inch or 28 cm paper).

If the line-printer option is chosen, an additional option is invoked. This determines if a tabulation of x-y-and z values is listed. This list contains:

1. Row or column LABEL
2. Row or column SEQUENCE NUMBER
3. x-data value
4. y-data value
5. x-coordinate in terms of character positions
6. y-coordinate in terms of character positions
7. FLAG either PLOTTED or NOT PLOTTED
8. SYMBOL used for z-value transformed to interval scale

SCAT provides options for plotting either rows or columns in a single plot sequence. A plot sequence is defined by the selection of one of each of the master options (1-5) above for one or a selection of rows or columns to be plotted (same options - different x and y's). Sequences and combinations of rows or columns to be used as either x-y axes or z-values may be automatically generated by entering the appropriate parameters for the row/column numbers. Depending on options, there are up to 6 parameters that may be nominated, two each for x-axis, y-axis and z-values. The pairs of values correspond to the *sequence number* of the first and last row or column to be used in the plot or series of plots. The following examples outline the various combinations available:

1. A SINGLE pair of rows/columns, eg. plot ROW 1 against ROW 3.

1st X col to be plotted	= 1
last X or 1st Y col to be plotted	= 3
1st Y col to be plotted	= 0
last Y col to be plotted	= *

2. ALL pairs in range 1->X where X=M for columns and X=N for ROWS, ie for ROWS, this will produce ALL COMBINATIONS (N*(N-1)*0.5) of rows in the DATASET:

1st X col to be plotted	= 0
last X or 1st Y col to be plotted	= *
1st Y col to be plotted	= *
last Y col to be plotted	= *

3. All COMBINATIONS of pairs of rows or columns in the RANGE 2->8, ie $N*(N-1)/2$ where N = total number of selected rows/cols. For example, to plot ALL combinations of rows or columns in the range 2 to 8:

1st X col to be plotted	= -2
last X or 1st Y col to be plotted	= -8
1st Y col to be plotted	= *
last Y col to be plotted	= *

4. FIRST SET of rows or columns (x-axis) against ALL of a SECOND SET of rows or columns (y-axis). The first from the first set will be plotted against ALL rows or columns in the second set, then the second from the first set against ALL in the second set and so on. For example, plotting columns 1 to 3 against columns 10 to 20 implies:

1-10,1-11,1-12.....1-20, 2-10,2-11,2-12.....2-20,
3-10,3-11,3-12.....3-20 (ie 63 plots in all).

1st X col to be plotted	= 1
last X or 1st Y col to be plotted	= 3
1st Y col to be plotted	= 10
last Y col to be plotted	= 20

5. PAIRED SEQUENCES of a FIRST SET against a SECOND SET.
This option plots the first row or column in the first set against the first row or column of the second set, the second in the first set against the second in the second set etc. For example, if the first set was defined as columns 1 to 3 and the second set, columns 11 to 13 then the plots 1(x)-11(y), 2-12 and 3-13 would be produced. The parameters would be:

1st X col to be plotted	= -1
last X or 1st Y col to be plotted	= 3
1st Y col to be plotted	= 11
last Y col to be plotted	= 13

6. One row or column as *z-values* on a given x-y pair. Using this option, a single row or column may be used to annotate a single plot of an x-y pair of rows or columns. For example to interpolate 9 as z-values on a plot of column 1(x) versus column 2(y):

1st X col to be plotted	= 1
last X or 1st Y col to be plotted	= 2
1st Y col to be plotted	= *
Sequence number of row to be used as z-value	= 9

7. A contiguous sequence of *z-values* on a single x-y pair. This option goes one stage further than the previous option because it facilitates the plotting of a contiguous range of z-values on a single x-y base. One plot is produced for every z-value (row or column) in the selected range on the same x-y row/column pair. For example, to produce 3 plots, one each for columns 3 to 5 corresponding to z-values on columns 1 (x) and 2 (y), use the following sequence:

1st X col to be plotted	= 1
last X or 1st Y col to be plotted	= 2
1st Y col to be plotted	= *
Sequence number of row to be used as z-value	= -3
Final row to be used as z-value	= 5

SCAT could be used for the following purposes:

1. x-y plot of site locations
2. x-y plot of site locations with an additional column plotted as a z-value
3. x-y plot of ordination vectors
4. x-y plot of ordination vectors with an additional column plotted as a z-value.

Some examples of attributes that could be profitably plotted as z-values may be:

1. An *extrinsic* attribute (not included directly in analysis)
2. A *group* of numbers from a classification
3. An *ordination* score

REFERENCES

- Belbin, L., Faith, D.P. and Minchin, P.R. (1984). Some algorithms contained in the Numerical Taxonomy Package NTP. CSIRO Division of Water and Land Resources Technical Memorandum 84/23.

OPTIONS

Input File Format

There are two different file formats that are acceptable to SCAT

- 1 = Standard data file
- 2 = Ordination file (from SSH, DCOR, SERE, PCA)

The standard data file is the file specified by the parameters in *PATN.PRM* and contains only the data matrix (no headers or labels). The ordination format is that produced by any of the modules-

- 1. SSH
- 2. DCOR
- 3. SERE
- 4. PCA

This type of file contains-

- 1. A title (from *PATN.PRM*)
- 2. A heading
- 3. Ordination vectors
- 4. Row labels
- 5. Summary of ordination eigenvalues, stress, ordering etc.

Selection of ROWS or COLUMNS

As with a number of other modules in PATN, SCAT provides the option to plot columns against columns or rows against rows (not rows against columns). The options are:

- 1 = Operate on ROWS
- 2 = Operate on COLUMNS (DEFAULT)
- 3 = STOP SCAT

SCAT will always return to this option once a plotting specification has been defined and the associated plots have been filed or listed to the terminal. To stop SCAT, option 3 must be selected. Once the cycle has been terminated, SCAT will detect if hard-copy plots have been filed during any cycle and will request what should be done (TYPING, PRINTING or just save).

Specifying the output device

SCAT permits three different output devices. They are:

- 1 = Standard line printer
- 2 = ANSI video terminal supporting 132 column mode
- 3 = Video terminal supporting an 80 column mode

If the line printer option is selected during any SCAT cycle, the file *-.sct* will be created and used to store any subsequent plots for the line-printer.

If the 132 column terminal option is selected, SCAT will expect that the terminal being used has ANSI escape sequence support and transmit escape sequences that will switch the terminal to and from 132 column mode. If the 80 character mode terminal is selected, SCAT will assume any standard terminal supporting a width of 80 characters. No escape sequences will be used.

In the case of either terminal selection, the plots will be output directly to the terminal being used and *not* filed. That is the plots, once displayed, will be lost. Each plot will remain on the screen until the <RETURN> key is pressed. NOTE that the scaling of the two forms of output are such that proportionality is always maintained. SCAT uses:

1. 101 characters in the x direction and 61 in the y direction for line printer plots and
2. 101 characters in the x direction and 21 in the y direction for 132 column video output
3. 80 characters in the x-direction and 21 in the y direction for 80 column video mode.

Choosing the ANNOTATION TYPE

SCAT provides four types of annotation for the symbols to be used to represent the location of a point:

- 1 = Plot counts of values at coordinates
- 2 = Density plot mode (plot on 4 scale -+*# the density at coordinate)
- 3 = Plot sequence number of row at coordinate (default)
- 4 = Plot value of 3rd (row or column) - extrinsic attribute ?

The first option counts up the number of points landing on each character position in the plot. A count of zero is treated as a blank while non-zero counts are represented by the following table:

1 = 1	11 = B	21 = L	31 = V
2 = 2	12 = C	22 = M	32 = W
3 = 3	13 = D	23 = N	33 = X
4 = 4	14 = E	24 = O	34 = Y
5 = 5	15 = F	25 = P	35 = Z
6 = 6	16 = G	26 = Q	>35 = *
7 = 7	17 = H	27 = R	
8 = 8	18 = I	28 = S	
9 = 9	19 = J	29 = T	
10 = A	20 = K	30 = U	

The second option provides a rough grey scale interpolation of the first annotation option. SCAT uses a four density scale-

' '	= background
'-'	= first level
'+'	= second level
'*'	= third level
'#'	= fourth level

The third option (DEFAULT) plots the sequence numbers of the points at their character location. The rule used for plotting when more than one value lands at the same character location is 'first in gets plotted'. If the tabulation option associated with the printer output option is selected, this is not a problem for the location of each point is noted in terms of character coordinates. In addition, each point is flagged as being either:

1. PLOTTED or
2. NOT PLOTTED

The last option provides for plotting a value at the point location as defined by and additional row or column (ie a z-type value). For example, if the data matrix was:

```
1 1 2
2 2 4
4 4 8
5 5 8
7 2 1
9 1 2
```

and the first column was plotted against the second, with the third column as the z-value, then:

```
at location 1-1 the value 2 might be plotted
at location 2-2 the value 4 might be plotted
at location 4-4 the value 8 might be plotted
...and so on (see section on 'Symbol Selection' below).
```

Selecting the SCALE TYPE

There can be two types of scaling used to plot data, depending on the nature of the rows or column values:

- 1 = Scale x and y axes independently to the maximum range
- 2 = Scale both axes according to the maximum range (proportional scaling) (default)

When the data in the rows or columns to be plotted is on disparate scales and all you want is an overall distribution of points, the independent scale option should be selected. Using this form of scaling, the resulting plot will always contain a data point in the first and last column and first and last row.

When both rows or columns contain data on a similar scale or it is desired to directly compare the plots regardless of values, select the proportional scaling option. With this form, the row or column containing values with the greatest range will define the size of both axes. On the axes with the maximum data range, points will be located in the first and last character position.

ROW and COLUMN selection

SCAT provides six different methods for detailing the plotting to be performed. Many of the methods detailed below provide a rapid way of producing a range of plots with a single specification. The table below gives the number and order of specification of the row or column sequence numbers to be entered to the prompts:

```
n1 - First x row/column
n2 - Last x or first y row/column
n3 - First y column
n4 - Last y column
n5 - First z column
n6 - Last z column
```

If columns were selected, the possibilities include entering the values-

n1 n2 <RET>. A single plot using column n1 Vs column n2.

-n1 -n2 <RET>. Plots all pairs of columns in the range n1 through n2.

n1 n2 n3 n4 <RET>. Plots column n1 Vs n3, n1+1 Vs n3+1 through to n2 Vs n4 (n2-n1+1 plots).

-n1 n2 n3 n4 <RET>. Plot columns n1 through n2 Vs columns n3 through n4. ie n1 Vs n3, n1 Vs n3+1, ...n1+1 Vs n3 ... n2 Vs n3, n2 Vs n3+1 ... n2 Vs n4.

n1=0 <RET>. Plots all pairs of columns defined in the current data or ordination file. ie for m columns there are -

$m*(m-1)/2 \quad \text{plots.}$

n1 n2 <RET> -n5 n6. Plot n1 Vs n2 as a base map using columns n5 through n6 for determining symbols.

As an example, using the values -

<pre>n1= 1 n2= 2 n5= -5 n6= 7</pre>

for the last case above, would produce a set of 3 plots all using column 1 versus column 2 as a base. On the first plot column 5 values would be used to determine the type of symbol to be plotted at the position defined by columns 1 and 2. On the second plot, the same base would be used (columns 1 and 2) but column 6 would be used to determine symbols. The last plot would also use columns 1 and 2 for the x and y axes while using column 7 to determine symbols. If 'n5' is a positive integer, a single map will be produced using the values in n5 to determine symbols.

Symbol Selection

Selecting annotation option 4 allows for the possibility of using an additional row or column of data to determine the symbols to be used at the locations defined by the rows or columns for the axes. For example, if ordination scores and group numbers have been added as additional columns to the right-hand side of the data, plots can be made using the ordination scores as the x and y axes while using group numbers for the annotation.

For example, each row of data will be located by its ordination scores on the scatter plot, while the symbol plotted at that position will be determined say by group number. In this way, scattering of groups or other attributes can be displayed on different base maps (geographic, ordination or whatever).

SCAT determines the range of values in each column to be used to define symbols and partitions the range into the number of classes the user selects. The user has the option to accept the default symbols (1-9, A-Z and * as noted above) or to supply an alternate set. For example, if you decide to partition the values 1 through 100 into 5 classes, the 5 default symbols would be

```
1-20    = '1'
21-30   = '2'
.....
81-100  = '5'
```

while supplied symbols may be something like -

```
1-20    = ' '
21-30   = ' .'
.....
81-100  = '*'
```

MASKING DATA:

MASK

OUTLINE

MASK is a general purpose module designed to **mask** and/or **re-order** rows and columns of a data. Masking is the operation of selectively deleting (or extracting) rows and/or columns of the data matrix to create a new output data file equal to (re-ordering), or smaller (masking) than the input data.

Re-ordering can be achieved by nominating the order by use of sequence numbers of rows and columns in the output matrix. Re-ordering and masking of rows/columns can be achieved simultaneously. MASK requires that the rows and columns to be **retained** in the data be nominated. Data is considered to be masked **in**.

Mask Styles

It is important to realise that MASK has two styles of masking-

1. Direct and
2. Indirect.

Direct masking is where the decision whether to retain or delete rows or columns of data is based on some extrinsic criteria (external to MASK). For example, directing MASK to retain certain rows is considered a direct mask.

Indirect masking occurs when the decision to retain or delete rows or columns of data is based on the data itself. This is termed intrinsic (to the data). For example, elimination of rows having the attribute altitude greater than 1,000 metres is considered an indirect mask.

If direct masking is used to retain all rows and columns and indirect masking is bypassed, the output file, other than potentially having a different name and format, will be identical to the input file.

Order of Operation

Masking operations may be **iterative** in the sense that once some columns have been deleted (either directly or indirectly), some rows may be candidates for indirect masking. This may in turn feed back so as to make further columns candidates for indirect masking, and so on. MASK takes a pragmatic approach. The columns are masked first, then the rows are masked on the columns that remain. No iteration is performed. If some cycling is required, MASK will need to be re-activated on the output data matrix.

Updating

Any alteration to rows and columns will be reflected in new label files (*-.rlb,-.clb*). In addition, a 'saved parameter file' (*-.prm*) containing parameters reflecting the output file will be created. The *-.prm* file can be retrieved to *PATN.PRM* using PRAM, if the MASK output file is required in **PATN** at some later time. Under no circumstances should the output file have the same name as the input file if no updating is requested. If this occurs:

1. *PATN.PRM* will point at the new file anyway (the latest version of a file is the one assumed)
2. the latest cycle of the data file may not match the parameters and
3. the latest cycle of the row and column label may not match the parameters.

If you do not intend to use the file resulting from MASK immediately, it should always be given a new BASE or ROOT name.

REFERENCES

none.

OPTIONS

Direct Masking

Direct masking is where rows and columns are selected for retention in the output data matrix. For example, directing MASK to retain rows 1, 3, 5, 7 and 9 is a direct operation because MASK assumes that some **extrinsic** criteria is involved.

The direct mask will be requested first for rows and then for columns.

MASK provides 3 options which amount to the entry of a string of free format integer values representing sequence numbers of the row and columns to be *retained*. The three options are:

- 1 = read the sequence numbers from the terminal (default)
- 2 = read the sequence numbers from an ASCII file
- 3 = retain all rows or columns (1 to N, or 1 to M)

The first option expects to read the direct mask from the terminal. The second is recommended when a record of the masking operation is to be kept or used again or elsewhere in **PATN**. This option is also the appropriate one when more than 10-20 numbers are to be entered. It is easier to enter and check values using a file and editor than to find you have miss-typed a number at the terminal. The last option allows the retention of **all** rows or columns, but automatically generates all required sequence numbers.

The *sequence numbers* detailing the rows and columns to be retained are expected in FORTRAN *free format* with consecutive values being range coded with the upper value set as a negative number. For example, to retain rows 1,3 and 5 to 10 inclusive and to re-order the rows so as to place row 3 AFTER row 10, the coding would appear as:

1. 1 5 -10 3 and then:
2. any printable non-numeric character
3. a Control-Z and depending on system:
4. A carriage return <CR>

The printable or control-z character on the end of the input is to represent an End Of File sequence. A negative sequence can also be used to re-order rows or columns, the following is legal -

1, 3, 10 -5

Indirect Masking

Indirect masking is where rows or columns of the input data are retained or deleted on some *intrinsic* criteria. Intrinsic in this sense meaning that the data itself will arbitrate whether rows or columns are retained in or deleted from the output matrix. MASK provides for two indirect criteria:

1. A row or column may be retained or eliminated depending on a *particular value* in a selected column and row, and
2. A row or column may be deleted from the output matrix on the basis that either:
 - a. the *number of non-zero values* it contains is less than a user defined threshold value or
 - b. the *sum* of the data values in that row or column

Rows may be included or deleted on the basis that they contain a value in a selected column that is either inside or outside a user defined **range**. For example, sites may be eliminated on the basis that they are outside the range 1000 - 2000 metres on the attribute 'altitude'. Three parameters are required:

1. a column sequence number
2. a minimum value
3. a maximum value

The following rules apply:

1. If the column sequence number is positive, the row is retained if the column value falls between (and not including) the maximum and the minimum values.
2. If the column sequence number is negative, the row is retained if the column value is greater than the maximum value, or less than the minimum value.

Masking rows or columns based on sums

In the second situation, while some attributes may have been present in a larger parent data set, the same attributes may have no 'presence's' (values equal to or greater than one) in the current input data set. In this case, nominating a column sum (as defined by the number of non-zero values) less than '1.' will automatically delete these attributes from the output data set. The same can be true of rows of the data matrix where say, after direct or indirect elimination of attributes that were 'rare', badly coded or contained errors, some rows had no relevant attributes describing them.

The limit can be based on either of two values:

1. the *number of non-zero values* in the row or column, or
2. the *sum of all values* in the row or column

Select the most convenient form for your purposes and note that the same option must apply to both rows and columns.

MASK requests two values -

1. a row sum minima and
2. a column sum minima.

The column check is performed before the row check. If the column value, based on the selected option is less than the specified minima, it will be deleted from subsequent scanning and output. At the next step, if the direct mask includes a row, it will be retained only if its sum is greater than the specified value.

Masking Rows using Missing values

If you wish to eliminate from the output data file, *all* objects that have *any* missing values, this option can be selected. This option is useful for eliminating all missing data from a set of data in situations where subsequent predictions of the missing values may be made.

TRANSFORMING DATA:

TRND

OUTLINE

TRND may be used to transform or standardise values in rows or columns of the current PATN database. The options can be divided into two types -

1. transformations
2. standardisation's

Transformation is used in **PATN** to refer to the alteration of a data value independent of all other data values. *Standardisation* alters values which depend on at least some other values in the data. The first 6 options provide standardisation's while subsequent options are transformations.

TRND will continue to request transformation/ standardisation options until EXIT is selected. This type of operation enables independent operations to be applied to different rows and/or columns of the data matrix. For example, a transformation could be applied to ROW 1 -15 followed by a standardisation to the same or different columns (or rows). TRND produces the following files -

1. an output data file and optionally
2. new row and column label files,
3. a *-.prm* file and possibly,
4. altered parameters in *PATN.PRM*.

The saved parameter file *-.prm* file can be restored to *PATN.PRM* if the modified file is required in **PATN** at some later time. Under no circumstances should the output file have the same name as the input file when no updating is required. If this occurs:

1. *PATN.PRM* will point at the new file anyway (latest cycle is always accessed)
2. the latest cycle of the data file will not match the parameters
3. the latest cycle of the row and column label files will not match the parameters

REFERENCES

- Burr, E.J. (1968). Cluster sorting with mixed character types. I. Standardisation of character values. Aust. Comput.J. 1:97-99.
- Clifford, H.C. and Stevenson, W.C (1975). An introduction to numerical classification. (Wiley).
- Belbin, L., Faith, D.P. and Minchin, P.R. (1984). Some algorithms in the Numerical Taxonomy Package; NTP. CSIRO Division of Water and Land Resources Technical Memorandum 84/23.

OPTIONS

Row / Column Selection

The first option determines if you want TRND to -

- 1 = operate on *rows* of the data matrix,
- 2 = operate on *columns* of the data matrix or
- 3 = **stop**.

Which rows or Columns?

TRND accepts the selection of row or columns by sequence number using a MASK style of format. For example, columns 1 to 15 are nominated as

```
1 -15 e
```

Input of sequence numbers is *free format integer*. Values can be entered over as many records as required, can be in any order. Descending sequences can also be used. For example,

```
51 -20 101 202 e
```

implies row or columns 20, 21, 22, 23...51, 101 and 202. The input sequence is completed by any non-numeric character (micro-PATN) or by the key sequence <CTRL><Z> on minis. See MASK for further details on this format.

The notation used in the following sections is:

```
Dij          : the Data value for the ith row and jth column of the data matrix
Dij new      : the new value of Dij
Dij old      : the old value of Dij
i.           : the whole of the ith row of the data matrix
.j           : the whole of the jth column of the data matrix
Sum i. or .j : the SUM of the values in the ith row or jth column
SOS i. or .j : the SUM of the SQUARES of the values in the ith row or jth column
```

Option 1 $D_{ij}/\text{SUM } i.\text{or}.j$

The first six options of TRND are standardisation's. TRND scans the complete data matrix in preparation for the alteration of any row or column value.

This option results in proportions. The selected row/s or column/s are summed. Each value in these row/s or column/s are divided by this sum, ie:

$$D_{ij} \text{ new} = D_{ij} \text{ old} / \text{Sum } i. \text{ or } .j$$

New values will be in the range zero to one. If applied to data on different scales, a single scale will result.

Option 2 $D_{ij}/\text{SQRT (SOS } i.\text{or}.j)$

The divisor in this option is the square root of the sum of the squares of the values in the selected row or column. This standardisation appears to be appropriate for use on the columns of non-presence/absence data in preparation to the application of the association measure TWO-STEP in the command ASO.

It is also likely to be appropriate for other combinations of non-presence/absence data and association measures. The standardisation is:

$$D_{ij} \text{ new} = D_{ij} \text{ old} / \sqrt{(\text{SOS } i. \text{ or } .j)}$$

Option 3 $D_{ij}/\text{Max } i.\text{or}.j$

This option proportions each value in the row or column to the **maximum** value in that row or column, ie:

$$D_{ij} \text{ new} = D_{ij} \text{ old} / \text{MAX } i. \text{ or } .j$$

Option 4 $(D_{ij}-\text{MIN } i.\text{or}.j)/\text{RANGE } i.\text{or}.j$

This option subtracts the minimum value of the selected row or column for each value in that row or column and divides by the range of values in that row or column.

This has the effect of placing all values in a selected row or column on the scale zero to one. This form is usually termed range standardisation -

$$D_{ij} \text{ new} = (D_{ij} \text{ old} - \text{Min } i. \text{ or } .j) / \text{RANGE } i. \text{ or } .j$$

This is the *recommended* standardisation procedure. It should be used for the equal weighting of attributes prior to the use of association measures such as Bray & Curtis and Kulczynski. Other association measures such as the Gower Metric have in-built standardisation (it uses range).

Option 5 $(D_{ij}-\text{MEAN } i.\text{or}.j)/\text{SD } i.\text{or}.j$

This is usually called normalisation, because the values are transformed to normal deviates. The new values are in terms of the number of standard deviations less than or greater than the mean value for that row or column. For example a value of 1.0 implies one standard deviation greater than the mean. The formulation is:

$$D_{ij} \text{ new} = (D_{ij} \text{ old} - \text{MEAN } i. \text{ or } .j) / \text{SD } i. \text{ or } .j$$

Option 6 $D_{ij}/(2*VAR \text{ } i.\text{or}.j)$

This standardisation is attributed to Burr (1968) and was often used as a column standardisation prior to the application of the Euclidean distance. In this context, it was used as a re-scaling procedure because Euclidean distance is un-standardised. It is not generally recommended these days. The formulation is simply each value divided by two times the selected row or column variance -

$$D_{ij} \text{ new} = D_{ij} \text{ old} / 2.0 * \text{VAR } i.\text{or}.j$$

Option 7 LOG10 ($D_{ij} + \text{constant}$)

The following options are transformations; any value in the data can be altered without reference to any other data value. This option has two sub-options,

- 1 = LOG base 10 or
- 2 = LOG base e (2.718..)

This transformation is independently applied to each value in the selected row/s or column/s of the data matrix. A value of 1.0 is usually added to the data value prior to taking the log to ensure positive data and that the LOG of zero is not attempted.

This transformation can be used for three common purposes -

- 1. transformation of skewed data to a more normal form,
- 2. to counteract higher levels of error with higher data values or
- 3. to re-scale data values to values more suitable for association.

The intensity of skewness will determine the type of transformation to be applied to approximate normality. A reasonable 'rule of thumb' is -

small skew-use SQRT
moderate skew-use LOG
strong skew-use -1/D

The general purpose transformation option can achieve either the first or third cases. Generally speaking, linear interpolation is the preferred method of data re-coding.

Option 8 Zeros -- > Constant

This option transforms all zero values to a user defined constant value. This has been used in the past to circumvent a problem associated with the Canberra Metric association measure. Any non-zero data value, when compared with zero, results in a Canberra Metric association value of 1.0 (maximal dissimilarity). Bill Williams suggested a fix that amounts to transforming the zero values to 1/5 the smallest value in the matrix.

Option 9 Non-Zeros --> Constant

This Option has been designed to re-code zero/non-zero data to presence absence form. The user enters the value of the constant to be used and all values in the matrix that are not zero take this supplied constant value.

Option 10 $D_{ij} = P \cdot D_{ij} + Q/D + R^S + T$

This is a simple method for a variety of common type transformations. The user defined constants P , Q , R , S and T default to values that have no effect on the input values. The formulation is-

$$D_{ijnew} = (P * D_{ijold} + Q * 1 / D_{ijold} + R)^S + T$$

The constants and purpose are:

1. P - Multiplier factor - default of 0.0
2. Q - Reciprocal factor - default of 0.0
3. R - Multiplier factor - first additive factor - default of zero
4. S - Power factor - default of 1.0
5. T - Second additive factor - default of 0.0

Option 11 Linear Interpolation

This is a very useful option for general data re-coding. For example, if numbers of birds were counted and recorded at sites, it is likely that **categories** of the numbers would be required for pattern analysis and not the original counts (along with the errors). Instead of 0 to 5,000, a scale of 0-5 is probably the type of coding that **PATN** will best utilise. Integer values may be optionally produced.

A number of transform PAIRS are required in the form -

First old value	First new value
Second old value	Second new value
Third old value	Third new value
.	.
.	.
kth old value	kth new value.

Values are transformed in a linear fashion from the old to the new scale. For example, given the above situation, any value corresponding to the 'first old value' will be transformed exactly into the 'first new value', while any intermediate values will be transformed to intermediate points in the new scale. For example, given the two transformation pairs:

1.0	10.0
2.0	20.0

an 'old' value of 5.0 will be transformed into a 'new' value of 50.0 while a value of 1.5 will be transformed to 15.0.

Option 12 Ranking

This option converts *real* data values into a rank either by row or by column. That is, it transforms interval or ratio data in the default data matrix into *ordinal* values.

Two sub-options govern the type of ranking that will be performed. The *first* sub-option creates ranks by using the *average* rank of tied values. This is equivalent to sorting the data and using the average **position** from the sorted data. Consider an example:

1	1	2	3	4	4	would be ranked as:	1.5	1.5	3	4	5.5	5.5
---	---	---	---	---	---	---------------------	-----	-----	---	---	-----	-----

the ranks for say ten values may not contain '1' or '10'.

The *second* sub-option creates ranks by using the next sequential ranking, starting at rank '1' for the smallest value. Unlike option 1, this option creates *integer* values as ranks. Tied values are ranked identically. This option is equivalent to sorting the values and using the NEXT sequential whole number, starting from '1' for the smallest value. Consider the same example as above:

1 1 2 3 4 4 would be ranked as: 1 1 2 3 4 4

TRANSFORMING ASSOCIATIONS: TRNA

OUTLINE

TRNA is the counterpart for association matrices to TRND for data matrices; a command for performing transformations and standardisation's on association matrices. *Transformation*, here implies that some options modify values in the input association matrix independently from all other values. *Standardisation's* imply that new values are generated on the basis of other values in the association matrix. The options available, where d is the individual association values and D is a complete association matrix, are -

1. Re-code > threshold --> missing
2. Re-code > threshold using shortest path algorithm
3. Re-code > threshold using a minimum spanning tree
4. ABS (d)
5. $(d-d.min)/d.range$
6. constant-d
7. $(P*d+Q/d+R)**S+T$
8. $ABS(D_1-D_2)$
9. $P*D_1 + Q*D_2 + R*D_3.....ZZ*D_Z$
10. Gower Transformations
11. Encode or check spatial adjacency coding
12. Ranking
13. Cophenetic correlation
14. Mantel randomisation test between two lower-symmetric matrices.

The purpose of transformations (1-3 and 10) is to improve associations for subsequent analysis. In some cases, corrections may be used to enhance the quality of estimate of association or to transform the associations into a Euclidean form. Other options (5, 8, 9 and 13) may be used to compare and contrast different matrices.

If the output file is given the same name as the input file, the old input file will be effectively lost. It is therefore wise to use a new name for the resulting output association matrix.

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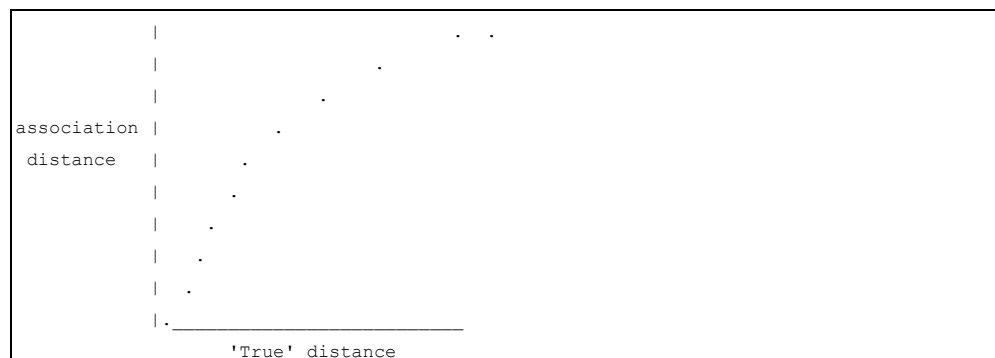
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OPTIONS

Option 1 $d > \text{Threshold}$ -> Missing

Current research on measures of association and ordination has shown that many association measures provide a good estimation of resemblance when objects have attributes in common. Once no attributes are shared, the value of association is 1.0 or maximum dissimilarity is obtained, regardless of other factors.

A diagrammatic representation of this is -



No current clustering method can recover the large association values exactly. Many existing ordination methods such as PCA, PCoA and ALSCAL84 make things worse by dealing with squared association values rather than the values themselves. This only tends to emphasise the larger association values making the problem worse.

One alternative is to re-estimate these larger associations by using a heuristic. The first option is designed to leave re-estimation directly up to the ordination method, currently multidimensional scaling (SSH). The two other methods presented as option 2 and 3 are the Shortest Path (SP) and Minimum Spanning Tree (MST). The theory is to use the smaller (accurate) associations to re-estimate the larger associations.

Unlike the following options, this option simply flags values of association greater than a user supplied threshold. The threshold value can be in the form of any of the three -

1. An absolute value
2. A proportion of the maximum D value
3. A proportion of the range of D values

This in itself would be of little significance except for the possibility of subsequent techniques being used to estimate the missing values from the non-missing. One such technique is the multidimensional scaling program SSH. This method can use a combination of linear and ordinal regressions that enable an estimation of the missing values.

SSH expects missing data to be coded as less than zero.

Option 2 D > Threshold --> Missing (Network)

The shortest path method is computationally expensive (around N^3), but is not limited to using Minimum Spanning Tree pathways of the next option. Logic dictates that it will if anything, overestimate the larger distances by being restricted to smaller steps through the network. The combination of the original and shortest-path corrected values may be effective on providing a lower and upper-bound for the association values.

This method attempts to find the shortest pathway between rows having an association value above a user-defined threshold. The algorithm used was published by Floyd in the ACM algorithms, 1962. It first sets the large values in the association matrix as missing and then *attempts* to find the shortest pathway connecting the two points without using any missing (large) distances. This is not always possible. Failure is encountered with matrices containing many large association values.

The shortest path algorithm is based on the summation of distances via intermediate objects. For example, given the objects 'A', 'B', 'C', 'D', 'E' and 'F', if the association A-F has been set missing, the algorithm may re-estimate A-F by a path such as the shortest of A-C-F, A-B-C-D-E-F, A-E-F etc.

In some instances, where curvilinearity in an ordination is obscuring the overall structure, over-estimation using the 'shortest-path' may be preferable to indeterminacy using missing association in SSH. This amounts to some stretching of large distances in the former and a lack of control of peripheral data points in the latter.

Option 3 $D > \text{Threshold} \rightarrow \text{Missing (MST)}$

The minimum spanning tree (MST) is defined as the path of minimum length that connects all objects and has no circuits (see MST).

The idea is to re-estimate distances above a user-defined threshold by summing the MST path lengths between the offending pairs of objects. The MST algorithm is fast ($N \log N$) therefore this represents an efficient solution. Problems may arise however when two objects are separated by very long MST paths and only moderate association. This implies that, if anything, the MST will always tend to over-estimate association values.

Given the example used in option 2, and the following schematic representation of the minimum spanning tree of the same data:



then the re-estimate of the association A-F would be constrained to be the sum of the distances:



This is reasonable, but appears not to be as reasonable as the 'shortest path' algorithm. If the MST has a few small side chains, then it would generally approach the alternate technique, while MST with many branching nodes (see MST), may give rise to path lengths that are overly inflated.

Option 4 - ABS (D)

This option is relatively straight forward and it results in the output of the absolute values of the input association matrix.

Option 5 ($d - D_{min}$) / d_{range}

Selecting this option results in re-scaling all association values between zero and one. This may be a handy feature for transforming a number of different association measures onto the same scale before using option 8 to compare two matrices.

Option 6 X-D

This option is intended to transfer between similarity and dissimilarity-type association measures. The majority of association measures available in **PATN** are dissimilarity in type. A minimum value of zero corresponds to two objects that are (in terms of the measure) identical and a value of one corresponds to maximal difference. Similarity measures are the complement of this.

This option requires the user to enter a constant (X). If the input values are scaled between zero and one, then a value of one (1.0), for (X) would be fine.

Option 7 $(P \cdot D + Q \cdot 1/D + R)^{S+T}$

This option corresponds to that found in TRND. This formulation is an attempt to allow a reasonable amount of flexibility for generalising transformations. A wide variety of transformations are possible by nominating five constants in the formula -

$$D_{\text{new}} = (P \cdot d + Q \cdot 1/d + R)^{S+T}$$

where d = input association value and P, Q, R, S, T are user defined constants.

All of the constants default to values that will have no effect on the output values. The constants and purpose are:

1. P - Multiplier factor - default of 0.0
2. Q - Reciprocal factor - default of 0.0
3. R - Multiplier factor - first additive factor - default of zero
4. S - Power factor - default of 1.0
5. T - Second additive factor - default of 0.0

Option 8 ABS $(D_1 - D_2)$

This option has been designed to compare two association matrices by taking the absolute value of the difference.

This option is useful when comparing two different association measures on the same set of data. It could also be used to compare the same, or different association measures on two sets of attributes on the same set of objects. The resulting difference matrix, in standard form, can be analysed in the same way as any other association matrix.

The range standardising option above is recommended prior to the use of this option.

Option 9 $P \cdot D_1 + Q \cdot D_2 + R \cdot D_3 + S \cdot D_4 \dots$

This method is designed to provide the flexibility necessary when dealing with *mixed attribute data*. In this situation two problems arise -

1. Different measures of association are appropriate for different attributes in the data set, independent of data transformation and scaling. This is not uncommon.
2. It is unreasonable to assume that independent weighting of each attribute will result in an association value which is scientifically justifiable. In other words, some attributes, for reasons of number or scaling, will dominate the measure of association.

The first problem requires different data sets with the same objects and different attribute-types. ASO can take each independently, and produce an association matrix order N .

The second problem requires some weighting of the resulting association matrices as noted above. The current option, in conjunction with range standardisation (option 5), can achieve this.

There are plenty of complex situations, for example, soil profiles with a different number of attributes for each horizon. Double weighting is required to weight each horizon equally in addition to groups of attributes for each horizon that require equal weighting. This option can accept multiple association matrices, each with an independent weighting, and produce an overall 'average' association matrix.

The number of input matrices is requested and then, for each matrix, a constant value to be used as a multiplier. These constants are automatically proportioned. It is not necessary to calculate proportions, any scale may be used as long as all values are relative to each other.

Option 10 Gower Correction for PCoA

This option is to perform the Gower correction to enable association measures, other than the Pearson's Product Moment Correlation Coefficient to be used in Principal Component Analysis (PCA). The combined transformation and PCA is usually termed Principal Coordinate Analysis (PCoA).

There are two separate but interrelated tasks. The first is a transformation on associations to similarity form. The second is a standardisation to Euclidean form.

The aim of the Gower correction is to permit the use of most association measures in ordination techniques, ie it converts the association inferred from most proximate values into Euclidean form. The correction is -

$$D^* = D - R_m - C_m + G_m$$

where

D^* = corrected distance
 D = original distance
 R_m = row mean
 C_m = column mean
 G_m = grand mean

For non-Euclidean association measures, the usual transformation prior to the correction has been to complement the measure. For Euclidean type associations (Minkowski, Chi-squared) a different transformation is suggested by Gower. This is -

$$-0.5 * d^2$$

ie, Euclidean distance needs to be squared, halved and its sign changed. For squared Euclidean distance, squaring is obviously not required.

If a true Principal Component Analysis is required, then it is assumed that ASO has been used to produce an association matrix containing product moment correlation coefficients. TRNA is still required to copy the *-.aso* file to the *-.gow* file and a set of diagonal values (1's) is created in *-.dia*. In this situation therefore, no transformation (step 1) or standardisation (step 2) is required.

A number of different transformations are available. The standard Gower transformation is the second ($-.5*d**2$). Some work needs to be done to re-evaluate the range of association measures in relation to the transformation to Euclidean form.

```

1 = -0.5 * D (for squared Euclidean distance)
2 = -0.5 * D ** 2 (for most other association measures)
3 = 1.0 - D (research transformation)
4 = no transformation (for the correlation coefficient)

```

Output will be a file with the extension .GOW that is designed as input into PCA. An additional file -.dia contains values for the diagonal of the association matrix.

Option 11 Encoding Adjacency

This option is used in conjunction with the hierarchical agglomerative classification module FUSE to produce a classification of spatially related data that is spatially (geographically) constrained. This generally means that fusions are **constrained** to be between pairs of objects and groups of objects which have been declared to be spatially adjacent.

The coding of adjacency is simple. Objects that are to be considered adjacent (on whatever basis) have their corresponding association value made negative-

$$D_{out} = - D_{in}$$

FUSE can then detect and maintain adjacency during the fusion process.

Input for this option is a list of adjacent objects for each object in the data set, minus one. Because adjacency is a symmetric factor (if A is adjacent to B, then B must be adjacent to A), only a lower or upper symmetric matrix, minus the diagonal, is required for complete enumeration. The coding is in *integer free format* in the form:

Object number - Number of adjacent objects - List of adjacent objects.

The numbers can either be separated by one or more blank characters or a single comma. The coding for any one object may continue over as many records as desired. The adjacency coding for each new object must begin on a new line. An efficient rule is to code adjacency for each object, only those other objects that have a *higher sequence number*.

For example, for 10 objects -

1	2	3	4	
2	2	4	5	
3	3	4	5	6
4	2	7	8	
5	2	6	7	
6	2	8	9	
7	2	8	9	
9	1	10		

This is interpreted as object 1 has 2 adjacent neighbours; 3 and 4. Object 2 has two; 4 and 5, and so on. The number in the first column is always less than the list of objects it is adjacent to.

It is possible to have a set of objects that are not completely connected in terms of adjacency (disjoint). A number of spatial groups may be defined; all objects in the groups connected together in some way or another, but having no inter-group connection. In this case, the adjacency is incomplete and FUSE will have to drop the adjacency constraint when the next fusion has no adjacent pairs of objects or groups of objects.

Sub-option 1 Encoding Adjacency

This option reads an adjacency coding in free integer format from a file and use it to encode adjacency in the default *-.aso* file. Selecting this options will also, automatically invoke the second option.

Sub-option 2 Checking the Adjacency Connections

This option is used to examine all the connections implied by the adjacency coding of an association matrix. This results in the production of a standard *group-definition file*. Each group in the file contains objects that are connected by the adjacency network. If more than one group is defined, there are disjunctions in the coding. FUSE can handle this by dropping the adjacency constraint when this level of the hierarchy is reached (no adjacent fusion is possible).

Sub-option 2 is invoked automatically if sub-option 1 is selected to read adjacency.

This option cannot be used when the method of producing the association can result in negative values, for example, the Correlation Coefficient.

Option 12 - Rank Associations

If you believe that anything more than the rank order of association values in the *-.aso* matrix are superfluous, then ranking before classification may be an option. This option effectively reduces ratio-values to ordinal-values. It implies that magnitude of difference is not important, only that there is a difference and that $a > b$ or $b > a$.

This option may be used in conjunction with ASIM (ANOSIM) to "make the test as non-parametric as possible" (Clark & Green, 1988). Personally, we know that for some association measures, ratio-accurate information exists to a level around 0.9 (see SSH). Therefore, this transformation for ASIM is seen as unnecessary. The choice is (as always in PATN) yours!

Two sub-options govern the type of ranking that will be performed.

The FIRST option creates ranks by using the AVERAGE rank of tied values. This is equivalent to sorting the data and using the AVERAGE position from the sorted data. Consider an example:

1	1	2	3	4	4	would be ranked as:	1.5	1.5	3	4	5.5	5.5
---	---	---	---	---	---	---------------------	-----	-----	---	---	-----	-----

In this case, the ranks for say ten values may NOT contain '1' or '10'.

The SECOND option creates ranks by using the next sequential ranking, starting at rank '1' for the smallest value. Unlike option 1, this option creates INTEGER values as ranks. Tied values are ranked identically. This option is equivalent to sorting the values and using the NEXT sequential whole number, starting from '1' for the smallest value. Consider the same example as above:

1	1	2	3	4	4	would be ranked as:	1	1	2	3	4	4
---	---	---	---	---	---	---------------------	---	---	---	---	---	---

Option 13 - Cophenetic Correlation

The cophenetic correlation is defined as Pearson's product moment correlation between as input association and the implied ultrametric matrix derived from a dendrogram.

This measure is akin to a *stress* that is generated in trying to hierarchically organise a set of objects given the input association measure. It is similar in nature to the *average deviation* in ALOC the stress value in SSH or the eigenvalues in PCA/PCoA. Seeing that the measure is a correlation, the higher the value the closer are the implied ultrametric-distances from the dendrogram to the input association measures.

Option 13 seeks the *input* matrix from the default *-.aso* from ASO, GASO etc. and the *ultrametric* matrix as *-.ult* from the last option in DEND. Both of these files are in PATN binary format and cannot be read or printed without using ASO to convert them to ASCII. The output is a single correlation value to the terminal.

Option 14- Mantel Test

The Mantel test (Mantel, 1967; Manly, 1991) is a randomisation-type test that is designed to estimate the degree of resemblance between two distance-type matrices. Two standard lower-symmetric matrices (binary *-.aso* files) are compared by some reasonable measure. In most of the studies, Pearson's product moment correlation coefficient (*r*) is used. After establishing the relationship between the two matrices, one is randomised. In **PATN**, the order of the values is permuted using a random number generator. The two matrices are again compared and the result noted.

The process of permutation of the same matrix is repeated from 100 to 10,000 times. The degree of resemblance of the original estimate can then be statistically validated by noting how many times the correlation of a random permutation exceeded the original estimate. Say for example 100 permutations are performed and three of them equalled or exceeded the original correlation. This would be interpreted as a significance level of 0.03 (3/100).

Some thought is required as to the validity of the permutation and the number of randomisations. If all arrangements of the objects in one of the matrices are equally likely, then the test as explained is valid. In some situations, a problem can be coerced into this form. A wider range of questions than may be initially imagined can be addressed using this method. The reader is encouraged to scan Manly (1991) and Legendre (1993) to see how some interesting biological questions have been formulated using the Mantel test. As to the number of permutations that may be necessary, Again, Manly (1991) has a general rule of thumb, for a 0.05 level of significance, he suggests 1,000 and for 0.01, 5000! He points out that this is probably only necessary when the result is close to 0.05 or 0.01. As noted in most good statistical texts, one should not be **driven** by 0.05 or 0.01 barriers. The idea is to gauge each result on its merits; limits such as 0.05 and 0.01 are just handy guidelines!

TRNA option 14 will automatically read the default *-.aso* file (called matrix 1) and request a second (default of *-.as2*: matrix 2). The first option is for the use of either *mean-difference* (1: default) or *correlation* (2) to be used as the method of estimating resemblance between the two matrices. NOTE: higher resemblance means lower (1) and higher (2).

If average-difference is used, **both** matrices are standardised by subtracting the matrix minimum and dividing by the matrix range, ie

```
d.new=(d.old-d.min)/d.range
```

If the correlation coefficient is used, then **both** matrices are normalised, ie:

```
d.new=(d.old-d.mean)/d.standard deviation
```

This is only required once and the standardised versions are **not saved** over the original files. The two matrices are then compared and the result noted. The user is requested to nominate the total number of comparisons to be made. Comparison of the two matrices is made after each randomisation and three measures are noted:

```
1. if the measure is < original value
2. if the measure is = original value
3. if the measure is > original value,

and if using 'r':

4. if the absolute test value is >= absolute original value.
```

Summaries of the comparisons are listed in percentage form on completion along with the original comparison value. In addition, the mean difference or mean correlation and the associated standard-deviation is output. TRNA will note the % completion of the randomisation process. Be realistic in the number of randomisations if you have a slow machine and large order matrices! If you don't see 10% completion after 10 hours processing, you know what to do...ask for less or get a faster machine. TRNA is *extremely* efficient in permuting and comparing so don't bother to write a faster one!

If you have some interesting uses or amendments for this option, please let me know (e-mail, fax, phone, or even write a letter) so I can pass them onto other users.

ASSOCIATION BETWEEN OBJECTS: ASO

OUTLINE

ASO is one of the most important modules in PATN, being the main way of generating association measures between all pairs of rows in the data file. The term *association measure* is used in **PATN** to include other more specific terms such as -

- proximity
- distance
- dissimilarity
- similarity
- affinity
- correlation
- resemblance

Structure

ASO produces a lower symmetric matrix of association values stored by row in an unformatted file *-.aso*. An order five matrix (based on a five row data matrix) would be in the form -

[illegible]

ASON may be used to re-structure and re-format various other forms of association matrices into and out of PATN. In some instances, the association between an object and itself cannot be assumed to be zero (dissimilarity). In this case, a file -

-.dia

will be created to contain the diagonal values.

The time ASO takes to run is proportional to the square of the number of rows. Large data sets may therefore take considerable time to process. Consider using ALOB/ALOC for large data sets (these examine object-centroid distances, not object-object distances).

ASO uses the fact that the association measures for continuous or numeric data can in most cases, be used to produce the more common presence/absence (1/0) association measures. The continuous and presence/absence counterparts are given below.

Standardisation

Another significant issue when dealing with measures of association is *standardisation*. For reasons of comparison and simplicity, most of the published association measures are standardised to produce values in a limited range. In other words, most formulations of association measures have an in-built standardisation. The one obvious exception is the Minkowski series. This does not however suggest that data standardisation is never required, it means that some thought is required to analyse the response of the measure that is to be used with the data coding. In some circumstances, the data may still require some transformation prior to the application of the association measure. One of the important features to examine is the response type as outlined below.

Warning: a separate standardisation is usually necessary with an analysis of attributes.

Terminology

The terminology used in the formulations using non presence/absence data below is:

$$D_{ik} - D_{jk}$$

where D_{ik} is the data value of the i th object row and k th attribute.

For presence/absence data, it is usual to base the formulation of association on the following two-way table -

		ROW 2	
		1 0	

	1	A B	
ROW 1		-----	
	0	C D	

Association Type

The headings for each option below includes four summary features of the measures. The *first* details the general **type** of measure by using the coding -

D - dissimilarity type (smaller values = increased similarity)

S - similarity type (larger values = larger similarity)

Range of Values

The *second* code shows the **range** of the values taken by the particular measure of association. For example -

```
0 -> 1 implies from 0.0 through to 1.0
```

The measures of association used in **PATN** are generally of the *dissimilarity* type. The smaller the value of association, the greater the affinity between the pair of objects. The minimum value taken is usually 0.0, inferring that the two objects are identical. The maximal value is 1.0 for all measures except the Minkowski series (option 4). Because these are un-standardised, there is no upper limit. The only other odd measures are Pearson's Product moment correlation coefficient and the rank order equivalent, Spearman's Coefficient. These measures vary from -1.0 for perfect inverse correlation through zero (no correlation) to +1.0 meaning perfect positive correlation. NOTE: the great majority of **PATN** options that use association measures, assume they are the 'standard' dissimilarity type.

Response Type

The *third* coding gives some indication of the response of the measure to differences in values at different points in the attribute range. This is not applicable to measures that are specifically designed for presence/absence data. Two types of response are available-

I - Interval and
R - ratio

The interval response implies that the values of association are unaffected by comparisons at different positions in the range of the attributes. For example, simple difference (= Manhattan distance):

```
6 - 1 = 10 - 5 = 100 - 95 = 1000 - 9995
```

is interval. Differences in values are independent of position of the values in the range. Ratio responses on the other hand, provide different values of association for the same difference depending upon position in the range of the attribute. That is, the distance away from zero is significant. Using the above example, it is seen that the Canberra Metric (option 2) is ratio-

6 - 1	not=	10 - 5	not=	100 - 95	not=	1000 - 9995
-----		-----		-----		-----
6 + 1		10 + 5		100 + 95		1000 + 9995

In such instances, the dissimilarity values decrease for the same difference at larger data values. In many instances, this type of response is appealing because, higher values are rarer, ie matches at the higher level should contribute more to similarity than matches at the lower attribute levels. At this time, all ratio response association measures in ASO have this type of *polarity*. If the opposite polarity is required, use TRND to Re-code.

Symmetry

The fourth symbol refers to whether the presence/absence form of the measure is *symmetric* or *asymmetric* in respect of 0-0 matches. *Symmetric* in this sense means that 0-0 matches will be considered as adding to the overall similarity between the two objects, while 'asymmetric' is taken to imply that these comparisons are not taken into account.

The decision hinges on whether both objects can be considered as more similar by both failing to have an attribute present. For example, should two sites be more similar because they both fail to contain a particular species? The absence of the species at both sites may be due to a number of reasons. It could be that the species was absent because the same circumstances were present or absent in both sites. In this case, the matching of the zeros is significant and a symmetric approach would be reasonable. On the other hand, the species may have been absent from both sites for independently different reasons. There is some argument in this case that the sharing of the absence should not increase the similarity of the two sites, therefore 'asymmetric' measures should be used. The code used is -

0x0	-	0 - 0	matches excluded
0i0	-	0 - 0	matches included

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OPTIONS

Option 1 Bray & Curtis (D|0->1|R|0x0)

This is the default association measure in **PATN** because it has consistently performed well in a variety of tests and simulations on different types of data (See Faith, Minchin and Belbin, 1987). On that basis it is to be recommended for data that requires some type of *ratio* response.

For continuous data the measure of Bray and Curtis is formulated as-

$$D = \sum |D_{ik} - D_{jk}| / \sum (D_{ik} + D_{jk})$$

For presence/absence data, the association measure is attributed to Czekanowski and is -

$$D = 1 - 2A / (2A+B+C)$$

This measure is *not* a simple derivative of the Jaccard measure below. The '2A' in the numerator is the joint *richness* and the denominator is the total richness.

Option 2 Canberra Metric (D|0->1|R|0x0)

This is an intuitively simple and appealing measure. It is formulated-

$$D_{ij} = (1/(M-Z)) * \sum ((D_{ik}-D_{jk}) / (|D_{ik}|+|D_{jk}|))$$

where Z = the number of 0-0 matches.

While it has a non-linear response, it has a single drawback for non-presence/absence data that contain a number of zero values. For any comparison of a non-zero value with zero, the Canberra Metric results in an association value (for that attribute) of 1.0. For example:

$$\frac{1 - 0}{1 + 0} = 1 \quad \text{and} \quad \frac{100 - 0}{100 + 0} = 1$$

It is unreasonable to say that '1' is as different from zero as '100' is. To get around this problem, Bill Williams (*pers.comm.*) suggested that the zero values in the data matrix should be replaced by one fifth of the smallest non-zero value in the matrix. For the above example, this would result in:

$$\frac{1 - 0.2}{1 + 0.2} = 0.66 \quad \text{and} \quad \frac{100 - 0.2}{100 + 0.2} \approx 1.0$$

The absolute value signs on the denominator are attributed to Adkins and result in a Canberra Metric value of 1.0 when attributes of different signs are compared.

For presence/absence data the Canberra metric is transformed into the dissimilarity form called **Jaccard** -

$$D = 1 - A / (A+B+C)$$

Option 3 Correlation (S|-1 -> 1|0i0)

This is Pearson's Product Moment correlation coefficient. As such, it varies from '-1' implying perfect complementary association where high values on one attribute linearly correspond to low values on the other attribute, through zero (no relationship) to perfect linear association. As a similarity measure, it cannot be used by the majority of subsequent analysis measures without some transformation using TRNA. The single exception to this is where this coefficient is used to obtain a principal component analysis.

The correlation coefficient is -

$$r = (N \cdot SP_{jk} - SS_j \cdot SS_k) / (N \cdot SS_j^2 - SS_j \cdot SS_j) \cdot (N \cdot SS_k^2 - SS_k \cdot SS_k)$$

where SP_{jk} = Sum of products between j-k and
 SS_j = Sum of squares of j

For presence /absence data, it becomes -

$$D = (A \cdot D - B \cdot C) / \sqrt{(A+B)(A+C)(B+D)(C+D)}$$

The correlation coefficient is not recommended for any Pattern Analysis, not even shape comparisons (see Chi-Square distance).

Option 4 - Minkowski (D|0->?|0i0)

For convenience, the Manhattan series, including the Manhattan metric, the Euclidean distance and other similar measures, are coded into a generalised form -

$$D = (\sum |D_{ik} - D_{jk}|^p)^{1/q} / M$$

where the values of p and q are constants that are requested and M is the number of non-missing attributes. **Warning:** the associations are *standardised* by M because of the need to accommodate **missing data**. If a comparison on any attribute is skipped due to either object having a missing value, the divisor (M) is reduced by 1.

Typical values of p and q, and the corresponding measure are -

If $p = q = 1$ = Manhattan metric
 If $p = q = 2$ = Euclidean Distance
 If $p = 2, q = 1$ = Squared Euclidean distance

The presence/absence equivalent formulation corresponds to the Simple Matching Coefficient -

$$D = 1 - (A+D)/(A+B+C+D)$$

Option 5 Gower Metric ($D|0 \rightarrow 1|I|0i0$)

The Gower metric is probably one of the better measures to use when dealing with continuous data requiring, or displaying (if the processes are known), a truly linear response. For example, if the histogram of associations (ASON) showed a uniform or symmetric shape, this method would be appropriate.

It is a simple formulation, basically Manhattan distance that is range standardised. For continuous data this is -

$$D = (1/M) * \sum |D_{ik} - D_{jk}| / \text{RANGE}(k)$$

while for presence/absence data, this corresponds to the Simple Matching Coefficient -

$$D = 1 - (A+D) / (A+B+C+D)$$

Option 6 Two-Step ($D|0 \rightarrow 1|R|0x0$)

This measure is unique. It applies more strictly to the association of attributes than of objects and has been designed to address the problem that species classification usually resulted in a frequency bias. Species having a similar frequency of sites were placed together in the classification.

TWO-STEP is based on the Bray and Curtis measure. The first step of TWO-STEP applies this measure in a true asymmetric form, ie the relationship between species A and B may not be the same as the relationship between species B and A. The basic philosophy of the first step is 'if I don't occur in a site, then I can't say anything about another species (that may or may not)'. For species i and j at site k, where $i > 0$ -

$$D_{ij} = \text{Bray and Curtis (continuous data) or} \\ D_{ij} = B / (2A+B) \text{ for binary data}$$

and for the j-i comparison, when $j_i > 0$ -

$$D_{ij} = \text{Bray and Curtis (continuous data) or} \\ D_{ij} = C / (2A+C) \text{ for binary data}$$

This results in a square matrix that is asymmetric with the diagonals having a value of 0.0 (species are identical to themselves). This matrix is output from the TWO-STEP option and may be used in the multidimensional scaling program SSH.

Because the majority of analysis options in **PATN** (and elsewhere) require a symmetric matrix, a second step is added. The solution that TWO-STEP uses is a simple one; treat the asymmetric matrix as though it were a *data* matrix and apply some appropriate measure of dissimilarity. This methodology imparts a significant gain. Unlike all other measures, the comparison of two species is based on the similarity and differences of their 'responses' to all other species. It is not based on just the one single comparison.

The response required for a comparison of the values in the rows of the asymmetric matrix requires a non-linear response. The difference in small values (associations) are valued more than differences in moderate values. Comparing two high values imparts little information, for it implies the statement "just because we are both a long way away from this third species, it doesn't mean that we are closely related".

The polarity of the response is therefore the reverse of the usual, where higher values are weighted more. There are two alternatives with the second step -

1. standardised Manhattan difference and
2. a polar response equation -

$$D_{ijk} = 0.5 * (|D_{ik} - D_{jk}| + \text{MAX} (D_{ik} + D_{jk}))$$

One further detail is that the direct comparison (i-j) is weighted equally to the off-diagonal comparisons. The final measure is then -

$$D_{ij} = \frac{(D_{ij} + D_{ji})}{2} + \frac{D_{ik}}{(M-2)}$$

This means that the direct comparisons are up weighted in relation to the indirect comparisons.

Option 7 General Binary (D|?|?|?)

This option provides for coding of a wide range of presence/ absence association measures based on the usual two-way table nomenclature -

$$D = \frac{e*A + f*B + g*C + h*D}{i*A + j*B + j*C + l*D}$$

where A, B, C and D are defined as usual and the constants: e, f, g, h, i, j, k and l are entered by the user.

Using this formulation, the Jaccard coefficient is simply programmed by using the following coefficient assignment:

```
e = 1
f = 0
g = 0
h = 0
i = 1
j = 1
k = 1
l = 0
```

while the coefficient of Russell and Rao would be coded as:

```
e = 1
f = 0
g = 0
h = 0
i = 1
j = 1
k = 1
l = 1
```

Option 8 C-Coefficient (D|0->1|R|0i0)

The C-coefficient can be seen in some respects, as a compromise between the binary symmetric/asymmetric measures in terms of the relationship with the 0-0 matches. The formulation for continuous data is -

$$D_{ij} = \sum [0.5 * (|D_{ik} - D_{jk}| + \text{MAX}(k) - \text{MIN}(D_{ik}, D_{jk}))]$$

The use of the continuous form of the measure may require prior transformation using TRND to subtract the minimum value of each column (attribute) from each column value. The presence absence form is -

$$D_{ij} = 1 - (A+D/2) / (A+B+C+D)$$

Option 9 - Kendall/Russell & Rao (D|0->1|L|0x0)

The presence/absence form of this measure is attributed to Russell and Rao and is formulated as -

$$D_{ij} = A / (A+B+C+D)$$

This coefficient is only interested in the number of matched characters. It has been used by pheneticists and cladists in conjunction with UPGMA for generating phenograms.

A quantitative equivalent to this was formulated by Kendall as -

$$D_{ij} = \sum \text{Max}_k - \sum (\text{Min}(D_{ik}, D_{jk}) / M$$

The first term is the sum of the Maximum value of each attribute. **Warning:** the associations are **standardised** by M because of the need to accommodate **missing data**. If a comparison on any attribute is skipped due to either object having a missing value, the divisor (M) is reduced by 1.

Option 10 Chord Distance (D|0->1|R|0x0)

The original chord distance measure varied from 0 (identical) to the square root of two. It has been modified for **PATN** to range from 0->1 for consistency -

$$D_{ijk} = [1 - Q_{jk} / (Q_{jj} * Q_{kk})^{.5}]^{.5}$$

where

$$Q_{jk} = \sum h(D_{hj} * D_{hk})$$

$$Q_{jj} = \sum h(D_{hj})^2$$

$$Q_{kk} = \sum h(D_{hk})^2$$

and $h=1, \dots, n$

This measure corresponds to the square root of the cosine theta measure as used in PATN (complement of the cosine theta). It may be used as a *shape* (Vs *size*) coefficient, comparing the variables in terms of ratios rather than absolute value differences. In this regard, Chord distance is very similar to Chi-Square distance and somewhat similar to the correlation coefficient and the Cosine measure. The better size measures appear to be Chord and Chi-square.

Option 11 Spearman Rank (D/S|-1->1 0-1|I|0i0)

This measure can be formulated as the application of Pearson's product moment correlation coefficient to data that has been transformed into ranks. Sub-options permit both correlation and distance formulation. The correlation formula, as based on ranked data is -

$$r_o = 1 - 6 [\sum (A-B)^2 / M (M^2-1)]$$

(range -1 to +1)

and as a distance measure -

$$D_{ij} = 3 [\sum (A-B)^2 / M (M^2-1)]$$

(range 0 to 1)

where A,B are the attribute values converted to ranks.

Option 12 Binary Rank (D|0-1|R|0x0)

This coefficient was derived from users responding to a questionnaire requesting the ranking of association on a test set of binary data. The result of the survey was consistent and quantifiable. The A's were shown to contribute primary association, the D's lesser association, while increases in B+C's contributes to dissimilarity. An example of ranking from least similar row pairs to most similar row pairs for 10 attributes was -

```

A  B+C  D
-----
0 -10 - 0 least similar pair
0 - 9 - 1
0 - 8 - 2 -----> 0 - 0 - 10 |
1 - 9 - 0
1 - 8 - 1 -----> 1 - 0 - 9 |
2 - 8 - 0
2 - 0 - 8 -----> 2 - 0 - 8 |
.
.
4 - 6 - 0
4 - 5 - 1 -----> 4 - 0 - 6 |
.
.
9 - 1 - 0
9 - 0 - 1
10- 0 - 0 most similar pair

```

The formulation of this response corresponds to -

$$D_{ij} = 1 - (A / (A + D + 2 * (B+C))).$$

This maintains a monotonic association value by first ranking pair-wise comparisons by their A value and internally via increasing D value. This results in the formula -

$$D_{ij} = 1 - (A/T) + D / T \cdot (T-A+1)$$

where T = total number of attributes.

Another interesting feature of this coefficient is that it breaks the tied values associated with the coefficient of Russell and Rao. In terms of self comparison, the coefficient will produce different values depending on values of A, B, C and D. The result is that the coefficient can be considered to be 'row conditional' in multi-dimensional scaling.

Option 13 Profile (D|0->1|I|0i0)

This measure is designed for the comparison of *profiles* (Faith, Austin, Belbin and Margules, 1985), curves or two-dimensional attributes, as against the more usual, un-linked (1-dimensional) attributes. This formulation has greater significance when the data is *spiky*. Smooth data will operate reasonably with an independent attributes approach.

When using profile-type attributes, the *number of attributes* (columns) parameter is set to the number of profile attributes, not the total number of classes. For example, given 3 profile attributes with 10 classes each, the **PATN** parameter would be '3', while ASO (profiles option) will request the number of classes in each of the three profile attributes (in this case - 10, 10 and 10).

The horizontal **scale** for these attributes can be ordinal, interval, or ratio. The vertical scale can be nominal, ordinal, interval, or ratio. The formulation for this option between two objects a and b with a single profile attribute having N classes is -

$$D_{ab} = \sum_{i(1 \text{ to } N)} \cdot \sum_{j(1 \text{ to } N)} V_{ij} \cdot (a_j - b_j) \cdot (a_i - b_i)$$

where $V_{ij} = \text{fn } (e^{P(N-|i-j|)})$

The V_{ij} function, through the user parameter P controls the nature of the class separation decay weighting; the contribution to association at a given class from other classes depends on the number of classes separating them. As can be seen from the formula, V_{ij} is based on an exponential decay curve.

Additional parameters associated with this option are -

1. Circular or non-circular. If the profile attributes are the months of the year, the circular method accepts January to be as close to December as it is to February. Non-circular would assume 11 'units' of difference between January and December but only 1 'unit' of difference between January and February.
2. The number of classes per profile. Don't forget, a profile = **PATN** attribute.
3. A value for P. This controls the size of the neighbourhood, ie a small value gives a large neighbourhood. A value of 0.1 is a reasonable start. A value of 0.01 provides only a 1% decay while a value of 1.0 approaches the Gower Metric (option 5), proving independent (un-linked) attributes.

4. The range for standardisation of each class. The choice is either a constant or the actual range. Using a constant will maintain the shape of the curves across all objects. Each class however will have a weight that is dependent on data values in that class. Using the actual range effectively changes the shape of the curve but will equally weight each class.

Option 14 Chi-Square Distance (D|0->1|R|0x0)

This measure is commonly associated with correspondence analysis and reciprocal averaging. While this is true in principal, the ordination algorithms employed in such analyses often involve transformations to data or to the resultant vectors that lose the connection between the input Chi-Square dissimilarity and the output ordination distances. The Chi-Square distance is -

$$D_{ij} = \sqrt{ \sum_k [(1/C_k) \cdot (D_{ik}/R_i - D_{jk}/R_j)^2] }$$

where

$$C_k = \sum_{\text{COL } k}$$

$$R_i = \sum_{\text{ROW } i}$$

$$R_j = \sum_{\text{ROW } j}$$

Chi-Square distance may be used as a *shape* (Vs *size*) coefficient, comparing the variables in terms of ratios rather than absolute value differences. For example, the following two objects are identical in terms of Chi-squared distance-

Object 1	1	3	5
Object 2	2	6	10

Both have the same *shape*, but are different in *size*.

Chi-Squared distance is very similar to Chord distance and somewhat similar to the correlation coefficient and the Cosine measure. The better *shape* measures appear to be Chord and Chi-square. The best *size* measures would be Bray and Curtis, Kulczynski and Gower Metric, depending on application.

Option 15 Cosine/Ochiai (D|0->1|R|0x0)

The measure as formulated in PATN corresponds to the complement of the cosine of the angle between non-centred rows. The complement has been taken so as to transform the original similarity measure into a dissimilarity form. It is formatted as -

$$D_{ij} = 1.0 - \sum (D_{ik} \cdot D_{jk}) / \sqrt{(\sum (D_{ij} \cdot D_{ij}) * \sum (D_{ik} * D_{ik}))}$$

The binary form, which is the coefficient of Ochiai is formulated as -

$$D_{ij} = \frac{1 - A}{\sqrt{(A+B) * (A+C)}}$$

It may be used as a *shape* (vs *size*) coefficient, comparing the variables in terms of ratios rather than absolute value differences. In this regard, it is similar to Chi-Square and Chord distance and the correlation coefficient. The better *size* measures appear to be Chord and Chi-square. The correlation coefficient is not recommended for any Pattern Analysis (not even shape comparisons)!

Option 16 Yule (binary) (D|0->1|R|0x0)

This dissimilarity measure is for use with presence/absence data that are coded in the standard form with '0' for absence and '1' for presence. The response of the measure in some situations is not appealing. For example, if either 'B' or 'C' is 0.0 then the association value is zero while if either 'A' or 'B' is zero, the value of the measure is 1.0.

Other than in the above circumstances, the measure has been shown in simulation studies (Faith, Minchin and Belbin, 1987) to have a good recovery of 'true' distance when the attribute distribution curves are monotonic.

$$D_{ij} = \frac{B * C}{A * D + B * C}$$

Option 17 Kulczynski (D|0->1|R|0x0)

The Kulczynski coefficient was originally formulated for presence/absence data as:

$$D_{ij} = \frac{0.5 * (B + C)}{A+B \quad A+C}$$

which has proven (Faith, Minchin and Belbin, 1987) to be, along with Bray and Curtis, a good measure of association for ecological applications. Because of this superiority, two continuous forms of the measure have been designed and are included in this option. They are firstly, the symmetric form -

$$D_{ij} = 1.0 - 0.5 \left[\frac{(\text{Min}(D_{ik}, D_{jk}) + \text{Min}(D_{ik}, D_{jk}))}{\sum D_i \quad \sum D_j} \right]$$

and the symmetric form as

$$D_{ij} = 1.0 - \frac{(\text{Min}(D_{ik}, D_{jk}))}{\sum D_i}$$

and

$$D_{ji} = 1.0 - \frac{(\text{Min}(D_{ik}, D_{jk}))}{\sum D_j}$$

GROUPED ASSOCIATION:

GASO

OUTLINE

GASO was designed for *mixed* attributes. It is not generally distributed with the PC version of **PATN** because it is open to considerable abuse if you are not confident. GASO works on the same principal as ALOB (the large version of ALOC). GASO is a *complicated* version of ASO in the sense that it permits the aggregation of attributes into groups that will be *equally weighted in the determination of pair-wise association between objects*.

The previous version of GASO permitted the use of a number of association measures to be used on different sub-sets of attributes. This has been discontinued. Ideally, the contributing- weight of each measure should be equal. The only way to achieve this is to first calculate all the associations, range standardise them and then add them in a consistent manner. My solution to this was to permit one association measure only. If you require the use of mixed associations, it is probably better to aggregate the different *types* of attributes into separate data sets. Then apply the appropriate association measure using ASO to each set in turn. The different association matrices can then be standardised and added in TRNA. While it takes a little longer, you are made aware of types of operations you are performing and hopefully minimising the possibility of error.

OPTIONS

Number of attribute groups

GASO assumes the presence of a **-.gas** file detailing which group each attribute belongs to. For M attributes there will be M integers in a file. If all numbers are the same, it is equivalent to running ASO using either Bray & Curtis or Gower metric. The integers in this file must begin with '1' and increment to the number of attribute groups you desire. For example if there are 3 groups, the only group numbers permitted are '1', '2' and '3'. These are to be found in **-.gas** in ascending sequence. For example, for ten attributes this would be correct:

1 2 1 1 3 3 2 1 2 2 while the following would not: 2 1 1 1 3 3 3 3 3

Association measure

The choice at moment is between the Bray & Curtis (Czekanowski) and the Gower metric (simple matching coefficient). More will be added as the need and feedback arises

'MINIMAL-SUBSETS'

OF OBJECTS:

MSET

OUTLINE

The concept of a 'minimal-set' comes from the work of Kirkpatrick (1983) and Margules, Nicholls and Pressey (1988). The concept was to derive a *minimum* set of sites (objects) that contained a specified number of replicates of all species (attributes). MSET embodies but two of a multitude of algorithms which can be used to answer the question: "What is identity of the minimal number of sites needed to reserve/preserve each species k-times". MSET assumes the raw-data is a site-by-species matrix containing presence/absence entries. In the real world, this presumes that each *site* is a comprehensive and systematic sampling of a class of species within a finite area.

MSET currently contains two options, one attempting to maximise *site-rareness* and the other *site-richness* in order to achieve a selected level of representation or replication with the minimal number of sites. As with most **PATN**-algorithms, MSET is heuristic; it *iterates* on a simple *rule-of-thumb*. In MSET, the rule may be to select the site which has the highest site-richness that *complements* what has already been sampled. MAXD is yet another interesting alternative that selects subsets of objects based on intrinsic criteria that you may care to examine.

Algorithms such as MSET are developmental. As a class, MSET is a 'greedy-type' of algorithm. As such, it cannot guarantee *optimality* (see Underhill). In other words, there can be no guarantee that MSET will produce the absolute minimal number of site that will replicate each species k-times. The algorithms in MSET appear to be both simple and efficient. I have emphasised the former as the simple formulae below demonstrate. Other approaches, such as linear-programming or integer-programming may be able to detect a global solution, but do so at a cost of increased complexity to set-up and increased computational time.

An Example

PATN assumes that you have a standard data file (*.dat*) with rows as sites and columns representing species. Entries in the matrix are presence (coded as '1') and absence (coded as '0'). MSET will automatically interpret non-presence/absence data as 1/0 by assuming that a non-zero value is a '1'.

Assume that we want to establish a set of areas as a reserve network and that rare species are a significant issue. The first step would be to find the species that occurred *least* across all sites. Once found, the site that contained this species could be set aside as ‘reserved’. If there are ties in the rarity of a species (more than one site contained the rare species), we may wish to examine other qualities of the sites with identical scores of one or more rare species. For example, we may choose one of the sites containing the selected rare species that has *most species* or *most other rare species*. These are but two of a multitude of alternatives.

Once the first site is selected, we could simply repeat the process, again looking for the *next* rarest species. If we desire to replicate the reservation of the rarest species or move onto the next rarest species, the process is repeated. Iterations stops when replication level for *all species* in the sample is either met, or deemed to be unachievable.

Complementarity

A significant aspect of MSET is the use of *complementarity* (Vane-Wright *et al.*, 1991) that is built into the selection criteria when selecting the second and subsequent sites for the sample. The principle is to select a site that most *complements* the existing sample. For example, the sites-

0000000000000011111	and	1111111111111100000
---------------------	-----	---------------------

would be complementary in the sense that whatever the first site lacks, the second site has, and vice versa. In MSET, this means that species that have not yet been selected in the sample are increasingly *up-weighted* until they are. In a mathematical sense, two numbers are either a complement or not. In the context of MSET, there are however, *degrees of complementarity*. In the above example, swapping a single ‘1’ to a ‘0’ will decrease, but not eliminate complementarity. MSET has a continuous scale of complementarity; two identical sites have zero-complementarity while the example above has the maximum (of say one) complementarity.

Are such algorithms ‘pattern-analytic’? I see minimal set-type algorithms as *exploratory* and have therefore included it for completeness. MSET is experimental. Any feedback will serve to stimulate further thinking and more respectable versions.

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OPTIONS

For the options below, the more usual terms *object* and *attribute* are set aside for *site* and *species*.

Number of Times Reserved

This is the number of times that each species *must* occur in the sample sub-set. If for example, you wish *all* species to be located in at least three of the subset, then select '3' here. The default is '2'.

Site Richness

There are *two primary options*. If ties exist in the selected primary options, the alternate option is implemented in an attempt to break the ties. If ties remain, the site with the lowest sequence-number will be selected. In most cases, different primary options will produce different results on the same set of data. At the moment, both options can produce *efficient* samples in that the tests I have run suggest that no single option will always win. If you want the best, both options need to be run.

Site richness is initially defined as the number of species in a site. If site richness is the selection option, then the first site that will be sampled will be the one containing the most species. If tied values of site richness are encountered, then what is termed '*site-rareness*' will be examined (see below).

Complementarity is built into the selection of the second and subsequent sites by use of the formula for each species-

$$\text{Disparity} = \text{Number of Times Reserved} - \text{Number of times NOT reserved}$$

This means that the greatest positive disparity is given to those species that are outstanding from the current sample. Site richness is therefore calculated as the sum of these disparities for each species in the candidate site. **Site rareness is examined if site-richness values are tied.**

Site Containing Rarest Species

In the context of MSET, the *rarest species* is that which occurs in fewest sites. The *best rare site* is that site which contains the maximum summation of species with the highest rareness-index as defined-

$$\frac{(\text{frequency}_{\max} - \text{frequency})}{\text{frequency}_{\max}}$$

where frequency = number of sites a species occurs in
frequency_{max} = maximum species frequency for the data

This places species-rareness for each species on a zero to one scale where zero represents the most common species and approaches unity for rare species. Species-rareness is added across species that are present for each site to form an index of what in MSET is called *site-rareness*. As in the case of site-richness, site-rareness is calculated by using the disparity value (in the above section) as a simple method of incorporating complementarity into the selection of the second and subsequent sites. That is, site-rareness is the sum of-

```
disparity * species-rareness
```

for each species present in the candidate site. Interestingly, *average-site-rareness* does not appear to be as efficient as *(total)-site-rareness*. **Site richness is examined if site-rareness values are tied.**

Output

The output file is called *-.min* and contains a list of sites in the order they were added and with the index that was used for the selection (rarity or richness). Below that, the species are listed in order of subset frequency (how many times they occur in the sample). On the same line as each species, is the total frequency and the ratio of subset frequency to total frequency. At the bottom of the table, the number of sites in the subset is listed along with the summation of the selection value.

The sequence number of sites may be used as input to MASK by using the *-.min* as you would a *-.irm* but with the format specification of-

```
(10X,I5)
```

In addition, a *-.gav* file is created (similarly to GDEF option 9) that nominates which sites are in the sample (coded as group '1') and which are not (coded as group '2'). The format for the *-.gav* file is '(20I4)'.

Additional Constraints

With large or complex data sets there are likely to be situations where a number of potential reservation sites will be tied in relation to any combination of 'reservation indices'. Undoubtedly, the list of constraints or options could be extended in MSET to the point where ties would eventually be broken. Generalising MSET to accommodate species weights or include area, adjacency or other constraints, although relatively simple to implement, appears premature at this point.

Demo

The following interaction shows the keystrokes in PATN necessary to run a minimal-set solution on an existing data set. PRAM will be used to point to an existing data set. User input is in bold

```

PRAM
Title - Description of data & analysis .....
Random-variate data set generated from RAND
PATN internal database file name .....random.dat
Number of Rows (Objects) in data matrix.....      20
Number of Columns (Attributes) in data matrix..... 10
Number of Row GROUPS.....                          0
Number of Column GROUPS.....                        0
Missing Value.....                                -9999.
Logging (0=OFF_1=LIMITED_2=FULL).....                2

-----PARAMETER OPTIONS:
 0 = Alter ALL (1-7)
 1 = Alter TITLE
 2 = Alter PATN database FILE
 3 = Alter number of ROWS
 4 = Alter number of COLUMNS
 5 = Alter number of ROW GROUPS
 6 = Alter number of COLUMN GROUPS
 7 = Alter MISSING data value
 8 = Alter LOGGING level
 9 = SAVE PARAMETER FILE TO random.PRM
10 = RESTORE a parameter file
11 = STOP PRAM (I,D:0)                                ? : 10

RANDOM   PRM               152 15/03/95   12:52
Input file name (A43,D:<random.prm)           ? : random
11 = STOP PRAM (I,D:0)                         ? : 11

MSET
Reserve each species ? times (I,D:2)           ? : 2
-----SITE SELECTION OPTIONS:
 1 = Site with highest rareness index
 2 = Site with highest richness index (I,D:1)    ? : 1

-----<random.min
03/15/95  12:59:08.89 mset  Random-variate data set generated from RAND

ROW00018(      18) reserved with an index =    1.250
ROW00003(       3) reserved with an index =    2.000
ROW00004(       4) reserved with an index =    2.000
ROW00001(       1) reserved with an index =    1.500
ROW00016(      16) reserved with an index =    1.500
ROW00012(      12) reserved with an index =    0.7500
ROW00006(       6) reserved with an index =    0.5000
ROW00009(       9) reserved with an index =    0.5000
=====
              8 sites                                10.00

Species      subset-f    total-f    ratio
-----
COL00001(     1) is not present in data: check it!
COL00008(     8) =      1.0 /      1.0 =    1.0000
COL00007(     7) =      1.0 /      1.0 =    1.0000
COL00009(     9) =      1.0 /      1.0 =    1.0000
COL00006(     6) =      2.0 /      2.0 =    1.0000
COL00010(    10) =      2.0 /      3.0 =    0.6667

-----Press the RETURN key to continue -or- S <CR> to STOP.....  S
PRINT a copy of <random.min> (Y/N,D:N) ? :  N
Output file name (A43,D: <random.gav ) ? : random.gav
.....Current (default) format is : (20I4)
-----Enter output file FORTRAN format: ? : (20I4)
-----<random.gav
1  2  1  1  2  1  2  2  1  2  2  1  2  1  2  2
PRINT a copy of <random.gav> (Y/N,D:N) ? :  N

```

PRAM is used to point to the data set called *FRED*. MSET is then run with the default options (1. try to reserve each species twice and 2., use the rareness rather than richness approach). Two tables are produced in the file *random.min*. The first table contains a list of the sites required and the order they were added to the minimal set solution. The index used is a measure in this case of how many rare species the site contains. The total number of sites required in this solution is 8.

The second table list how well each species is reserved by comparison to how many times it occurs in the data set. The values in the latter table for say species 8 for example, suggests that it has been reserved once and it only occurs once.

A second file, contains a summary of the membership of the selected subset of sites. This file is in **PATN**'s *-.gav* format. Group '1' refers to those sites that have been selected as required for the minimal set solution. Group '2' refers to all other sites. For the above example, the first site is in group 1, the second site is in group 2, the 3rd and 4th sites are in group 1, and so on. This information can be appended as a column vector to the data set with module **MERG**. Merging in this case may be useful if you want to export the results for further analysis or plotting. For example, if you had geographic coordinates of the sites in a separate file, you could have this as your default data set (point at this data set using **PRAM** option 10), then merge the *-.gav* file from **MSET** (**MERG** option 3) and export it using **DATN** (option 2).

SUBSETS OF 'MAXIMALLY DIFFERENT' OBJECTS: MAXD

OUTLINE

MAXD is an experimental module that tries to find the k most different objects as a subset of N objects forming the complete data set. It does this by searching a standard (lower symmetric) PATN association (dissimilarity) matrix. There are currently three ways that a maximally different set of objects can be sampled from a population.

MAXD also provides for reading a set of sequence numbers of objects in a subset of the complete N taxa, and an association matrix, and then calculating the overall maximal difference according to the selected criteria.

The 'maximal difference' idea has a number of applications. For example, it could be the driving force behind the question "If a subset (only) of taxa can be reserved, which taxa should they be?" The argument goes that the subset should be those that have in some sense maximal (genetic?) differences. As another example, phase-one of the ALOC algorithm uses a simple form of maximal difference in an attempt to ensure that the multivariate space is adequately sampled by 'seed' objects.

The algorithms in MAXD could also be used as robust alternatives of the 'minimal set' algorithm as proposed by Margules, Nicholls and Pressey (1988; see the MSET module). The 'robustness' in this instance may be justified by making the reasonable assumption that using a matrix of association measures has integrated species information into a more realistic estimate of site difference. Species are usually equally weighted in most pattern analysis. However, retrospective estimation of species 'weights' after classification or ordination (from the association matrix) will reveal that species contribute differentially; some strongly revealing underlying gradients while others either noisy or seemingly 'marching to a different drum'.

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OPTIONS

Option 1

The first option is due to me and is very simple! It first finds the two objects separated by the great dissimilarity value. It next tests all candidates looking for the object that has *maximal total distance* to **all** of the already selected objects. This object is then entered and the remaining candidates are tested in the same way against all previously selected objects.

Option 2

The second option is simpler again. It looks for the candidate that has the largest distance to its closest already-selected object. That is, it finds the candidate object with the maxim distance to the closest of the objects already-selected.

Option 3

This option is due to the work of Dan Faith (Australian Museum, Sydney) and relates to a cladistics environment. The first step is the same as option 1. The second and subsequent steps look for that object 'j', which has a MAXIMUM of its MINIMAL distance to all other PAIRS of objects. Using 'k' and 'l' as a typical pair, the distance is calculated as

$$D = (D_{jk} + D_{jl} - D_{kl}) / 2$$

This corresponds to a taxonomic path length (number of character differences) of the added branch. The concept here is to find the taxa with maximal character difference to those already selected.

AGGLOMERATIVE HIERARCHICAL FUSION: FUSE

OUTLINE

This program implements many of the **SAHN** or Sequential-Agglomerative-Hierarchical-Combinatorial strategies (Sneath and Sokal, 1973). These strategies represent The best known (and loved) clustering strategies in most disciplines. Methods such as UPGMA have been well trialed over many years on a wide variety of data. The SAHN methods are however a 'mixed bag'. While I would generally be happy to use (flexible) UPGMA, on data sets containing less than 500 objects, I would be more circumspect with most others (there are however times and places where just about all may be useful).

Combinatorial Formula

FUSE contains two sub-strategies. The first is an implementation of Lance and Williams Flexible sorting formula. This provides the 'distance' between the new group just formed, call it 'k', comprised, of the objects 'i' and 'j' with some other group, or individual 'l' as -

$$D_{kl} = \alpha_1 * D_{il} + \alpha_2 * D_{jl} + \beta * D_{ij} + \gamma * |D_{il} - D_{jl}|$$

where D_{il} = is the distance between groups i and l, and

α_1
 α_2 ,
 β and
 γ

are user definable parameters that determine the nature of the strategy. Lance and Williams suggested that all known hierarchical agglomerative methods could be directly coded using this formula.

Monotonicity

Lance and Williams guarantee **monotonicity** of any strategy given that -

$$\begin{aligned}\alpha_1 &= \alpha_2, \\ \gamma &= 0 \text{ and} \\ \beta &= (1 - \alpha) / 2\end{aligned}$$

For most of the strategies, each successive fusion takes place at a larger, or equal value of association. Some of the options below are non-monotonic, meaning that a fusion can take place at a smaller value of association than some previous value. This is termed as reversal. It is a basic problem but does make the interpretation of the results difficult. This feature is coded for the options below as either -

M - monotonic (well behaved)
R - subject to reversals.

Both centroid options (option 7 - UPGMC, and option 8 - WPGMC) can produce **reversals** in the fusion table. A reversal occurs when monotonicity fails. Usually, each fusion is at an equal or higher dissimilarity value. Reversals occur when later fusions are at a smaller value than previous ones. In this situation, DEND will not draft the dendrogram structure correctly.

Homogeneity

The second SAHN strategy operates to maximise the **homogeneity** of the fusion groups. The essential difference of the new method is that it considers within group dissimilarities (heterogeneity) as well as between group dissimilarities. Its formulation is similar to Lance and Williams -

$$\begin{aligned}D_{kl} &= C_{il} * D_{il} + C_{jl} * D_{jl} + C_{ij} * D_{ij} - C_i * D_i - C_j * D_j - C_l * D_l \\ \text{where} \\ C_{il} &= (N_i + N_l) * (N_i + N_l - 1) \text{ and} \\ C_i &= N_i * (N_i - 1) \\ D_i &= \text{Group } i \text{ heterogeneity}\end{aligned}$$

Flexible UPGMA

Option 5 (UPGMA: Sokal & Michener, 1958) is now coded as **flexible UPGMA** and is capable of a range of β values akin to flexible WPGMA. The difference is that WPGMA allows for attribute space distortion (contraction if β is positive, dilation when β is negative) based on the WPGMA series where equal weight is given to groups. Flexible UPGMA is based on the UPGMA series; **equal weight is given to objects**, not groups. Scientifically, the latter is more fundamental.

Group Labels

FUSE uses a shorthand way of naming the old and new groups in the fusion process. All groups are identified by the row with the smallest sequence number. For example, a group containing rows 3, 5, 7, 8, 11, 15, 23, 579, 987 and 1011 would be referred to in the output file (-.fus) as 'ROW 3'. While the groups are limited to containing a single object, the row labels will be meaningful. Once two or more objects form a group, the row label will still refer to the object with the lowest sequence number in that group.

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OPTIONS

Contraction, Dilation and β

For convenience, codes are used on the headings below to detail two aspects of the various updating options. The first determines how the method *perturbs* the *multidimensional space* during the fusion process. As with any multivariate technique, the objects to be classified can be thought of as representing points in a space defined by attributes as axes which are at right angles. When limited to three attributes or less, the analogy is obvious; objects can be plotted in the space, depending on the value of the 3 or fewer attributes. Generalising to more than three dimensions is, in a geometric sense, hard to conceptualise, but mathematically simple. For example, the distance between two points in five-dimensional space could be represented as:

$$D_{ij} = \sqrt{ (x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2 + (A_2 - A_1)^2 + (B_2 - B_1)^2 }$$

The problem is that most of the algorithms below **distort** the space in some way as fusion progresses. Such distortions can be classed as -

1. contracting,
2. conserving or
3. dilating

Contracting strategies contract the space with each successive fusion. The objects and groups become closer as the agglomeration continues. This usually leads to *chaining*, a situation where for example-

A joins B then
 C joins A-B then
 D joins A-B-C then
 E joins A-B-C-D ...and so on

Chaining is most often thought of as an artefact to be avoided. In the case of contracting strategies it is an inevitable side effect. The *groups* produced by contracting algorithms are of a very uneven size. When chaining is seen in conjunction with conserving algorithms, it is a realistic demonstration of data *structure*. Lance and Williams noted such effects, and attributed it to 'group size dependence'. In contracting strategies, the bigger the group in terms of the number of objects it contained, the more it 'gobbled up' other objects.

Dilation is opposite to contraction where the objects and groups move away from each other with each successive fusion. This is akin to what is supposedly happening with stars in our universe; everything is moving away from everything else with the fastest movement occurring furthest away. As would be expected, dilating algorithms do not chain, in fact they have the opposite problem, they tend to produce equal size groups, regardless of the data. The dendrograms will appear to produce an even number of equal size groups at all levels in the hierarchy. This usually is not convenient, unless you want equal size groups. From a group size dependency point of view, dilating strategies force increased dilation to groups with increased members; 'the bigger you are, the faster (and further) you go'.

Conserving algorithms as may have been anticipated, conserve the space. They neither contract or dilate and the distance between objects, as measured by the differences between the input dissimilarities and the output dissimilarities (ultrametrics) is implied by the dendrogram structure. Unless there are compelling reasons for using either dilation or contraction, conserving strategies such as UPGMA with a β value of zero are recommended.

Another reason for using dilation may be as an attempt to counteract the underestimation of the larger dissimilarity values (see SSH). The contracting Nearest Neighbour algorithm may be justified on the other hand, when curvilinear or linear groups of objects is suspected.

The coding used below to categorise the various SAHN algorithms is:

C - Contracting
 Cs - Conserving
 D - Dilating
 V - variable (depending on parameters)
 ? - not applicable or unknown.

Adjacency Constraints

FUSE also provides the option of constraining the fusion of any two objects or groups to those which have been designated as being in some spatial sense, *adjacent*. Adjacency is coded using the command TRNA. This routine reads in integer free format, a set of adjacent objects for each object in the data matrix. The form of the input file to TRNA is -

```
1 3 4 5 6
```

which implies object 1 has 3 adjacent objects; 4, 5 and 6. This coding is symmetric; once object '1' is coded as adjacent to '4', the implication is that object '4' is adjacent to '1'. The method of encoding adjacency is to take the negative of the value representing the adjacent pair. When the adjacency option is 'turned on' in FUSE, this coding acts as a flag, limiting all fusions between objects and groups to those that are represented by a negative association values. FUSE, using the combinatorial formulas above, maintains the spatial contiguity of all groups at all levels in the hierarchy, by the **updating formula** -

```
If  $D_{i1}$  or  $D_{j1}$  is  $< 0$  then  $D_{k1}$  is set  $< 0$ .
```

If the adjacency constraint is used, FUSE defaults to using the furthest neighbour option (2). The reason for this is to circumvent the greatly increased possibility of *reversals*; the situation where a fusion occurs at a lower value of association than a previous one. There is a possibility that other *dilating* fusion strategies (Ward's Incremental Sum of Squares or Flexible UPGMA with larger negative β s) may be useful in conjunction with adjacency, but to date, insufficient research has been done.

Stress is measured at each fusion by taking the difference between the actual value of association used for the fusion (adjacency) and the overall best fusion. The formula for each fusion is -

```
Stress =  $|D_{\text{adjacent}} - D_{\text{best}}| / (D_{\text{adjacent}} + D_{\text{best}})$ .
```

This is similar to the formula used for the determination of the Canberra Metric in ASO. The stress is constrained to produce stress values between '0.0' (no stress) and '1.0' (maximal stress).

FUSE provides, a user defined maximum stress that can be used to *relax* the stress associated with any fusion. It works in the following way. A stress value of '.5' will constrain the classification to producing contiguous groups at all fusions so long as the association used is less than 50% above (worse than) the overall best fusion value. A stress limit greater than '1.0' will force complete adjacency, while any value less than '1.0' may, if the stress exceeds this value, allow non-adjacent groups to fuse.

It is possible for adjacency to form a disjoint network. In this situation, FUSE will drop the adjacency constraint at the point where no adjacent fusions are possible and write a warning message.

Output of Associations

FUSE can output the remaining lower-symmetric matrix of inter-group associations at any point in the fusion process. This is a good way of obtaining inter-group associations because the process is totally 'compatible' with the chosen fusion strategy. For example, selecting UPGMA as the FUSE option, it would be less efficient and less 'compatible' to subsequently use GSTA to get a set of group means and then apply ASO to get inter-group associations. The use of group centres from GSTA would be analogous to the use of the centroid option in FUSE.

FUSE asks for a level to produce an inter-group association matrix. The name of this file is obtained by appending a 'G' to the existing root, or replacing the 8th character with the letter 'G' if the root is already 8 characters long. For example, if the data file is -

fred.dat,

the inter-group association file will be

fredg.dat.

In addition to the association file, a parameter file and a file containing ROW labels is created. Note: similar to DEND, the labels produced are those of the object with the lowest sequence number in their respective groups. The group numbers, as listed by GDEF are labelled 1->NRG (Number of Row groups parameter) and the labels used here and in DEND will correspond to the first object listed in each group. The reason for this is that it is easier to relate to the sort of group it is by an object, rather than an unrelated term such as 'GROUP-1'! Further processing of the inter-group associations is facilitated by the associated parameter and label file.

Replying with 0 will exclude this option while a value between 2 -> N-1 will result in a file that can be used as an *-aso* file.

Option 1 Nearest Neighbour (C-M)

This is the first of the *single linkage* methods, so named because the updating formulae referred to at the beginning of the section only require a single distance between the new group and all other groups. As the name suggests, it is the nearest distance that is used, with the minimum of the associations being D_{il} and D_{jl} . The parameter values for the flexible formula are -

α_1	=	0.5
α_2	=	0.5
β	=	0.0
γ	=	-0.5

While Jardine and Sibson (1971) suggest that this is the only justifiable fusion algorithm, most others would disagree. It is intensely contracting, but has the advantage of being one of the very few clustering/classification techniques that can detect non-spherical clusters. For example, the chaining effect links objects along narrow corridors. Once the contracting properties are recognised, the nearest neighbour strategy may be a useful technique.

If some of the association values are missing and they are coded as larger than the largest actual association, nearest neighbour will still present, given the above considerations, a reasonable classification.

Option 2 Furthest Neighbour (D-M)

The furthest neighbour algorithm is the second of the 'single link' methods. It is the opposite of the other single link technique, nearest neighbour. It uses the furthest distance when deciding on the association value to be used when computing inter-group 'distances'. That is, it will take the maximum of D_{il} and D_{jl} . The parameter values for the flexible formula are:

$$\alpha_1 = 0.5, \quad \alpha_2 = 0.5, \quad \beta = 0.0, \quad \gamma = 0.5$$

As assumed, the technique is strongly space dilating.

Option 3 (User) Flexible (V-V)

This option provides the user with the opportunity to choose any value for α_1 , α_2 , β and γ in the flexible formula. The values used for these parameters must be constants.

As noted in the codes, this formula is capable of dilation, contraction or conservation on the one hand, and monotonicity or non-monotonicity on the other, all depending on the values supplied.

Option 4 Flexible WPGMA (V-M)

This provides the general formulation of Lance and Williams flexible formula with the constraints -

$$\alpha_1 = \alpha_2 = (1 - \beta) / 2, \quad \gamma = 0$$

This is referred to as WPGMA by Sneath and Sokal. This stands for 'weighted pair group arithmetic averaging' implying that the groups are weighted equally. As noted above, *either* the objects or the groups can be weighted equally. This option takes the latter position. The user supplies a β value in the range '-1' (where the measure is intensely dilating), to '+1' (providing an intensely contracting strategy). A β value of zero is space conserving, only in the sense that it is conserved in relation to equally weighted groups.

The initial flexibility of control over dilation-contraction made this formulation attractive in the early days. It is not generally recommended however, because the algorithm varies the weighting of objects as fusion proceeds by weighting objects inversely with group size. The UPGMA equivalent ($\beta=0.0$) is suggested.

Option 5 Flexible UPGMA (V-M)

This is similar in theory to WPGMA, the difference being that objects are weighted equally throughout the fusion process. This means that groups are weighted proportionally to the number of objects they contain. This would be a fundamental hypothesis. The parameter values for the flexible formula are -

$$\alpha_1 = (1 - \beta) * \frac{N_i}{N_i + N_j}$$

$$\alpha_2 = (1 - \beta) * \frac{N_j}{N_i + N_j}$$

$$\gamma = 0.0$$

Space distortion is controlled by the β parameter. A value of '-1' results in strong dilation, '+1' in strong contraction while '0' is space conserving. A conservative choice may be β equal to zero, however, research has shown that a slightly negative β appears to be better able to recover known partitions (see Belbin, Faith & Milligan, 1992).

Option 6 WPGMC (C-R)

WPGMC is a term used by Sneath and Sokal to represent 'Weighted pair group centroid'. Lance and William refer to this method as 'median' because the new association value is based on the median of the candidate to the newly formed group.

This method is the centroid equivalent of WPGMA, where groups and not objects are weighted equally. This method is attributed to Gower (1967) and, because groups are equally weighted, it tends to counteract the problem of loss of identity of small groups when fusing with much larger groups. The values for the flexible parameters are -

$$\alpha_1 = 0.5$$

$$\alpha_2 = 0.5$$

$$\beta = -0.25$$

$$\gamma = 0.0$$

Both of the centroid fusion schemes are non-monotonic; they may, depending on the data, cause reversals in the hierarchy.

Option 7 UPGMC (C-R)

Sneath and Sokal use UPGMC to represent 'unweighted pair group centroid'. It is in terms of weighting, equivalent to UPGMA in equally weighting each object.

This weighting, being applied to centroids can produce 'loss of identity' of a small group when fused with a large group; the centroid being wholly within the point distribution of the larger group'. The WPGMC method has been designed to counteract this effect by weighting groups to be fused as having equal weight, regardless of size. Lance and Williams refer to this method as 'centroid'.

While the method is geometrically simple, being based on the centroid of a triangle, reversals can readily occur. Consider three objects or groups that are located at the apices of a triangle that is almost equilateral. Once the two closest objects/groups fuse, the distance of the new group (located at the centroid or mid-point of the old groups) to the remaining object/group is less than the distance in the first fusion.

The values for the flexible parameters are -

$$\begin{aligned}\alpha_1 &= \frac{N_i}{N_i + N_j} \\ \alpha_2 &= \frac{N_j}{N_i + N_j} \\ \beta &= -\alpha_1 * \alpha_2 \\ \gamma &= 0.0\end{aligned}$$

Option 8 Ward's Sum of Squares (D-M)

Ward (1963), based this method on fusing the two objects or groups that minimised the increase in the sum of squared 'distances' of each object in the new group to the centroid of the new group. While a number of authors (Belbin, 1984) have pointed out that the method should only be applied to the association measure 'squared Euclidean distance', many other metric and non-metrics have been used.

The technique has been extremely popular in geography. Although the method sounds reasonable, it is the most intense dilation option available in FUSE. The consequence of this is that it forces hyper-spherical clustering on any data set. The groups, as defined by any line cutting the hierarchy (dendrogram), will always contain about the same number of objects. This will be true regardless of the data.

If no 'natural' clusters exist, or if clusters are truly hyper-spherical and well separated, then this method will not result in what could be termed 'mis-classification' of objects. In the majority of circumstances however, mis-classifications must be expected. It is not recommended, except for the situations noted.

Option 9 Homogeneity (C-M)

While all other fusion strategies rely on the association between groups, either based on all of the objects in the group or some representation of them (centroid, average etc.), the homogeneity method uses in addition, within group associations. The method is combinatorial.

The homogeneity of a group is defined as the average of all pair-wise associations between the objects making up the group. This value is updated during fusion in the diagonal of the association matrix. The formulation is-

$$\begin{aligned}D_{kl} &= C_{i1} * D_{i1} + C_{j1} * D_{j1} + C_{ij} * D_{ij} - C_i * D_i - C_j * D_j - C_1 * D_1 \\ \text{where} \\ C_{i1} &= (N_i + N_1) * (N_i + N_1 - 1) \text{ and} \\ C_i &= N_i * (N_i - 1) \\ D_i &= \text{Diagonal value (Heterogeneity) of group } i.\end{aligned}$$

MONOTHETIC DIVISION:

MDIV

OUTLINE

MDIV is based on Bill William's ASYM algorithm (Williams & Bunt, 1981). MDIV is an acronym for Monothetic Division; a classification strategy that divides the data set into 2 sub-groups (a dichotomy) using only a **single** attribute at each division.

The original ASYM program was designed only for presence/absence data and dichotomised the parent group into two daughter groups, one for the presence and the other for the absence of the split attribute. I have generalised the algorithm in MDIV to accommodate *continuous data*. In addition, Bill Williams did not realise that his measure of group heterogeneity based on the sum of Jaccard measures could produce reversals. To circumvent this problem, MDIV uses a different approach for both calculating association and heterogeneity.

Division Vs Agglomeration

Most would agree that, in relation to clustering or classification algorithms, divisive algorithms, would in theory, be superior to agglomerative. The reason for this is twofold. Firstly, efficiency. It is usually only a small number of groups that are required. While agglomerative algorithms such as FUSE, join all objects to the point where they are in a single group, divisive algorithms don't. For example five splits by a divisive algorithm on of a data set of 500 objects is far superior than 499 fusions of an agglomerative algorithm when about six groups are required.

The second advantage is that, with fewer calculations required, divisive algorithms have less opportunity for error. In the above example, 5 iterations are required against 499 for the agglomerative counterpart. Computers are 'high-speed morons', don't give them a chance to foul it up by compounding small errors; always minimise the calculations required. This is great in theory.

Number of groups

One drawback of divisive methods is that the resulting groups cannot be automatically sub-divided without re-running the program. Fusion to a smaller number of groups may be done using GDEF. It is therefore prudent to produce **more** groups (=more splits) than you believe are required for the final configuration.

Binary Keys

Dichotomising on a single attribute produces clusters that are very simply defined. The cost of such simplicity is usually overall cluster optimality. For example, early splits are likely to separate objects that may have considerable overall similarity. Being able to produce simple *binary keys* that can classify any objects on a small sub-set of attributes is however attractive. For example, a field worker, noting the presence and absence of key species at a new site, can quickly classify the site into one of the pre-determined clusters. With *polytheistic* methods (those using all attributes), this ceases to be a simple procedure (but can still be achieved).

Method

The MDIV algorithm uses two association measures, Manhattan Distance as the measure of heterogeneity and the Bray and Curtis (Czekanowski) measure for the determination of the attribute used for splitting a group.

For each group, the sum of all pair-wise Manhattan associations between each attribute with all other attributes is calculated. For the same set of pair-wise comparisons, the sum of all Bray and Curtis measure is accumulated for each attribute. The attribute with the lowest average Bray and Curtis association is taken to be near the centre of multidimensional distribution and is used for splitting objects from their parent group. For presence/absence data, attributes that are either present in all rows or absent in all rows are called 'non-effective' and cannot contribute to the analysis; they contain virtually no 'information'.

One new group is produced that contains all rows that have a presence (1) on the split attribute and the other daughter is comprised of all rows with an absence (0) on the split attribute. For continuous data, non-effective attributes are those on which the group has already been split. With this type of data, a split threshold is determined by a best sum-of-squares criterion.

NOTE: MDIV determines what type of attributes are present (P/A or continuous) by scanning the complete data set looking for any values not equal to zero or one. If any such values exist, the data set is treated as being made up of continuous attributes.

If *missing values* are detected while scanning the data, attribute means are substituted. This needs to be done because MDIV uses the sum rather than the average of association values. Such substitutions are noted at the beginning of the run.

Homogeneity

A measure of homogeneity or heterogeneity is required to determine the next group to split (and as a scale for plotting the divisions). The original ASYM algorithm in MDIV uses a function of the sum of the (effective) association values between attributes for this purpose. For example, if all the rows in a group were identical, the attribute associations would sum to zero. Increasing heterogeneity would be reflected in increasing dissimilarity between attributes. In Williams ASYM paper, the Jaccard association values ranged from 0 (no similarity) to 1 (identical), the heterogeneity is measured as -

$\frac{\text{Number of Effective Associations}}{\text{Sum of Effective Associations}}$
--

The value on the left cannot be exceeded by the value on the right. A simpler derivation of this measure may be obtained by simply summing the dissimilarity values (ie sum of dissimilarity = number of effective attributes - sum of similarities). Williams believed this measurement of heterogeneity would monotonically decrease with successive divisions. This does not appear to be the case. Reversals in heterogeneity do occur with the Jaccard. For example, take the data set

1111
1111
1111
1000
0100
0010
0001

The heterogeneity using the Jaccard measure is 1.95 (Bray & Curtis=1.786). If attribute 1 is used to split the bottom 3 objects from the top three, the heterogeneity of the bottom three objects is 3 greater than the parent! Examination of this problem lead to the use of Manhattan distance to measure group heterogeneity as that is an alternative measure of heterogeneity that appears to be monotonic.

Unlike PDIV (Polythetic Division), MDIV stores only the data in memory, not the pair-wise association matrix. MDIV can therefore classify larger data sets than PDIV.

PATN produces a standard table for all hierarchical classification methods, both agglomerative and divisive. This is for compatibility purposes with DEND and other subsequent commands that read this table. The standard table is in an agglomerative form. To trace the division history, start at the bottom of the table and read up.

Split Attribute

Unlike Williams ASYM algorithm, a different measure of association is used for calculation of heterogeneity and the split attribute. Ideally, the Bray and Curtis/Czekanowski should be the foundation of both (see Faith, Minchin & Belbin, 1987). As with the Canberra Metric/Jaccard in ASYM, the Bray & Curtis/Czekanowski can however produce reversals in heterogeneity. This is the situation where a daughter group has greater heterogeneity than the parent (see paragraph above). As may be expected, the use of the minimum of the sum of Manhattan distances for each attribute did not work well. The Bray & Curtis/Czekanowski was substituted to good effect; results from a number of tests produced what appeared to be very reasonable results. The minimum sum of associations points to an attribute that can be interpreted as being the attribute which can best predict the remaining attributes.

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OPTIONS

The only option in MDIV is the number of groups to be produced (one more than the number of splits).

POLYTHETIC DIVISION:

PDIV

OUTLINE

My idea for an effective divisive clustering algorithm was based on a method by MacNaughton-Smith for *polythetic-divisive clustering*. I have tried to create PDIV as a divisive equivalent to the FUSE strategy UPGMA (Unweighted Paired Group arithMetic Average).

Any dissimilarity-type matrix is accepted as input. PDIV sums the dissimilarities for each object and group of objects. At each iteration, the group with the greatest average dissimilarity will be divided. The object in this group with the greatest average dissimilarity to all other members is transferred to a new daughter group. All connections with its old group are severed. The next step examines the average dissimilarities of each row in the old group to both the old group and the newly formed daughter group, ie the value -

$$D_{old} - D_{new}$$

is calculated for each object. The object with the greatest positive difference is transferred to the daughter group. This process is repeated until no objects in the old group have an average dissimilarity closer to the daughter group than their present group (no positive differences). At this stage, the definition of the daughter group has stabilised, and it becomes another candidate group for subsequent splitting.

The PDIV algorithm is appealing from a conceptual point of view. Generally, the divisive type techniques are more *efficient* than their agglomerative counterparts, because the number of divisions is always less than the number of fusions. In addition, the Divisive methods attain greatest *accuracy* at the level of a few groups (limited iterations). The agglomerative methods on the other hand usually have to perform a large number of iterations until the level of aggregation matches the desired level of summary.

One drawback of divisive methods is that the resulting groups cannot be automatically sub-divided without re-running the program. Fusion to a smaller number of groups may be done using GDEF. It is therefore prudent to produce **more** groups (=more splits) than you believe are required for the final configuration.

The output from PDIV emulates the FUSE program so that DEND can be run to produce a dendrogram but the order of printing the "fusion" table is the reverse of that used to generate the classification, ie from the bottom of the table to the top. To get a dendrogram you will have to "fool" DEND into accepting that there are really -

Number of Splits + 1

rows. Use PRAM to alter the number of rows. Unlike FUSE, PDIV should not be used to split data down to groups containing only a single object.

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OPTIONS

The only option is the number of final groups (one more than the number of splits)

NON-HIERARCHICAL

CLASSIFICATION:

ALOC

OUTLINE

ALOC stands for *allocation*. It implements what I think is a cute and simple non-hierarchical clustering strategy. The algorithm presented is theoretically able to accommodate very large data sets, orders of magnitude larger than the usual hierarchical schemes. It is largely due to the need to calculate only object-group associations and not all object-object associations.

Results (Belbin, 1987, Belbin & McDonald, 1993) suggest that while the algorithm is simple, it provides results at least comparable with the best hierarchical strategies implemented in FUSE (flexible UPGMA $\beta = -0.1$). **ALOB** is a version of ALOC that has been specifically pruned to handle very large data sets. Details of the differences between the two versions are outlined below.

ALOC is based on the allocation of objects to a set of *seeds*. These seed objects are so named because they form the initial configuration from which the classification grows. Seeds represent a set of pre-defined groups but may be -

1. stereotypic or representative objects,
2. objects selected at random,
3. centroids from a prior classification or
4. a group definition for the N objects.

In (4), the group definitions or a group affinity vector are read and a set of group centroids are determined (phase 3 below). In this situation, iteration continues from phase 4, bi-passing phases 1-2.

Phases

1 Seed Generation

ALOC has two very different methods for the generation of seeds. The normal approach is to simply use the first object in the data file to create a starting configuration that ALOC can iterate on. The alternative is to get ALOC to read either a standard PATN group-definition file (-.gdf) or group-affinity-vector (-.gav) and create a set of seeds (centroids) from this. If prior groups are used, say seven, then seven seeds, defined by the centroids of the objects forming the groups, will automatically be created.

The relationship between a set of groups and a set of seeds that may be generated via these groups may not be 'stable'. For example, groups generated from seeds derived from a group definition from ASO, FUSE and GDEF may change when run through ALOC. ALOC is suggesting that its definition of 'optimal-groups' is different to FUSE. This makes sense. If a 'pure' comparison of a set of objects with some prior groups is required without ALOC 'getting in the way', 'zero iterations' and 'skip phases 2 and 3' should be selected. In this way, the basic tabulation of objects to a fixed suite of group centroids can be obtained.

Allocation radius

The number of groups produced by ALOC (and ALOB) is controlled by the use of an allocation radius. This is an association value (between zero and one) that is used as a threshold for the generation of additional seeds. The process works as follows: each object in the data set is compared to each seed. If the resulting association is **greater than** the allocation radius, the object itself becomes a seed and the number of groups increases by one. The process continues until all objects have been compared with seeds. If a large threshold is used (say 1.0), ALOC will not generate any additional seeds. If a small threshold is used, many seeds (groups) will be generated. Think of the process as 'seeding' points within the multivariate space that have at least a given degree of separation.

ALOC reports the identity of any additional seeds that are created to the terminal and the *-.als* file. It then requests any alteration of the threshold value. If a new value is provided, phase 1 is re-initiated. In this way, any number of groups may be quickly established before going to the following phases.

This general approach provides a cute solution to the problem of how to ordinate a large number of objects. The seed objects generated in this way form an efficient summary of the gradients. A data set containing just these seed objects can easily be generated by MASK, using an ALOC *-.als* file which contains the sequence numbers of the seeds.

NOTE: The difference between the use of prior groups versus data to define seeds is that in the first case, the seeds are *artificial*, in the second case, they are actual objects.

Selecting the number of groups

The allocation radius controls the number of groups. ALOC & ALOB permit the selection of a fixed number of groups by selecting an allocation radius > 1 . If the value is > 1 , then it is assumed to be the number of groups required. ALOC/ALOB use a 'branch and bound' approach to modify the allocation radius so as to produce the selected number of groups. With some data sets, this will not always be possible! There will be occasions where the structure of the data will limit the options on the number of groups that can be produced.

The algorithm has a maximum on the number of attempts it will try. In these circumstances the actual number of groups will differ from the target by a few groups either way. The best idea in this situation is to manually select an allocation radius that will produce more groups than required. Once produced, these groups can then be fused via the *-g.** data set and the new group definitions created with GDEF option 5.

2 Fixed Allocation

The *second* phase begins with object 1 and sequentially *allocates* each object to the nearest seed as generated in phase 1. The seeds remain *fixed* in multidimensional space for this step which provides a degree of order independence from the data. **NOTE:** this step is skipped for group definition or group affinity vector input to ALOC.

At the end of phase 2, all objects have been allocated to one (and only one) group.

3 Centroid Re-Definition

The *third* phase dismisses the seeds and calculates the *group centroids* based on the objects that were assigned to each group in phase 2. From this point on, the *stereotype* position of each group is defined by its centroid; the arithmetic average across all attributes.

Use of a pre-defined group definition or group affinity vector, either from PATN or some other classification enters at phase 3.

4 Iteration

The *fourth* phase is *iteration and re-location*. This phase begins only if the number of iterations requested is greater than 0. Each iteration sequentially extracts the object from the group to which it currently belongs and allocates it to the group with the nearest centroid. Groups may disintegrate down to a single object, by the process of re-allocating them to other groups. Groups can never be deleted, they will always retain a single object (an object cannot be further from itself than other objects). While convergence is assured only in the case of metric associations, ALOC always appears to converge.

Seeds

PATN provides a variety of options. A random sample could be generated by SAMP. It could also be used to generate a smaller sample data set for analysis by ASO and FUSE, GDEF, GSTA or PCA. MASK could extract row representative objects. NNB may be used to detect objects in dense areas to act as seeds. The attribute space may also be partitioned up into a limited number of equally sized cells. A study of a number of different methods (Belbin, 1987) has suggested the contrary of what might have been expected, that ALOC is reasonably insensitive to seeds. This being the case, the best method of seeding ALOC is to use the first object in the data set. This object can be read from the data file; it does not need to be copied to a special seed file.

A special case occurs where you may want ALOC to either continue to 'optimise' a previous classification or perform a comparison between objects and a set of previously defined groups. In either case, a group-definition or group-affinity-vector may be read and used, in conjunction with the default data file, to generate a set of centroids. These centroids are the seeds. The re-grouping or simply the distance of the objects to seeds (with or without iteration) may then be performed. If zero iterations are requested, ALOC will request if phases two and three (see above) are to be performed. *NOTE: if phases 2 and 3 ARE run, the definition of the seeds may be altered.*

Final Groups

On completion of iteration, ALOC calculates the association between each of the final group centroids using the nominated association measure and outputs this file as *-g.aso*. The judicious use of seeds and allocation radius should be used to generate a larger number of groups than desired for the final summary of the data. If ALOC is used to provide just enough groups, then it is impossible to examine sub-group structure.

If additional groups are provided, they can be subsequently FUSEd using the additional files created by ALOC-

- g.dat* the group centroids data file (PATN format)
- g.aso* the inter-group association matrix
- g.prm* the parameter file
- g.rlb* row labels for *-g.dat*

All that needs to be done to classify the groups is to use PRAM option 10 to input the *-G.PRM* file and then use FUSE, DEND and possibly GDEF. Remember that the GDEF will produce groups of groups (not objects). Option 5 of GDEF may then be used to merge objects in the ALOC group definition file from the group definition file. To provide a final group definition file of objects using the groups of groups file (*-G.gdf*), GDEF option 5 is used while PRAM is operating on the *-g.dat* file, not the *-dat* file.

NOTE: the final group definition file (*-gdf*) may not be the same as one that may have (optionally) be used to initiate ALOC's seed configuration. Differences may occur depending on the relationship between the data and the prior classification, the option on fixed allocation and seed re-calculation, and number of iterations. For example, if a classification from FUSE is used to initiate ALOC and all phases in ALOC are requested, it is more than likely that the output group definition will NOT match the input one (don't forget to give it a different name. Don't overwrite the original!).

As a product of ALOC, the file *-.alo* will contain the iteration history and allocation radii, the stress and optionally, at the end of the file, the 'fuzzy' affinity list. This is a list of the final associations in terms of the mixed attribute association of each object to each centroid. In an ideal situation, each row should be close (zero distance) to only one cluster and far from all others.

ALOB

ALOB is similar to **ALOC** except that it has been designed to handle very large data sets (greater than 5,000). Over 100,000 objects can be accommodated by ALOB. All you need is a reasonably fast computer (1 mip plus), virtual or a lot of 'real' memory and/or infinite patience. The differences to ALOC are that *ROW labels* are not used in ALOB, only sequence numbers, and that ALOB uses a GASO-type approach to the estimation of object-centroid association (see 'Attribute groups' below).

All *input* parameters and files are identical to ALOC, but the output from ALOB is reduced. The *-.alo* file contains only the generation of additional seeds at phase 1 and the summary of the number of re-allocations and heterogeneity at the end of each iteration.

For the sake of brevity, the composition of the groups is not stored in the traditional *group definition* format but in *group-affinity* format in the file *-.gav*. For example, if 10,000 objects were being classified, the *-.gav* file would contain 10,000 corresponding integers. The first value corresponds to the group number for the first object, the second number to the group number of the second object, and so on. See the section on GDEF for a more comprehensive definition of this format.

With the potential for very long execution times using ALOB, the group affinity vector is written to a *-.gav* file firstly, at the fixed allocation phase, and subsequently, at the end of each iteration. If the system crashes and any time after the fixed allocation, the *-.gav* file is then available to restart ALOB given the configuration of object and groups at the last allocation before failure of the system. If the system crashes, simply use the option to read a *-.gav* file at the beginning of ALOB and let it continue, re-allocating until the stopping rules are activated (or it crashes again!).

The inter-group association matrix and the group centroids are filed as in ALOC.

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OPTIONS

Association Measures

There are two different approaches that ALOB/ALOC have taken in relation to association measures. The latest version of ALOC provides a simple choice between three different measures of association-

CONTINUOUS	PRESENCE/ABSENCE
1 = Bray and Curtis	Czekanowski
2 = Kulczynski	Kulczynski
3 = Gower Metric	Simple Matching coefficient

Prior to this, it was possible to select an appropriate measure of association for each attribute independently. I felt a little uneasy about the implications of this approach for a range of situations. For example, it was possible to mix Euclidean distance (unstandardized) with say the Gower Metric (standardised). Even mixtures of Czekanowski and Gower Metric may produce problems due to lack of standardisation. For example, while the latter two measures range from zero to one, there is nothing to suggest that .5 on both measures would be considered to exhibit an equal 'dissimilarity'. In other words, if the range of *actual* (as against *potential*) values of each measure could be determined, some standardisation of weighting may be established. In ALOC/ALOB, this is virtually impossible. I would suggest the use of the Bray and Curtis for *ratio* presence/absence data, Kulczynski for other *ratio* data and Gower Metric for *interval* data.

If you really want to maintain compatibility with the old version of ALOC/ALOB or really need the ability of handling mixed association measures and weighting, then please ring or write to me and I will send the older version to you. I have retained the following details from the old version just in case. In this version the measures were-

CONTINUOUS	PRESENCE/ABSENCE
1 = Bray and Curtis	Czekanowski
2 = Canberra Metric	Jaccard
3 = Minkowski Series	Simple Matching Coefficient
4 = Gower Metric	Simple Matching coefficient
5 = Circular Gower	

All measure, except the Circular Gower are defined in ASO. The 'circular Gower' measure has been coded as a modification to the Gower Metric to enable a meaningful association between compass bearings. It is formulated as -

```

Dif = Dik - Djk
if Dif: 180.0 then Dif = 360 - Dif

Dij =  $\frac{D_{if}}{180.0}$ 

```

Attribute Grouping (***** ALOB only *****)

ALOB assumes the presence of a *-.gas* file detailing which group each attribute belongs to. For M attributes there will be M integers in a file. If all number are the same, it is equivalent to ALOC. The integers in this file must begin with '1' and **increment** to the number of attribute groups you desire. For example if there are 3 groups, the only group numbers permitted are '1', '2' and '3'. These are to be found in *-.gas* in ascending sequence.

Each attribute must be assigned to a group. The simplest case occurs when each attribute is treated equally. In this case all attributes can be given a group number of '1'. Group numbers must be assigned sequentially. If there are NRG groups there must be groups defined from 1,2,3,...NRG-1,NRG. An additional requirement is that group 3 cannot appear before groups 1 and 2 in the coding. For example -

```

1 1 2 2 2 3 4 4 4 ..... is legal
1 1 3 2 3 3 4 4 4 ..... is illegal: group 3 is defined before 2!

```

Maximum number of iterations

ALOC, by its nature, is an iterative procedure. It takes an initial set of defined groups which may be in the simplest case, one group containing the first object, and iterates around a simple procedure to improve the homogeneity of the groups. If ALOC is being used in its traditional sense to create a classification from scratch, the number of iterations should be greater than 5. The larger the data set, the more iterations that are usually required before some type of optimality is achieved.

One iteration involves extracting each object from its present group, determining the closest group and allocating it into it. An object may be returned into its original group (not a re-location: see section below) or it may be relocated into a different group. Normally, the number of relocations is used as the stopping rule for ALOC. This means that the number of iterations should usually be set large enough not to be activated as the stopping rule.

In some circumstances, the iterative phase is not desired, for example, to determine the association between a set of objects and some prior classification. In this case, setting this parameter to zero may be appropriate. If this occurs, ALOC then skips asking about the 'minimum number of relocations' and requests if phases two (fixed allocation) and three (re-calculation of seeds based on the fixed allocation) should also be skipped (see below).

Skip phases two & three

This option is presented only when the number of iterations (see above) is set to zero. This situation may occur when it is decided to compare a set of objects with either a set of pre-defined seeds (possibly from some type of classification) or the seeds implied from either a group-definition (-.gdf) or a group-affinity-vector (-.gav). Phase two and three should be skipped when no alteration to the defined seeds is required before the tabulation of object-group associations (see below).

An example of the use of zero iterations and skip phases two and three options is where distances of a set of objects to a fixed (or implied fixed from a group definition) seeds is required. In other words "How well do these set of objects fit these groups"? In this case, the only output of interest is the tabulation of object-centroid distances ('fuzzy-affinity table').

Minimum number of relocations

This is the second of the *stopping-rules* controlling the iterative phase of ALOC. If the 'maximum number of iterations' was set to zero, this option is skipped.

A *relocation* is defined when, in the iterative phase, an object is moved out of the group that it was in, and moved to some other group. Ideally, it is better to see that after extraction from its assigned group, all objects would *gravitate* back into the same group. Oscillating situations can however occur where objects are endlessly switched back and forth between two or more groups. In this case, the minimum number of relocations will never reach zero.

It is this parameter that should be activated if the iterative phase is required. In other words, sufficient iterations should be performed so that it is this parameter that stops the iterative phase. Use the default of zero normally for this parameter.

Object-centroid tabulations

Three options are provided that control the nature and extent of the listing in `-.alo` of object-centroid associations. The FIRST option skips any tabulation.

Option TWO can be used to tabulate the distance from each object to the group that it has been assigned. This assignment has come about as a consequence of all the prior phases of ALOC (seed generation, fixed allocation, seed-re-calculation and iteration) or a subset of these. For example, to tabulate the distance of all objects to a prior classification using `-.gdf`, `-.gav` or a set of seeds, phases one-four may all be skipped.

Option THREE is the default and is useful for analysing the stability and 'fuzziness' of the derived groups. The association between each object and its closest group centroids is calculated. These are tabulated by increasing distance (diminishing level of association).

Affinity Cut-off

If the 'fuzzy-affinities' are requested by option two or three above, ALOC will request a further parameter: the cut-off value used for limiting the number of closest groups listed for each object. This list is ordered by increasing dissimilarity. The closest group the each object is listed first, the second closest group is listed second and so on. The cut-off value can be used to *limit* the list either by-

1. a specific number of closest groups by using a value of ≥ 1 OR
2. listing only those groups $<$ the supplied dissimilarity value when the value is < 1 (a variable number of groups for each object).

For example a value of '0.8' will list for each object, only those groups less than 0.8 distant (in terms of the selected dissimilarity value selected).

TABALO

This is a stand-alone utility that provides a number of summary measure from the 'fuzzy-affinities-table' optionally produced from ALOC. I don't have much time for 'fuzzy'-clustering algorithms; I believe the use of ALOC is more efficient and effective. After all, groups produced by any algorithm from a wide range of 'natural-data' are somewhat 'artificial'. In other words the group 'boundaries' are likely to be 'rubbery'. This fact can be determined from this table (as well as a dendrogram or ordination). Just because ALOC assigns an object to a single group it doesn't mean that it isn't close to one or other groups. The affinity table potentially (depending on your cut-off selection) provides a comprehensive list of the affinities (hence the name) of all objects to all groups. Scanning the affinity table and using TABALO and some common sense gives you information that is more than comparable with any fuzzy algorithm.

TABALO reads the .alo file looking for the table and produces a file looking like-

OVERALL STATISTICS:				

Mean number of centroids in affinity table.....				
				3.0000
Mean of all object-centroid distances in table..				
				0.3423
Mean of all objects-closest centroid distances..				
				0.1900
Minimum object-closest centroid distance.....				
				0.0000
Maximum object-closest centroid distance.....				
				0.3000
STATISTICS FOR EACH GROUP				

GROUP	N=	MEAN	MAX	DENSITY
NUMBER		Ob-Cen	Ob-Cen	(#/2*MAX)
-----	-----	-----	-----	-----
1	1	0.000	0.00	infinite!
2	8	0.238	0.30	13.333
3	1	0.000	0.00	infinite!

The first part of the listing contains measures that relate to all groups. The first TWO values in table (mean number and mean association) are dependent on the option used to limit the number of centroids listed in the ALOC affinities table. For a complete set of statistics, a fixed limit equal to the number of groups should be used rather than a threshold on dissimilarity level. The following THREE options are dependent only on the closest centroid and should be insensitive to the selection option.

The second part of the table provides statistics by groups. Note that if the group contains only a single member, the radius is zero and therefore the density is infinite. Again, if a variable number of centroids are listed in the ALOC table, the listed maxima may not be strictly correct. Simply re-run ALOC with the fixed number of centroids equal to the number of groups.

TWO-WAY INDICATOR ANALYSIS: TWINSpan

OUTLINE

TWINSpan is a polythetic division clustering algorithm based on using the first ordination axis from reciprocal averaging (correspondence analysis). As it is one of the Cornel programs, it is not fully documented here. For further details, please read the TWINSpan manual.

MULTIDIMENSIONAL

SCALING:

SSH

OUTLINE

SSH stands for Semi- Strong- Hybrid multidimensional scaling (Belbin, 1992), an ordination program that permits combinations of ordinal (monotone), interval or ratio scaling. See the Users Guide for a definition of these terms. SSH implements an alternate and improved version of hybrid scaling as defined by Faith, Minchin and Belbin (1987). The original Hybrid algorithm was based on the KYST program developed at Bell Laboratories by Kruskal *et al.*

Research has suggested that SSH is superior to principal components/coordinates, correspondence analysis/reciprocal averaging and other multidimensional scaling programs such as ALSCAL84. SSH fits output distances to input distances rather than squared input and output distances. The latter could be considered to overly weight the larger input distances and these appear to be the least reliable. In addition, multidimensional scaling as SSH and KYST implement it, permits a great deal more flexibility in data handling than other ordination methods. SSH has a number of significant differences from the KYST algorithm of Kruskal -

1. SSH uses the Guttman & Lingoes **strong** monotone regression rather than KYST's weak monotone regression.
2. SSH uses a single symmetric matrix without a diagonal while KYST permits multiple input matrices of varying shape (asymmetric, symmetric or rectangular).

SSH is optimised toward hybrid-type scaling while KYST permits greater flexibility with greater computational costs. If you need the greater flexibility of KYST, copies may be obtained (US\$150) as "Second MDS Program Package" (nominate ASCII tape format) from-

Computer Information Services Group,
Room 2F128
AT and T Bell Laboratories,
600 Mountain Ave,
Murray Hill,
New Jersey, USA 07974
TELEX: 138650, answer back: MUHL

and for a copy of technical report on KYST, write to Joseph B Kruskal, Room 2C281

Algorithm

It is beyond the scope of the this Manual to provide all of the details of the mathematical basis relevant to the SSH algorithm. An outline should however provide sufficient information for confident use. For anyone serious about understanding how multidimensional scaling programs work, Coxon (1982) and the papers by Kruskal and Wish (1978) and Shepard (1962) are strongly recommended.

Basically, SSH *assumes* an initial configuration of objects in the user selected number of dimensions. It then produces the inter-object association matrix as measured from this initial configuration between all pairs of objects (in the user selected dimensions). The next step is to try and predict the *input* association measures (dependent variable) from the configuration (*output*) associations by regression. One of the features is that the regression can be *interval* -

```
Y=a.x+b,
or ratio-
Y = a.X
```

or *ordinal* (monotone) or any combination of interval/ratio with ordinal. In relation to the association measures, the ordinal regression simply implies that if one association value is bigger than another in the input association matrix, then the same relationship must hold true in the output associations. These are derived from the distances between the objects in the user-defined dimensions.

The Guttman *strong* monotone approach (1968) says that if two values are *different* on input, they should be different on output. Kruskal's weak monotone regression (1964) on the other hand, implies that input differences are permitted to be tied on output, ie -

```
Strong: If  $A_i > B_i$  then  $A_o > B_o$  (Guttman)
Weak : If  $A_i > B_i$  then  $A_o \geq B_o$  (Kruskal)
```

For those who are familiar with Shepard diagrams (see Coxon, p.54), weak ordinal (monotone) regression permits the steps to have *vertical* components while strong does not. Using the terminology of Coxon, SSH takes in fact a *semi-strong* approach to the ordinal fit; it will break tied input values but will not equalise different input values. This is exactly what is required by examination of the input association values.

With association measures such as Bray and Curtis, we can expect a lot of *tied values* of 1.0 when objects have nothing in common. We want to be able to break these ties to permit some of them to get larger but maintain differences if there are any. The semi-strong monotonicity criterion does just that.

Stress

Once the regression is determined, *estimates* of the *input* association values from the configuration distances (usually called $\hat{d}$) can be calculated. This leads to an estimate of **stress**, the miss-match between the 'distances' between all points in the 'current' configuration and the best estimates of the same values. Stress in SSH is formulated as -

$$\text{Stress} = \sqrt{\frac{\sum ((D_{\text{out}} - D_{\text{est}})^2)}{\sum (D_{\text{out}})}}$$

where D_{out} is the distance from the current configuration and is equivalent to an **ultrametric distance** from DEND and D_{est} is the estimate from the regression(s). As can be inferred from the formula, the larger the disparity, the larger the stress. The estimated maximum value is close to 1 while the minimum value is 0.

Given a set of associations, stress can be thought of as representing a three-dimensional undulating *surface* where the vertical axis represents the value of stress while the x and y axes infer a configuration of the objects. SSH aims to find a minimum value of stress by usually starting at a random point on the surface, finding the direction of greatest slope, and proceeding downhill (down the fall-line).

Optima

A good analogy of the major limitation in multidimensional scaling programs such as SSH follows from this illustration. Consider the stress surface to be topography. A paratrooper is blindfolded and dropped somewhere on the surface and given the order "find the lowest point in the area"! If he doesn't break a leg on landing, the only solution is to walk (carefully) down slope from the landing point. A point will inevitably be reached when movement in *any* direction will be up-slope. This valley will be A minimum elevation, but may not be **THE** minimum elevation. The former is referred to as a *local* minimum and the latter as a *global* minimum. This feature is the reason for use of the multiple random starts built into SSH.

The minimisation of stress is achieved, as noted, by determining the direction of steepest descent, and moving the points (objects) in the configuration space accordingly. The SSH recipe is overall -

1. accept or create an initial configuration (user defined dimension)
2. calculate the distances between points
3. form the regression(s)
4. calculate the estimates of distance
5. calculate stress from disparities
6. if stress is satisfactory, -----> output configuration & STOP else
7. calculate the slope and move points
8. go to step 2.

The concept of multidimensional scaling is elegantly simple. The gremlins are the *local* minima. Because the stress 'surface' is not known, except in a very local area, one usually can not guarantee that the current minima is the global or absolute minimum.

Iteration: the calculating of the regressions and stress, then moving the points is simple in concept but mathematically demanding. The key issue is the *step size*; how FAR should the points move in one iteration? This SSH algorithm uses the same formulation as KYST. Basically, the step size starts large and diminishes as the minimum stress is approached.

Rotation of Axes

SSH, unlike KYST and other multidimensional scaling algorithms, **does not rotate the final axes to principal components**. If this is desired, PCA may be used on the SSH results. Alternatively, PCR may be used to rotate the SSH axes to *simple structure*. This is a term applied in psychometric applications to optimising interpretability of ordination results by orthogonal rotation to maximise variance or some other criteria (see PCR). The PCR rotation algorithms rotate axes in **pairs** while the PCA approach optimises one axis at a time.

Large Data sets

Ordination algorithms are generally very memory and time intensive algorithms. Attempting an ordination of hundreds of objects is an especially difficult proposition on a micro. The number of attributes is irrelevant in SSH; it doesn't use them directly, they are inherent in the association measure.

When analysing large sets of data, all objects do not need to be ordinated. If ordination is to be used to indicate gradients or trends, a number of alternatives exist. An effective alternative is to use one of the options in MAXD to sample the k-most different objects from a data set (50 should suffice in most circumstances to define the trends). Alternatively, ALOC phase-1, or a random sample may be used to achieve the same ends.

Warning

If there are *identical* objects in the data matrix and *ordinal* regression is selected, there is no guarantee that identical coordinates on the ordination vectors will result. There is nothing to be gained from duplicates; use SAMP to eliminate them.

REFERENCES

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- Shepard R N (1962) The analysis of proximities: multidimensional scaling with an unknown distance function. *Psychometrika* 27(2), 125-140. (important paper)
- Shifman S S, Reynolds M L & Young F W (1981) Introduction to multidimensional scaling. Theory methods and applications. Academic Press, New York, 413p. (lots of examples)
- Spencer R (1986) Similarity mapping. *Byte*, August, pp 85-92. (very simple introduction to multidimensional scaling)

OPTIONS

Number of Dimensions

Unlike principal components or other principal axis techniques such as correspondence analysis, multidimensional scaling programs require the selection of the dimensionality. Selecting for example, two dimensions, will mean that each objects will be positioned in a two-dimensional space that will, as close as SSH can achieve, fit the input associations. In selecting the number of dimensions, SSH effectively tries to squeeze m attributes into k dimensions.

It stands to reason that, the more dimensions you select, the easier it will be for SSH to position the objects and the lower the stress. Obviously, there is a trade-off. Selecting too few or too many dimensions will often mean that an unsatisfactory configuration will result. In the first case there are too many constraints for effective placement of objects in the space. In the latter case objects are unconstrained (too many degrees of freedom) and add superfluous noise. Kruskal's (1962, 1964) rule of thumb is still a reasonable guide-

Stress	Goodness of fit
-----	-----
> .2	poor
.15 > .2	be cautious
.1 > .15	fair (wish it were better)
.05 > .1	satisfactory
<.05	impressive
0	perfect !

The determination of dimensionality for MDS algorithms such as SSH has more significance than other ordination methods such as PCA, PCoA, RA, CA or DCoA. The gain however, is that MDS is generally designed to pack more information about the objects into the selected dimensionality than are the above methods.

Output of Euclidean Distances

SSH optionally writes all association values used in the analysis. The file `-.shp` (for Shepard) contains the values for the four association matrices:

1. The *input* association matrix
2. The *output* associations in the selected k-dimensions
3. The *estimates* of the input distances from the regressions,
4. The *order* of the association values in the three matrices.

All matrices are ordered on the basis of the ascending association values in the input matrix. Each is output as a single vector of $N*(N-1)/2$ values in FORTRAN (16F8.4) format. For example, for 10 objects, the `-.shp` file will contain 4 rows and 45 columns. This file has many applications. It can be used for the Shepard plot (input verses output associations) or it can be used to determine which objects have the poorest fit (greatest stress).

Maximum Number of Iterations

This is most basic stopping rule, limiting the maximum number of iterations of SSH to the value supplied, regardless of the stress. Varying the parameter can be useful when tracking the amount and direction of movement of objects with each iteration. The default value of 50 should suffice for most applications.

Minimum Stress Difference

This second stopping rule is generally the one that should be used to stop SSH iterating. It is based on the change in the stress value from one iteration to the next. The method does not use the difference between the stress of the current and last iteration. What is used is the difference in the *ratio* of stress to last stress. This amounts to stopping when little difference is seen in the decrease in the stress. The actual mathematics of how this stopping rule is implemented is somewhat more complicated. A weighted average of the stress ratio is also maintained by

```

sratav=sratat.666 * srat.333

      where
srat   = stress now/last stress           and
sratav = average stress ratio

```

The stopping rule is explicitly:

```

If (srat > (1-strdiff) and srat < 1)      and
(sratav > (1-10*strdiff) and sratav < 1)  STOP

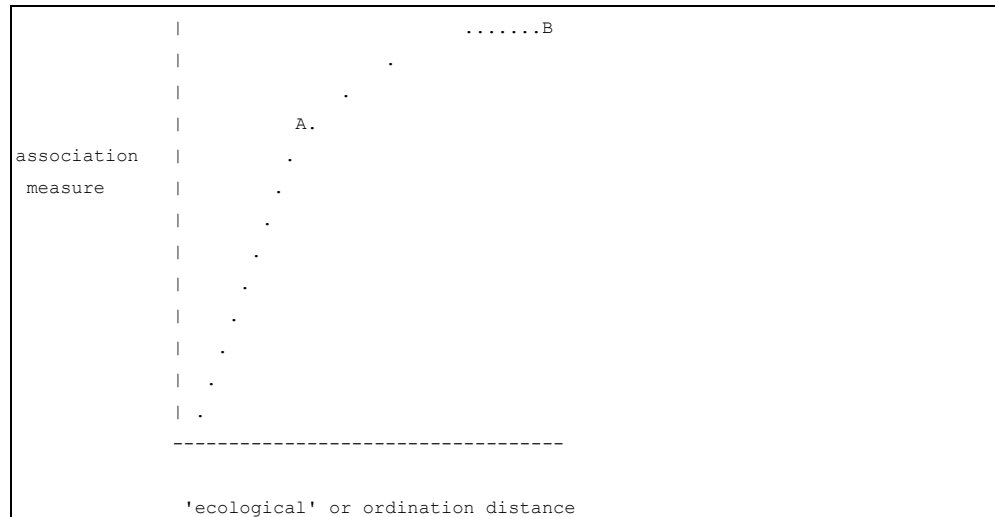
```

The need for this somewhat complicated approach is justified if the pitfalls of steepest descent algorithms are examined! Basically, it implies that we want to stop iterating only when the stress has had a chance to settle down. Premature exits usually imply a local minima has been found. The use of the above stopping rule provides an opportunity for escape in many of these situations.

Yet a third stopping rule exists. If the points are attempted to be moved by *very* small distances.

Interval/Ratio-Ordinal Scaling

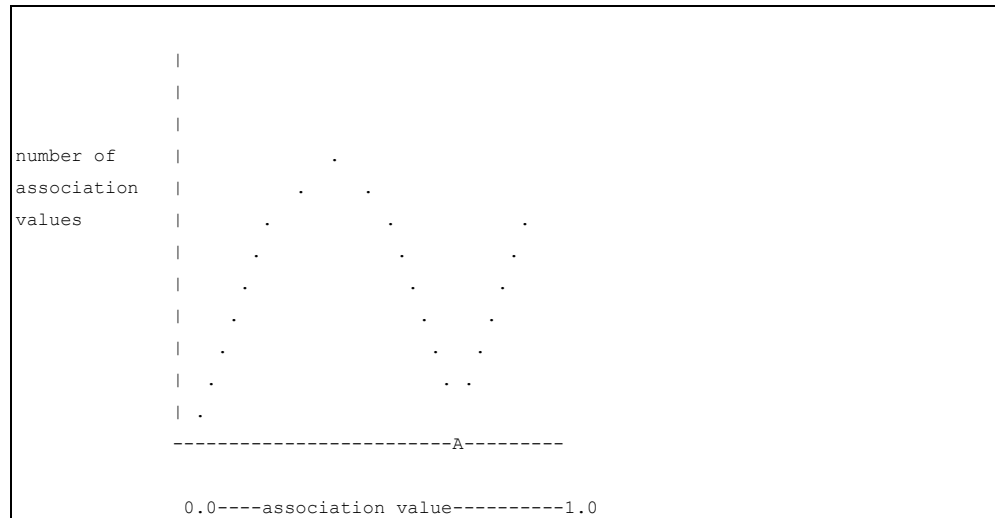
Linear regression options are interval or ratio. SSH permits any combination of interval or ordinal with ratio regression. Faith, Minchin and Belbin (1987) suggest that the best association measures available give good estimates of 'real' distances when the objects have much in common. When the objects have little in common however, the estimates are poor estimates of the 'true' distance. For example, with most ecological data, use of the Bray and Curtis (Czekanowski) or Kulczynski are appropriate. They have a response as follows -



In summary, the 'response' is part linear (to 'A') and part ordinal (between 'A' and 'B'). Given the above example, SSH is trying to generate the 'ecological' distances using regression. Use of interval or ratio regression alone would not be 100% effective. If values only up to 'A' were used, improved estimates could be obtained for the linear regression. Ordinal regression on the other hand, would provide a good fit above 'A' (with the semi-strong criteria), yet if applied to all 'distances', would miss the interval or ratio information below 'A'. The solution is therefore to use interval or ratio below 'A' and ordinal above. This concept is termed hybrid scaling.

SSH requests the location of 'A' from the diagram above. If using Kulczynski, Bray and Curtis, Czekanowski or *relativised Manhattan* distance, my suggestion is a value around 0.9. The best idea is to examine the histogram of association values from ASO (see ASON option 12). The shape of the histogram will be data dependent, but data with no well defined groups, will be unimodal. If many of the objects have little in common, a secondary peak at higher values of association will develop.

For example



The position marked 'A' on the above figure corresponds to 'A' on the first figure. It is this value that should be used as a cut point. While this discussion forms what appears to be the usual case, the cut value can range from zero to a very large value (>1). If you believe that your values of association are only ordinal correct, use a value for the cut that is less than the smallest association value. Similarly, if you desire, a ratio regression can be used over the complete range of association values by selecting a cut value greater than the largest association value. If you select an intermediate cut value, SSH will use interval or ratio regression on the values less than the cut and ordinal regression on those above it. Simple!

Hybrid scaling using the KYST multidimensional scaling program is subtly different. Two copies of the association matrix are prepared. The first matrix is treated as ratio below the threshold and missing above. The second matrix is treated as being all ordinal.

Ratio-Ordinal Cut Value

This is the value of input association that determines which association values are used in interval/ratio regression and which are used in ordinal regression. For values below the supplied value, interval or ratio (user selected) regression is used, while above the cut, ordinal or monotone regression is applied. A value less than the minimum association will apply only ordinal to the complete set. A value greater than the maximum will apply only interval/ratio. Anything in between will combine interval or ratio and ordinal regression to the appropriate values.

Examine the association values and select accordingly. The default cut is, from experience with the Bray Curtis style measures, 0.9.

Initial Configuration

SSH provides three options for the initial configuration -

1. use random coordinates,
2. read coordinates from an external data file
3. read coordinates from an external ordination file

Because of the local-minima problem, the use of multiple random starts is strongly recommended. The use of a PCA or other ordination start, while providing a fast estimate of a reasonable local optima, should not be trusted. For those of you who may be DECORANA fans, I recommend a DECORANA start! I guarantee that SSH will reduce the stress associated with the initial configuration, even if the result is a local minima!

SSH configuration option 1 uses the same random number generator as used elsewhere in PATN. It will generate uniform random coordinates in the range 0 -> 1.

SSH provides for multiple random starts by incrementing the initial seed value to produce another set of random coordinates. One idea I have found useful is to use the configuration from a previous SSH run as a starting configuration. The reason for this is that the step size often decreases too steeply to get out of local minima. By accepting the previous local minima as a starting configuration, SSH may be able to "jump out of the hole". Try it!

A problem first noted in KYST in relation to random starting configurations is negative regression. In this situation the relationship between the *input* and *output* associations is **inverse**. SSH detects a negative correlation if at least any of the associations are based on ratio regression (KYST doesn't). It appears that the general multidimensional scaling algorithm cannot guarantee to get out of this dilemma; most times it doesn't. The problem is that the final solution from such starts are often around 20% off the global minimum! If SSH detects the problem, it will throw away the current configuration and generate the next in the sequence. When you request K random starts, SSH will ignore those with inverse relationships.

For initial runs of SSH, option 1 is by far the safest. You may subsequently like to try 3 using a previous result or DCoA but be aware of local minima.

Number of Random Starts

When option 1 above is selected, SSH will request the number of different random starts to be used. The default of 10 is a reasonable lower limit. SSH will write the configuration with the lowest stress to the *-.ssh* file.

NOTE: unlike most of PATN, SSH will not ask if it is ok to overwrite the previous version of *-.ssh* and will automatically do it. So, if you have a favourite result that you want to keep, rename the *-.ssh* file to anything else before running SSH!

Random Number Seed

The usual initial seed for my random number generator in PATN. The same seed will always generate the same set of random numbers so be careful about this value in subsequent runs. Fractional numbers appear to work better than integers.

OUTPUT

SSH writes two files. The first *-.ssh*, is standard **PATN** ordination file showing the loading of objects on the selected dimensionality.

The second *-.shp* (for Shepard) file contains the values for the four association matrices maintained in SSH. The order of the association values is determined by ranking or sorting the input association matrix -

1. The *input* association matrix
2. The *output* associations in k-dimensions
3. The *estimates* of the input distances from the regressions,
4. The order of the association values in the three matrices.

Each matrix is written for N objects in the form of a single vector of $N*(N-1)/2$ values in FORTRAN (16F8.4) format. For example, for 10 objects, the *-.shp* file will contain 4 rows and 45 columns. This file has many applications. It can be used to plot the Shepard diagram (input verses output associations) or it can be used to determine which objects have the poorest fit (greatest stress).

PRINCIPAL COMPONENTS & COORDINATES: PCA

OUTLINE

PCA does **Principal Component and Coordinate Analysis**. This is one of a number ordinations methods. Alternatives include SSH, SERE and DCOR (DECORANA). Their aim is to find a small number of new attributes which account for a large proportion of the information in the data. If two or three such attributes can be found, a scatter plot of objects using these attributes as axes will summarise information in the data.

Geometry

The Principal Components approach is to find a **linear** combination of attributes for each new attribute so as to account for decreasing proportions of variance. The concept can be approached from a geometric point of view, the algebra may be followed in Digby, Sokal & Rohlf and Mardia (references below). Consider the objects as points in a multidimensional space with each of the original set of attributes forming axes, all at right angles to each other.

The first new attribute or *eigenvector* is located in this space so that it aligns with the longest projection of the objects. If the distribution of objects was cigar shaped, then the first eigenvector would be located along the long axis of the cigar. Using the first eigenvector as a new axis would enable each object to be given a score on this axis. This value would in the above case, give a reasonable location of the object in the space. It would also, by implication, account for the major variation in the data.

The second eigenvector must be located at right angles to the first with the constraint that it must account for the majority of the **remaining variance**. In the cigar example, this amounts to rotating the second axis at right-angles about the first until maximum overall extension of objects is located. The process continues until sufficient variance is accounted for.

Algorithm

PCA extracts all eigenvalues and eigenvectors from a **PATN** association matrix. The method used is Householder's reduction of a real symmetric matrix to tri-diagonal form, followed by the Q-L algorithm that extracts eigenvalues and eigenvectors. Only the requested number of eigenvectors are stored.

Principal Coordinates

If the Correlation Coefficient is used as the association measure (ASO), the use of TRNA (no transformation) and PCA will result in Principal Components Analysis (PCA). All other association measures must be transformed using the Gower Correction option in TRNA before use in PCA. The latter approach is termed Principal Coordinate Analysis (*PCoA*). This should not be considered as a different ordination technique, but rather principal component analysis of non-correlation coefficient by a pre-transformation and standardisation on an association matrix to Euclidean form (see TRNA).

Comparison of Methods

PATN contains a number of different algorithms (techniques) for ordination-

1. SSH (multidimensional scaling)
2. PCA (this module)
3. SERE (1-d ordination or seriation by parsimony)
4. DCOR (DECORANA: Correspondence Analysis or Reciprocal Averaging plus or minus de-trending)

Simulation studies on ordination techniques suggest that SSH is probably the most robust ordination method (Belbin, 1991). This is because principal-component methods fit squared dissimilarity values and not, as in SSH, the dissimilarity itself. This leads, in the case of the former, to an inflated importance of the larger values of dissimilarity, those that are believed to be the least accurate (see TRNA re-estimation options). The consequence of this leads to compacting the distribution of objects. SSH doesn't do this. While DECORANA should not be used for DETRENDING, using DCOR for correspondence analysis/ reciprocal averaging is, for biological data, an improvement over principal components/coordinates.

The new attributes from PCA may be merged as additional attributes to intrinsic or extrinsic data and may subsequently be used for -

1. scatter plots with ordination vectors as x-y (SCAT)
2. assessing the importance of ordination scores (GSTA)
3. sorting attributes on ordination values (SAMP)

The format of the ordination file is similar to an **PATN** data file with the eigenvectors forming the columns of the matrix. At the bottom of the file is listed -

1. eigenvalues in decreasing order of contribution,
2. the corresponding variances of eigenvectors
3. the cumulative variances of eigenvectors.

The eigenvectors are standardised so that the sum of squares of each vector equal the eigenvalue.

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OPTIONS

The number of eigenvectors and eigenvalues to be output to the ordination file. All are calculated but only the requested number are stored on the file *-pca*.

CORRESPONDENCE

ANALYSIS & DETRENDING:

DCOR

OUTLINE

DCOR is **PATN**'s pre-processor module for the Cornell program DECORANA (called DCA in **PATN**). This ordination program performs Detrended Correspondence Analysis (DCA) and its precursor, Reciprocal Averaging (RA: also invented as Correspondence Analysis by Benzecri). For details of the DECORANA algorithm, the reference, as well as a listing of the program, may be obtained by writing to the -

Section of Ecology and Systematics
Cornell University
Ithaca, New York USA 14850

Reciprocal Averaging is equivalent to a pre-standardisation of data prior to Principal Coordinates Analysis (PCoA), but can also be viewed as the alternate successive averaging of objects and attributes. In PCA (PCoA), the object scores are the **sum** of the set of attribute scores and vice versa. RA has object scores as the **average** of the attribute scores (and vice versa). Therefore PCA emphasises attribute-rich objects in comparison with RA.

Consider a set of sites (objects) and species (attributes). Sites are given an initial weighting, then species averages are re-estimated on the basis of -

	species			initial site	new site
	1	2	3	weighting (1)	weighting (3)
site 1	1	0	1	0	50
site 2	0	1	1	50	62.5
site 3	1	1	1	100	58.3
derived species weighting (2)	50	75	50		

The first calculation is the first re-estimation of species 1 based on the 3 initial site scores (0,50,100) and is-

$$= (1*0 + 1*100) / 2 = 50$$

Once these new species scores are determined, new site averages follow. For example, the second site score is calculated as

$$= (1*75 + 1*50)/2 = 62.5$$

Iteration proceeds until convergence is attained (little or no change in the site and species averages). If de-trending is requested, then each axis of the resultant ordination is de-trended against higher axes. This can have the desired effect of straightening out non-linear components but will also straighten out true non-linear distributions.

The TRNA options for re-estimating larger associations by using network distances is probably a more viable alternative to de-trending. This implies that re-estimation can be used in conjunction with ANY ordination technique. The best technique for handling poor estimates of the larger association for small sets of data is the SSH module.

PATN-DECORANA interaction

DCOR is a **PATN** pre-processor that -

1. interacts with the user to determine data type and parameters
2. creates an input batch stream for DECORANA,
3. runs DECORANA and then,
4. runs POST to un-bundle results from DECORANA for PATN.

The post-processing module POST is activated by the **PATN** supervisor to make available the files resulting from DECORANA in the same form as all other **PATN** options. Along with the output (history) file from DECORANA, standard ordination format files are produced for both rows and columns.

If the parameters nominated the file *FRED.DAT*, the DCOR processor will assume the presence of *FRED.ASO*, *FRED.RLB* and *FRED.CLB* and will produce the files-

<i>FRED.DCA</i>	row ordination
<i>FREDT.DCA</i>	column ordination
<i>FREDT.PRM</i>	column parameter file

DCOR has been updated to now request as the first option, the FORTRAN format of the data to be input into the file for001.dat for DECORANA. The default format is:

```
(I5,5(I5,F8.3))
```

Any legal FORTRAN format can be used (see the PATN Users Guide chapter on formatting if you need to brush up). The structure of this format is significant. It must contain a leading INTERGER as the record identifier and then a series of pairs of FLOATING-POINTS and INTERGERS; the first referring to the attribute number and the second to the value for that attribute.

NOTE: the maximum number of characters that comprise this format specification is 60. The reason for this change to DCOR is that it was possible (as one user found out) to overflow this format given large data values.

REFERENCES

- Hill, M.O. (1979). DECORANA, A FORTRAN program for de-trended correspondence analysis and reciprocal averaging. Department of Ecology and Systematics, Cornell University: NY, USA.
- Belbin, L., Faith, D.P. and Minchin, P.R. (1984). Some algorithms contained in the Numerical Taxonomy Package NTP. CSIRO Division of Water and Land Resources Technical Memorandum 84/23.

OPTIONS

While a brief summary is included here, the user is referred to the DECORANA manual for details on options available for this algorithm. Numbers in brackets are page references in the DECORANA manual.

Analysis type (16).

A value of ZERO indicates *Detrended* Correspondence Analysis (default) while a value of ONE results in basic correspondence analysis (= Reciprocal Averaging).

Downweighting of rare species (15)

By default, DECORANA divides species (columns) by the maximum value and sites by the totals. If this option is selected, species will be left un-standardised.

Re-scaling axes (16)

The DECORANA manual suggests the default value of zero which results in an application of re-scaling four times. Use of other values such as '-1' for no re-scaling or 'k' for 'k' applications of re-scaling is not recommended in the manual.

Number of de-trending segments (16)

The manual recommends a default value of ZERO. This results in 26 de-trending segments. Again, the DECORANA manual recommends that this default be adhered to. Quoting the manual, "if the value does need to be raised, then this will show up in the second axis having an appreciable systematic relation to the first axis (ie in the de-trending not having been fully effective). An alternate value of '50' (the maximum permissible) could then be tried."

ONE-DIMENSIONAL ORDINATION (SERIATION): SERE

OUTLINE

SERE is a module for one-dimensional ordination or **seriation** based upon parsimony (smoothness). As with most of **PATN** the algorithm is heuristic (iterates using a simple 'rule of thumb')-

Select any pair of objects as a starting configuration and then find which other object minimises the 'stress' of the one-dimensional arrangement on addition, either at the end of the sequence, or in the middle."

Stress is defined as the total amount of change across the sequence of objects. This is determined by addition of the inter-object association. For example the sequence -

ROW 1 ---0.1---ROW 2 ---0.2--- ROW 3 ---0.3--- ROW 4
--

amounts to $0.1 + 0.2 + 0.3$ units of change (0.6) while, re-arrangement may give:

ROW 2 ---0.2---ROW 4 ---0.3--- ROW 1 ---0.4--- ROW 3
--

which sums to 0.9 units of change. The first sequence is said to be more **parsimonious** than the second and provides, in the sense of the algorithm, a more optimal 1-d ordering of objects. Parsimony in SERE is 'a minimum of overall change along the gradient'.

The algorithm is simple. Find an object/position combination such that it minimises change, in terms of total association values, over the existing sequence of objects. Addition continues until all objects are included in the sequence. Take an arbitrary starting configuration using 5 objects A, B, C, D, E. If A and B initiate the sequence -

A - B

then C is added in the following configurations and evaluated.

C - A - B
A - C - B
A - B - C

D, E and F are examined in the same way. The lowest parsimony value is used to decide which of C, D, E, or F is added, and where it is added (to the left of A, in the centre between A and B, or to the right of B).

SERE is, by definition, *limited* to one dimension. If more than one gradient exists in the data, SERE cannot be expected to produce an optimal ordering; rather, it will blend all gradients into one.

The problem of finding the most parsimonious gradient is an *np-complete* problem. This means that the only way guaranteed of achieving the **global optimum** in terms of the algorithm, is by trying all possible combinations of sequences. A very large task, even for 'super-computers'. This complexity constrains the present algorithm to a heuristic approach. A number of options are provided to initiate or *seed* SERE.

SERE produces a standard ordination file containing objects in sequence order with a single set of coordinates based on association. The order of objects and the stress is also appended to the coordinate table. As with other ordination techniques (PCA, SSH, DCOR) the ordination vector may be merged to the right-hand-side of the data file.

REFERENCES

- Faith, D.P., Minchin, P.R. and Belbin, L. (1985). Parsimony and falsification in ecology: Towards an assumption-free approach to the study of species' response to gradients. *Studies in Plant Ecology*, 16, 31-32.
- Faith, D.P., Minchin, P.R. and Belbin, L. (1987). Compositional dissimilarity as a robust measure of ecological distance. *Vegetatio*, 69: 57-68.

OPTIONS

Seeding

SERE requires a pair of objects to be used to initialise or seed the algorithm. These two objects form the base on which all other rows are added, one at a time. The algorithm presented, while being relatively rapid for a single seed pair, becomes lengthy when a large number of seed pairs are to be used.

The seeding options are -

- 1 = Enter **one** seed pair (sequence numbers),
- 2 = a **proportion of all** pairs in nominated range,
- 3 = enter an object order to evaluate stress or
- 4 = STOP.

Option 1 will request the sequence number of the two objects to be used as seeds. It will perform the seriation, list the stress and the order and return to the option menu, at which time any of the options may be selected or re-selected.

Option 2 provides an efficient method of sampling a variety of combinations. It is a reasonable approach to finding the global minima. It requires as input -

1. a minimum value of association to use
2. a maximum value of association to use,
3. the proportion of pairs to try (zero to one) and
4. a random number seed value.

SERE scans the association matrix looking for all values within the user specified range. It then randomly selects a proportion of these for the corresponding seed objects.

Option 3 performs the opposite operation to OPTION 1. It reads in a sequence of objects from the terminal or a file in FORTRAN free integer format and evaluates the parsimony value.

Option 4 is used to stop SERE after a satisfactory solution has been obtained with any combination of the previous options.

MINIMUM SPANNING TREES:

MST

OUTLINE

MST is a module for calculating a network structure called a Minimum Spanning Tree (MST). It can be thought of as a set of lines, representing the pair-wise associations (from ASO) that interconnect all the objects with the following limitations -

1. all objects have at least one connection,
2. there are no circuits or loops and
3. the network or tree is minimal length

The MST is *unique* only for situations where there are no tied values in the association matrix. The MST provides a simple and effective means of displaying the relationships between objects in the form of a graph. There are similarities to the other major analysis techniques of clustering and ordination. The UPGMA clustering method is recommended because the fit between the input associations and the distances inferred from the dendrogram (**ultrametrics**) is optimal. Similarly, the goal of SSH is to optimise the fit between the input associations and the associations as measured between objects in the reduced dimensional space. The 'ultrametric-distances' from MST are constrained to be along the branches of the tree. Therefore, neighbours are accurately represented while the increase in the uncertainty is proportional to the increase in separation.

Scale & Ordination

The MST gives limited information about the spatial configuration of objects. The absolute position of all objects cannot be determined. Accuracy can only be maintained as the *length* between neighbours on the tree and not direction. It is unlikely in most real situations, that a MST could be drawn to scale. This is simply due to the multi-dimensional nature. But this fact highlights one of the strong features of the MST, it is complimentary to ordination.

The MST concentrates on an accurate representation of close objects. The further away the objects are on the MST, the less that can be said about The relationship between them. They may be close or far apart. Ordination is the opposite. Two objects that are well separated in an ordination plot must imply low affinity while those that are close in the displayed dimensionality may not necessarily be close. For example, two points may be close in two-dimensions but a third dimension may separate them to opposite extremes in the distribution.

Imposing the MST on an ordination plot is a practical procedure. The MST may confirm or deny close relationships by connections. It may also comment on the significance of the stress from say an SSH or PCA. If the MST has many crossed lines, the dimensionality may not be sufficient. Think about it.

The MST may also be used in a complementary fashion with ordination as a trend finder. Complex non-linear structures that may not be detected in the normal way may be disclosed by use of a MST. Such structures are a common problem with traditional ordination methods such as PCA/PCoA that are trying to fit underestimated 'distances' between objects. Curved, spiral or circular patterns are often difficult to detect and are not uncommon.

BOND

The MST concept may be taken one stage further by analysing the first and second nearest neighbour relationships using BOND. The MST and the BOND linkages generally coincide and the former, using ordination coordinates, furnishes an excellent summary of structure.

Display

MST outputs the linkage pairs in a table format that reflects the structure of the tree. If row "a" links to "b", and "b" to "c" and "c" to "d" etc. ("chaining"), then the links are printed indented to the right in the MST table. That is, the table is indented to the right when a new object joins the MST. When the MST returns to connect another link to an object already connected in the table, it does so at the same indentation point as the first connection of that object. For example, the first five linkages taken from the example below show:

1	ROW	1 (	1)	-	ROW	6 (	6)	:	0.2418
2	ROW	6 (	6)	-	ROW	4 (	4)	:	0.2233
3	ROW	4 (	4)	-	ROW	2 (	2)	:	0.2457
4	ROW	2 (	2)	-	ROW	9 (	9)	:	0.2361
3	ROW	4 (	4)	-	ROW	5 (	5)	:	0.2301

Where the first value on the line is the level of indentation, the second pair of values represent the pair of objects that have been linked and the third value is the distance (association) between them. Object 1 and 6 for example, are joined first exhibiting a 'distance' of .2418 units. Then, because object '4' is a new object to be linked to a previous one ('6'), the second line is indented to level 2. Object '2' then joins '1','6' and '4', followed in the same way by object '9'. Then, for the first time in the table, object '4' in addition to the linkages to objects '6' and '2', a new linkage to a new object '5' is required. At this point (the fifth row in the table), the new entry for object '4' is aligned with the previous object '4' entry (the third row in the table) at indentation level 3.

The configuration of the above sub-graph, not drawn to scale would be:



Using this method of printing, the structure of the network is more evident, and consequently easier to draft.

A second table in the output file is produced by MST. This table contains the number of MST linkages to each object in the input matrix. This list is ordered by increasing number of MST linkages and is intended to provide some concept of *connectedness*; those objects with a high level of connectivity (socialites)? are probably 'central' in the distribution of the points in multidimensional space. Those with low 'connectivity' are probably 'peripheral'.

Output of an Ultrametric Matrix

While MST does not provide an option for the output of the implied or *ultrametric* distances between all pairs of objects, as measured along links of the MST, TRNA option 3 does! These measures may be used in a number of different ways. They may be used to provide an upper limit for under-estimated associations (see SSH). They may also be used to compare the original and the *ultrametric* counterparts.

Ultrametric distances are strictly defined as the metric distances derived from a dendrogram but there are counterparts for any of the analysis options in **PATN**. For example, ordination techniques result in a reduced dimensional distribution of objects; the output Euclidean distances are those measured between all pairs of objects in this reduced dimensional space. With network/graph techniques, the 'ultrametric' distances are the pair-wise distances, as limited along the network connections.

REFERENCES

- Gower, J.C and Ross, G.J.S. (1969). Minimum spanning trees and single linkage cluster analysis. *Applied Statistics* 18: 54-64.
- Belbin, L., Faith, D.P. and Minchin, P.R. (1984). Some algorithms contained in the Numerical Taxonomy Package NTP. CSIRO Division of Water and Land Resources Technical Memorandum 84/23.
- Ross, D. (1984). TAXON package. CSIRO Division of Computer Research, Brisbane.

OPTIONS

none.

NEAREST NEIGHBOUR

LISTS:

NNB

OUTLINE

NNB lists the set of *k nearest-neighbours* for each object, sorted by ascending association value (decreasing proximity) from *-.aso* file. The advantage of such a tabulation is that actual association values are used rather than any amalgamation through processes such as classification (ultrametrics) or ordination (reduced dimensional distances). Data structures or associations resulting from FUSE, MDIV, PDIV or ordination methods may be analysed in the same way using this table. The finer detail available may provide additional insights about *structure*, *stress* and *mis-classification*.

If *adjacency* is coded into an association matrix, negative values are used to nominate an adjacency link between two objects (see TRNA). This is not a problem to NNB because absolute values are used for the tabulations.

NOTE: the number of neighbours printed must be at least **2** if BOND is to be subsequently used.

A second table provides the *average* distance for each object to all other objects. The table may also provide insights concerning the overall *structure*. The selection of seed rows for ALOC could be determined from *stereotypes*. The object with the lowest average distance must be close to the centroid of the distribution in multidimensional space. If the second object listed is not a close neighbour of the first, then it could form the seed of a second group, and so on.

There are two options as to how the average associations are tabulated. The **first** (sorted) lists objects in order of increasing average distance of all other objects. This option uses a two column format, written row-wise. The **second** option lists objects in a single column format in sequence number order (unsorted).

Redundancy?

If the distances to the first nearest neighbour are sorted (the first table in *-.nnb*), this provides a useful estimate of what I am tempted to call the degree of *redundancy* of objects within a data set. While redundancy may be a somewhat dangerous term in many applications (can anything really be redundant!), it may be an instructive philosophy to examine. As with most of my ideas, it is simple.

One of the objects listed as the closest pair is in a sense, a candidate for omission. Take the extreme example of identical objects. In terms of information, by definition, if you have one of the objects you have both. In relation to weighting in pattern analysis, this is not strictly true. For example, a different classification will be produced if identical objects are removed or added to a data set (assuming that all objects do not have an equal number of replicates).

If the cost of doing something to or with objects was a premium, which objects should be 'done to'? One answer may be to first eliminate most of the redundant objects. Another way of thinking about the data in this way is to find those objects that would least perturb any pattern analysis by their removal. An example may help to see the implications of this strategy. If a set of biological point surveys, which samples (areas) could not be left out (at any cost) as a consequence of their uniqueness of species composition? Pattern analysis I believe provides an explicit, and by traditional standards, an objective methodology to approach this type of problem.

Of course the method presented via 'closeness to nearest neighbour' is one of a number of possible methods of defining redundancy. Another would be for example. MAXD. Attribute sensitivity/redundancy is also addressed by SENS.

REFERENCES

- Belbin, L., Faith, D.P. and Minchin, P.R. (1984). Some algorithms contained in the Numerical Taxonomy Package NTP. CSIRO Division of Water and Land Resources Technical Memorandum 84/23.
- Tversky, A. (1983). Nearest neighbour analysis and point processes: applications to multidimensional scaling. *J.Math.Psych.* 27:235-50.

OPTIONS

NNB provides two options as to how many neighbours are listed -

1. A fixed number of neighbours, from the closest to the furthest in terms of the association measure in *-.aso*. Use 2 or more when BOND is to be used subsequently. For this option a value ≥ 1 is used; for listing the five closest neighbours, use '5'.
2. A variable number of neighbours, determined by a maximum value of association that the list can contain. The neighbours are ordered in the list by increasing value of association. For this option, use a value < 1 . For example, to list all neighbouring objects within a radius of 0.5 (association value), use '0.5'.

BOND ANALYSIS:

BOND

OUTLINE

Bond-strength was developed from an algorithm published by Bill Williams (1980). It analyses relationships between objects using only the first and second nearest neighbours from NNB. BOND is based upon attributing a score to first and second nearest neighbour links between pairs of objects. A first nearest neighbour link is allocated a value of 3, while a second nearest neighbour link is allocated a value of 2.

BOND weights the importance of the pair-wise links on the basis of neighbourliness. In a sense, this is equivalent to what is termed a 'row conditional' approach in SSH. Each object is treated independently and there is no global standardisation. For example, mutual first neighbour links are weighted equally, regardless of the 'distance' involved.

A table showing the possibilities between pairs of objects is readily generated as a consequence of the coding scheme -

Weight	Description
6	mutual first nearest neighbour
5	first one way, second the other
4	mutual second nearest neighbour
3	first one way only (>2 other way)
2	second one way only (>2 other way)

If the bonds are *drawn* on a two-dimensional representation of the objects, as from an ordination, a considerable amount of additional information is added. The best scheme is firstly to provide a good ordination base, then overlay MST links in pencil. Many of these links will be used by BOND. Coloured links or the use of double and single lines with arrows emphasise the strong and loose relationships in the data.

In addition to the network provided by BOND, a set of *groups* may be defined by drawing lines about the objects that are defined with bonds greater than '2'. Bonds equal to '2' will define weaker, inter-group connections. Because outliers must have a first and second nearest neighbour, regardless of the level of association, they will always be drawn to their nearest- neighbour group. If outliers exist (look at the association values from NNB), more groups should be defined. BOND therefore provides a reasonable lower limit to the number of groups.

The cut-off value of bonding strength for group definition is user-defined. Williams has suggested that groups be defined by bond strengths > 2 but values of 3-5 may be useful in some circumstances.

NOTE: Aspects of object redundancy may be determined either via output from NNB or BOND (see NNB section on redundancy for a discussion of this concept).

REFERENCES

- Williams, W.T (1980). Twonet: a new program for the computation of a two-neighbour network. *Aust.Comp.J.* 12:70.
- Belbin, L., Faith, D.P. and Minchin, P.R. (1984). Some algorithms contained in the Numerical Taxonomy Package NTP. CSIRO Division of Water and Land Resources Technical Memorandum 84/23.

OPTIONS

The only option provided in BOND is the cut-off value to be used for the generation of the group boundary. The default value of '2' should be used unless all implications are understood.

TRAVELLING SALESMAN

CIRCUIT:

SALE

OUTLINE

The classic travelling salesman problem attempts to optimise in terms of minimal distance, a route for a salesman needing to visit a set of 'cities' and return home. In other words, the salesman starts from any city and visits all other cities and returns to the starting location. The solution assumes that there are roads that interconnect all pairs of cities.

I have found that the travelling salesman circuit has been useful in delineating structures in an ordination that have not been detected by similar methods such as the use of minimum-spanning trees (see MST). At this stage, as with many PATN modules, SALE is experimental. Use it to see if it adds anything to assist interpretation. As ever, feedback would be appreciated.

Until a few years ago, this was regarded as a problem where an optimal solution could not be guaranteed, except for a small number of 'cities'. The problem is called 'Np complete' which means that the only way that an optimal solution can be guaranteed, is to examine the distance of each and every possible arrangement of N cities. Even for say a small number of cities, one requires a super-computer and a very large amount of computational time to guarantee an optimal solution. Simulated annealing is a modern 'heuristic method' (read 'rule of thumb approach'), that appears to provide optimal solutions in reasonable time.

"At the heart of the method of simulated annealing is an analogy with thermodynamics specifically with the way that liquids freeze and crystallise, or metals cool and anneal. At high temperatures, the molecules of a liquid move freely with respect to one another. If the liquid is cooled slowly, thermal mobility is lost. The atoms are often able to line themselves up and form a pure crystal that is completely ordered over a distance up to billions of times the size of an individual atom in all directions. This crystal is the state of minimum energy for this system. The amazing fact is that, for slowly cooled systems, nature is able to find this minimum energy state. In fact, if a liquid metal is cooled quickly or quenched, it does not reach this state but rather ends up in a polycrystalline or amorphous state having somewhat higher energy." Simulated annealing selects the best from among the set of possible solutions and performs random re-arrangements until no improvement can be made.

SALE reads the default association file *-.aso* and calculates the travelling salesman solution. Output is in the form of a file (*-.sal*) that is formatted the same as other ordination files. The order of the records in this file details the travelling salesman circuit. The distances are also listed between each pair of 'cities' (objects) in the travelling salesman solution and the total circuit distance. This should be minimal for a given set of objects.

In the classic problem, the locations would be spatial locations in say latitude and longitude or with the distances in kilometres or equivalent. In this implementation, whatever association that is found in the default file (*-.aso*) will be used as the distance. As with most of PATN, these associations are assumed to be dissimilarities / distances.

TSPN will look for the presence of a *-.sal* file, and if present, will create the connections for travelling salesman circuit for SPIN to place in 3d space as a separate layer. This layer can be switched off and on as required. See TSPN.

REFERENCES

Press, W.H., Flannery, B.P., Teukolosky, S.A. & Vetterling, W.T. (1986). Numerical Recipes. Cambridge University Press. 818p.

OPTIONS

none

PRINTING DENDROGRAMS:

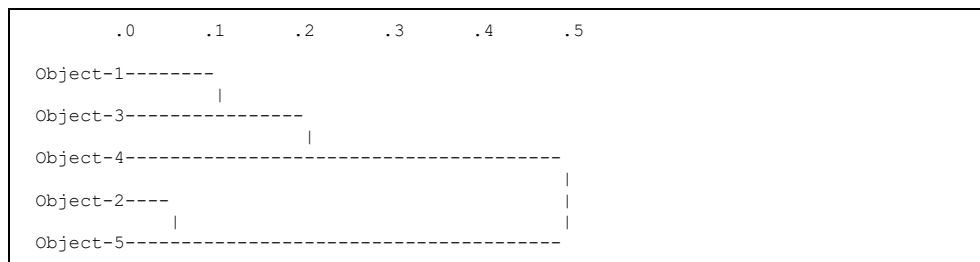
DEND

OUTLINE

DEND generates dendrograms or the binary tree structure representing fusion or division history from a hierarchical cluster analysis as produced by -

FUSE
MDIV or
PDIV

DEND produces these dendrograms on a standard line printer using one line for one or a group of objects, depending on options. This means that the dendrogram is drawn on its side with the x-axis representing decreasing association (increasing 'distance'). For example:



DEND uses a single line for each object in the display. In this example, for clarity, two lines are used for each object. The diagram represents, for an agglomerative technique, the following fusion *history*.

1. 'Object-2' fused with 'Object-5' at a dissimilarity value of around 0.05,
2. 'Object-1' fused with 'Object-3' at a value of 0.1
3. 'Object-4' fused with the group comprised of 'Object-1' and 'Object-3' at a value of 0.2
4. The group comprising 'Object-1', 'Object-2' and 'Object-4' fused with the other group containing 'Object-2' and 'Object-5' at a dissimilarity value of 0.5.

In the case of a division method, the diagram would be interpreted in reverse, ie right to left. For example, the first split occurred between objects (2) and (5) with the other group comprising (1), (3) and (4).

Base

For agglomerative methods, the dendrogram may be drawn with the base at any group level. The base is represented by a vertical line through the complete dendrogram parallel to the edge of the paper; all groups/individuals are *cut* at a single level of dissimilarity. Normally, the complete dendrogram is drawn, down to individual objects, but for a large number of objects, the dendrogram may be too large to readily interpret. In this instance, a number of dendrograms at different levels may be printed.

Is it legal to 'cut' a dendrogram along a line that is not parallel to the base? strictly speaking no. But in some circumstances, this may be a helpful or necessary action. For example, consider the classification of a group of heterogeneous objects that are made up of two 'logical' subsets. One group has a lot of heterogeneity while the other has little. A common example of this in ecology occurs when mixing 'dry sclerophyll' forests with 'rainforests'. Quite correctly from an analytical point of view, **PATN** perceives that a large proportion of the heterogeneity is in the rainforest samples (and acts accordingly). This may be distinctly unhelpful if you're interested in the eucalypts! Probably the best idea is to classify the two groups separately. Alternatively, cut the dendrogram on a line not parallel to the base (I won't watch!).

Rotation

All hierarchical methods in **PATN** do so either by the fusion of two objects or groups, or the splitting of a group into two sub-groups. This produces a binary tree. The intersection of horizontal and vertical lines define what could be termed *nodes* where rotation may take place into and out of the page. Consequently, while the left to right ordering is immutable (well... as noted in the previous paragraph, not totally so), the up-down ordering of rows is somewhat arbitrary. DEND, for example, uses a scheme where the order of objects is determined by starting at the top (right) of the dendrogram and rotating the group containing the object with the *lowest sequence number* to the top of the diagram. This process is continued to the base. Row 1 will therefore always be the first object in the dendrogram. Alternate schemes are possible.

Ordering of Objects

DEND can read a preferred ordering from the *-.fus* file. The arbitrary nature of the standard ordering can be improved to highlight certain relationships. For example, an ideal way of producing an 'optimal' one dimensional ordering, as used by DEND, is to use the seriation program SERE. The multidimensional scaling program SSH could also be used.

The order of objects as output by any ordination algorithm can be appended to the fusion file using the following format -

```
stress threshold .....
(blank line)
.ORDER
(blank line)
1 2 5 6 7 4 3 2 1 .....
```

This means..leave one blank line after the line containing the stress summary and average increment, write the title ORDER in the second through the sixth column, leave another blank line, then write the sequence detailing the order of objects desired'. Integer free format is used. As many records as necessary are read until N objects are input. **NOTE:** the order presented has to be a legal ordering; the sequence **must** be able to be generated by rigid rotation about any set of nodes. **An illegal structure will not be drawn correctly.**

WARNING: DEND will NOT draft dendrograms correctly for FUSE options that produce reversals in the fusion sequence (Eg. WPGMC). A reversal occurs when the usually monotonic increase in association with fusion reverses, ie, it decreases instead.

The dendrogram order, as determined by DEND or read from FUSE is output to the file *-.irm*. This ordering of the objects may be useful to re-order the data set with MASK. Once this is done, DATN can be used to list the data in this order; somewhat similar to TWAY. It may also be handy to re-order the *attributes* before using the presence/absence option of GSTA. In this way, the attributes are listed down the page in dendrogram order with their associated constancy values across object groups. This file contains a set of N integers in FORTRAN (13I6) format. This may generally (as long as the numbers don't exceed 99,999) be read into MASK in free format using '*'.

REFERENCES

- Gower, J.C and Ross, G.J.S. (1969). Minimum spanning trees and single linkage cluster analysis. *Applied statistics* 18: 54-64.
- Belbin, L., Faith, D.P. and Minchin, P.R. (1984). Some algorithms contained in the Numerical Taxonomy Package NTP. CSIRO Division of Water and Land Resources Technical Memorandum 84/23.

OPTIONS

Number of Groups

The dendrogram from FUSE, may be listed at any level between '2' and 'N' where 'N' is the number of objects. This defines the base level of the diagram. It is important to understand how DEND labels the *groups* on the dendrogram when the base is not 'N'. The approach taken is **not** to use an arbitrary set of labels such as -

```
group 1
group 2
group 3.....group k
```


This would mean nothing unless GDEF had already been used and some connection between group labels and the objects that they contain had been made. What **IS** used is the lowest sequenced object in each group displayed. For example, if a group contained the objects-

34, 21, 22, 43, 78, 102 and 2

then the group would be labelled with as '2'. This immediately gives at least some indication of what the group is.

Annotation

The width of the dendrogram drawn in characters. The overall width can either be 80 or 132 characters. The 80 character width option is suitable for output onto a standard computer terminal or when printing a small dendrogram with say less than 50 objects. The 132 width option is more suitable for when there are more than around 50 objects. In this case, the dendrogram is scaled to be as wide as is possible on standard 38cm printer paper; this gives extended resolution to the diagram.

Another possibility is to scale the dendrogram to 80 characters, and use up to an additional 50 characters for extended labels. This makes the total width 130 characters (80+50). The extended labels can be read from any file using any legal FORTRAN format. The intention here is to permit complex labels such as species names that require more than the standard **PATN** limit of 8 characters per label, to be printed on one of the most useful classification diagrams. If extended labels are to be used, the FORTRAN FORMAT of the label **MUST** include the width in characters of these labels. **PATN** requires the width and scans the format looking for

-

Axy or Ax

where 'x' or 'xy' represent the width in characters. For example, '(A25)' implies labels of 25 characters wide while '(A)' will assume 50 characters, the maximum width

Output of the Ultrametric Matrix

The *ultrametric distances* as defined by a dendrogram are those inter-object distances *implied* from the structure. In the example dendrogram above, the implied distances between objects 2 and 5 to all other objects (1,3 and 4) is around 0.21. This is the dissimilarity value where both were fused. Similarly, the ultrametric distance between objects 1 and 3 is 0.1. In this case the ultrametric and the *actual* (input) distance would be equal. The reason should be obvious!

Ideally, ultrametries and input dissimilarities should be identical but this is rare. Pattern analysis is like most other analysis tools; there is a cost involved in attempting to simplify the data structure. In statistics for example, this may be the unexplained variance.

In Pattern Analysis, an equivalent could be termed *stress*. In multidimensional scaling, the stress is in fact, a scaled difference between an 'ultrametric matrix' as defined by the Euclidean distances between objects in the reduced dimensional space, and the input dissimilarities. The same applies to other ordination techniques such as principal components/coordinates or even de-trended correspondence analysis. It should be noted that multidimensional scaling algorithms are designed specifically to minimise *stress* and on this basis, should be considered as a more holistic approach.

There is a basic counterpart to ultrametric distances in non-hierarchical clustering. In the case of ALOC, this is the 'average deviation' listed in the `-.alo` file (and the terminal during iteration). How well does a method minimise the differences between the 'ultrametrics' implied from the dendrogram and the input dissimilarities? It turns out that, at least in terms of *cophenetic correlation*, UPGMA ($\beta=0.0$) minimises stress (I am not sure about flexible UPGMA, $\text{bet}=-0.1$). This is simply the (Pearson's Product Moment) correlation coefficient between the two association matrices; input and ultrametric.

DEND optionally writes a copy of the ultrametric matrix to the file `-.ult`. This may be manipulated and used by whatever strategy you like using modules such as ASON and TRNA. For example, the cophenetic correlation could be calculated using ASON and ASO. An alternative could be to use TRNA to subtract the original and ultrametric matrices. The difference matrix could then be classified to determine the key areas of difference! You could even ordinate the differences!!

GROUP DEFINITIONS:

GDEF

OUTLINE

Group Definition

GDEF is a module designed to create and manipulate *group definitions* and associated (similarly composed or structured) data. A group definition is the term used by **PATN** to convey the group membership of a set of objects, as derived from some previously run classification technique. The options provided by GDEF are -

- 1 = Create group definitions from a fusion file (*-.fus*)
- 2 = Create or modify group definitions
- 3 = Create group comparisons
- 4 = Output group definitions in free format
- 5 = Fuse one set of group definitions based on a second set
- 6 = Order column groups by Kruskal-Wallis statistic from GSTA
- 7 = Order row/column group definition file from ordinations
- 8 = Create a *-.gav* file from a *-.gdf* or subset of a *-.gdf*
- 9 = Create a groups based on Environmental Representativeness

The group definition file (*-.gdf*) lists the groups and the objects within them-

GROUP:	1	4 MEMBER/S			
ROW	1 (1)	ROW	6 (6)	ROW	4 (4)
ROW	10 (10)				
GROUP:	2	2 MEMBER/S			
ROW	2 (2)	ROW	9 (9)		
GROUP:	3	4 MEMBER/S			
ROW	3 (3)	ROW	8 (8)	ROW	5 (5)
ROW	7 (7)				

Three groups have been defined above. Group 1 contains four members: objects 1, 6, 4 and 10. Group 2 contain 2 members: objects 2 and 9. Group 3 has four members: objects 3, 8, 5 and 7.

In addition to the 'standard' **PATN** format above, GDEF can read and write a free format version. The purpose of this file is to enable simple manipulation of the *order* or *number* of groups and/or the order of objects in the groups using any text editor. The format of this version is-

```

Number of groups
First group Number      Number of objects in First group
Sequence numbers of the objects in the group ....
Second group Number    Number of objects in the second group
Sequence numbers of the objects in the group ....

.....

Last group number      Number of objects in the last group
Sequence numbers of the objects in the group ....

```

A free-format example using three groups may look like this -

```

3
1 3
1 2 3
2 2
5 4
3 5
10 9 8 7 6 5

```

The format is similar to the 'standard' *-.gdf* file except that the number of groups in the file is the first value in the file and no labels are used, only sequence numbers. The only limitation in the format is that a new group must start a new record or line in the file.

In addition to free format group definition files, GDEF provides for interactive definition of file from the terminal either by group or by object. This is a useful method for accepting groups that have been defined externally to PATN.

Group Affinity Vector

An alternative *group defining* format is supported for input and output in a number of areas of **PATN**. It is referred to as a group affinity vector because it details the affinity of each object to its group. The group definition file defines groups by group membership while the group affinity vector defines objects by group membership. For 10 objects and 4 groups, an example of a group affinity vector could look something like-

```

2 1 2 2 3 1 4 2 3 3

```

The first number defined the group number of the first objects, the second number is the group of the second object and so on. The example informs us that 'objects 2 and 6 are in group 1; objects 1, 3, 4 and 8 are in group 2; objects 5, 9 and 10 are in group 3 and object 7 is in group 4'.

While the group affinity vector provides a *concise* format for defining group membership, it is not used in **PATN** to the same extent as the group-definition format. The reason is simply that the group definition format permits a far greater degree of flexibility. For example, an object can be in no groups. It can even be in two or more groups! While this seems a little obtuse, it isn't. The pain of **PATN** is also its advantage; it's flexible! For example, I recently had a need to use MERG option 6 with an unusual group definition where some uncertain species were in two groups while others were deemed as just too difficult to classify at all, and were omitted.

GDEF permits, as far as possible, input, output and translation between both forms. Given the limitations of the group affinity vector, this is not always possible.

Group Comparisons

The group comparison file (-*gcm*) contains a tabulation of the fusions subsequent to the level of group definitions. The rows in this file are organised in *pairs*. An example, if three groups were defined, subsequent fusion may have been:

```
1. group 1 with group 2 and then
2. group 1 and 2 with group 3.
```

This information is useful if it is necessary to compare the combination of groups 1 and 2 with group 3 using say GSTA. The group comparison file is based on the *binary fusions* and is coded in integer free format. For the above example, the comparisons would appear something like -

```
1 1
1 2
2 1 2
1 3
```

The first value on each line refers to the *number of groups* that follow. The succeeding values are the group numbers (from 1->NRG). For example, the table above is read as -

```
Line 1: First group has one member, it is group 1.
Line 2: Second group has one member, it is group 2
```

This defines the *first fusion* on the basis of the defined number of groups. The second fusion would be defined similarly -

```
Line 3: The first group has two members, they are groups 1 and 2.
Line 4: The second group has one member, it is group 3.
```

GDEF may be run as many times as desired, producing various group definitions and comparisons. The current -*gdf* file is assumed to be definitive.

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OPTIONS

Option 1 FUSE -> Group Definition/Comparisons

This option is used to define a set of groups from FUSE, the hierarchical agglomerative module. The hierarchy defined by FUSE does not define a single set of groups; it defines from 1 to N groups. The divisive clustering methods on the other hand, are rarely run to the point where the base of the hierarchy represents individuals, therefore groups are automatically defined by the use of either of the methods MDIV or PDIV.

Determination of a reasonable number of groups is largely subjective. It depends on both the nature (structure) of the data and the requirements of the user for the level of aggregation to include in the summary. Experience suggests that the average person finds difficulty in accommodating more than about 8 groups at once. Therefore, to some extent, communication is independent of data structure. If anyone asks me "how many groups should they select, I say 8! Or possibly $\sqrt{N}$ (I enjoy the reaction!).

The dendrogram (DEND) provides some objective input into the determination of 'natural' groups. Such groups are identified by being compact and well separated from each other. On the dendrogram such groups will appear as having many objects fusing early (small 'distances') then subsequent fusions take place at a significantly higher level. The **signature** for well defined groups is therefore a **gap** in the hierarchy; a period in the fusion 'history' where no fusions take place. If a (vertical) line is drawn at this level parallel to the base, groups are automatically defined. This technique will only work with the 'space-conserving' options in FUSE. Dilation methods, while ALWAYS appearing to give 'well defined' groups, on closer examination, are often artefacts of the technique.

There have been a number of attempts to define an **optimum** number of groups. They are either non-robust or dependent on statistical aspects that the data usually doesn't conform to. The best 'rule of thumb' is to seek a break in the hierarchy at around a reasonable number of groups, given knowledge of the data and requirements for summary and communication. The **default** is the square root of the number of objects. This is a good starting point. It implies that *information* increases with the square root of the number of objects, a reasonable assumption from experience.

Selection of number of groups may be assisted from other **PATN** modules. As noted in the USERS GUIDE, no one analysis technique should be used in isolation. In this regard, it is practical to have used, with any analysis -

1. the dendrogram (DEND)
2. NNB and BOND and
3. an ordination technique (eg. SSH).

The first has been discussed. BOND is a network algorithm that results in a set of group definitions, as determined from the 'bonding strengths'. While usually under-estimating the number of groups, it will give some indication of a lower bound. Similarly with the ordination methods. If 'natural clusters' exist, a good ordination method will make them obvious.

The second aspect of this option is the determination, from the same hierarchy, the fusions subsequent to the level at which the groups are defined. Selecting a different number of groups will define a unique set of subsequent 'comparisons'. This file is produced mainly for use with GSTA (group statistics), where 'groups of groups' may be compared. For example, if three groups were selected using this option, the next fusion may be group 1 with group 2, then finally, group 1 and 2 (together) with group 3. Using the output file *-cmp*, GSTA can then compare groups 1 and 2, taken together with group 3.

Option 2 Terminal/File -> Group Definitions

This option provides two different ways of entering a set of group definitions and two different sources for the input. This option is of most use when such definitions have been created outside PATN. It can however be used to re-read a set of definitions that have been output from GDEF and been subsequently modified using an editor. The definitions can be read in either of the formats:

1. Group by object or (group definition)
2. Object by group (group affinity vector).

The former is the *standard* way that **PATN** defines groups and simply refers to the fact that the groups are defined one at a time, by the objects they contain. For example:

Group 1 contains 5 objects and they are

The alternate method is a set of N numbers (=total number of objects) where the first value refers to the group number to which object 1 belongs, the second number to the group to which object 2 belongs, and so on. For example:

2 4 1 1 3

implies that 'object 1 is in group 2, object 2 is in group 4, objects 3 and 4 are in group 1, and object 5 is in group 3'. This format of input then amounts to a translation of a group affinity vector to a group definition format!

Either of the two input structures can be entered at the terminal or read from a file. If terminal entry is selected, **PATN**, depending on the form chosen, will prompt either for the membership of each group in turn, or the group to which each object belongs. If *-gdf*-style format is selected as input from a FILE, GDEF provides two options-

1. read a free format file into standard form or
2. read a reclassification file and apply it to the default *.gdf* file.

Reclassification

Option 1 is the mirror of GDEF option 4 (see below) while option 2 permits the reclassification of an existing standard group definition by a set of NRG numbers (NRG = number of groups defined). The reclassification file is expected to be in INTEGER free format. Take the example of a file-

1 2 1 3 3 4

GDEF would interpret this as:

'merge groups '1' and '3' into new group '1'. Leave group '2' as is.
Merge groups '4' and '5' into group '3' and rename group '6' as group
'4'.

The group numbers must be whole INTEGERS and go successively from '1' to 'X' where 'X' is the last group number (no missing numbers please).

The output file (-*.gdf*) is always in standard (fixed) format.

Option 3 Terminal/File -> Group Comparisons

This provides a rapid means of defining a set of group comparisons that can be used in GSTA. A set of group definitions must already exist in a *-.gdf* file before this options is selected. Any number and permutation of groups in each comparison may be used. As with all other 'fusions' and 'splits' in PATN, each comparison is two way (binary or dichotomous). Multiple comparisons must be done pair-wise.

This option first requests the number of comparisons to be made. Then for each comparison, requests the number of sub-groups in the first (super) group and the group numbers, then the same information for the second group. For example, the input would appear as:

```
Number of comparisons
First comparison:
Number of sub-groups in FIRST group
Sub-group numbers .....
Number of sub-groups in SECOND group
Sub-group numbers .....
Second comparison:
Number of sub-groups in FIRST group
Sub-group numbers .....
Number of sub-groups in SECOND group
Sub-group numbers .....
...and so on.
```

Option 4 Group Definitions -> Free Format

This option has been designed to facilitate the editing of the group definition file for subsequent re-entry to GDEF. It can however be used to create a *group affinity vector* (see section above), a form of group definition that may be useful when interfacing to other programs and packages.

In both cases, the output from this option is a FORTRAN integer free format file. While this format permits easy manipulation of the order and number of groups and the re-ordering of objects, only the group by member form of output will retain the order of appearance of the groups and the order of the objects within the groups. For example, the group by object form -

```
1 3
1 2      1
2 3
4 5      6
3 4
10 9     8      7
```


includes the information governing the *order of groups* as well as the internal *order of objects* within the groups. By comparison the object by group form for 10 objects -

2 2 1 3 3 4 4 1 2 2

gives no implicit ordering either of the groups, or the objects within the groups. All that it states is which object belongs to which group. For example, objects 1 and 2 are in group 2, objects 3 and 8 are in group 1, and so on. In summary, use form 1 (group by object) when the internal or external ordering is important.

Option 5 Expand Group Definitions

This option allows for the *expansion* of a group definition that contains groups of groups. This occurs when FUSE is used to agglomerate groups from say ALOC. ALOC may be used to produce more groups than will be required for final summary. To enable a more meaningful fusion of groups to occur, FUSE is used on the inter-group association matrix from ALOC (-g.aso). As with any FUSE run, GDEF is then used to partition the hierarchy at the selected level, producing a set of group definitions. In this particular instance however, the objects are groups. The basic problem becomes -

"how do I produce a final group definition that contains objects and not sub-groups ?"

This is what option 5 does! Begin by setting the data file to the group data set (as it should be if you have just run FUSE). For example, if 1000 objects were originally submitted to ALOC for classification and 20 groups resulted, and these 20 groups were fused to 10 by the use of FUSE then the parameters on entry to this option would be -

1. the group data file name (-g.*)
2. 20 Objects and
3. 10 Row groups

This option requires-

1. How many objects there were at the beginning (1000 in this case). The number of original groups (from say ALOC) is known because it is assumed to be equal to the number of objects currently defined (20 in the example).
2. The name of the label file for the objects and
3. The name of the group definition file that contains the sub-groups

The result is yet another group definition file (it would be wise not to overwrite the previous two) that, in the case of the example above, would show 10 groups and 1000 objects. It is usually not necessary or wise to create a label file corresponding to the groups, just use the object ROW labels for the FUSE of groups and subsequent GDEF. Complicated isn't it!

Option 6 Produce column .gdf ordered by Kruskal-Wallis statistics

This option produces a *column* group definition file *ordered* by *Kruskal-Wallis* statistical values from GSTA. The group definition will contain a single attribute group with attributes ordered by *ascending* Kruskal-Wallis statistic values. For GSTA to have generated these values, a set of groups referring to the objects must exist. The attributes used in GSTA may either be intrinsic (used in the analysis of objects) or extrinsic (not used). The larger the Kruskal-Wallis statistic, the more significant is the contribution of an attribute to the discrimination of object groups.

The group definition file produced can be used for generating a contribution order of the attributes in subsequent evaluation or tabulation modules such as TWAY.

Option 7 Produce .gdf ordered by row/column ordination

This option is similar to the previous one. It produces a group definition file containing a single group in which rows or columns are ordered by *scores on the first axis of an ordination* (SSH, PCA, RA, DCA, or SERE). A sub option provides a choice for selecting a group definition file from a row or column ordination.

The rows and/or columns in the data file may be ordered by (first axis) ordination scores in a two-way table (TWAY), providing a simple method of displaying the data matrix with gradient information superimposed.

Option 8 Subset classification -> .gav

This option turns a .gdf file into a .gav file. It can be used simply for translation **or** for the situation where MASK has been used to create a sub-set of objects that has subsequently been classified. All objects (rows) that were omitted in the masking form *a special group* in the resulting *group-affinity vector*.

MASK will automatically assume that if the requested *number of objects in the sub-set* is the same as the *total number of objects* (=N) then, a simple translation of the .gdf-type format file to a .gav-type format file is assumed. If this is not the case, then a slightly more complicated procedure is assumed. The option is run on the *master* or original data file; as it was before masking. At this time, LABN option 6 has to be performed to create an index file (-.irm) of the subset into the master file (see LABN). The reason for this is that the only effective way of matching objects from one set to another is by using the row labels. LABN option 6 is used when PRAM is pointing to the master set. The sub-set labels are read and the index number of the sub-set into the master set are written to the -.irm file.

This option requests the number of objects in the sub-set (for reading the index numbers from LABN) and the number of groups in the sub-set classification (for reading the group definition file). A standard group-affinity vector is output where the *omitted objects are classified as group zero*. This file is usually of a format that can be readily used for subsequent processing outside **PATN**; for example, plotting classification on geographic coordinates.

Option 9 Environmental Representativeness

This option was designed to implement a *reserve-design* strategy called *environmental representativeness* or *ER*. This approach is based on how close objects (in this case sites) are to group centroids (Belbin, 1993). The scenario is simple. If an object is close to its group-centroid, it is assumed to be a *good representative* of the group to which it belongs. This is equivalent to saying that the mean, median or mode is the best *single-parameter* summary of a set of values. The best *two-parameter* summary may be the minimum and the maximum, the mean and standard deviation or the median and the inter-quartile range.

Environmental Representativeness in GDEF uses two cut-points to form three categories of distances-to-centroid and three corresponding representativeness classes:

1. Most representative (< cut point 1)
2. Intermediate (>cut point 1 < cut point 2)
3. Least representative (> cut point 2)

It is assumed that the classifications programs ALOC or ALOB have been run with the option to produce a tabulation of each object to its *closest group centroid* (tabulation option 2 or 3 in ALOB or ALOC). GDEF option 9 reads this tabulation and produces a group affinity vector (-gav) file that re-assigns objects to groups based on the object-centroid distances and two user-defined cut-points.

An Example

The context in which ER was developed was to identify reserve areas within a set of environmental domains or an environmental regionalisation. The data consists of a classification based on ALOB (ALOC could have been used) of geographic areas based on climatic, soil and topographic information. The goal was to locate *sub-areas* within the domains (groups) that could be considered in some way to be *most-representative* of those domains. By definition, domains or regionalizations cover 100% of the area of interest. Only rarely can 100% of a region be set aside for a reserve. Therefore, a method is required to locate *within each domain*, that area which could be most-representative. But how much? That is the \$64 question. ER does not answer 'how much', it simply identifies the most-representative 10% or whatever percentage of a domain the user selects. Percentages are based on the assumption that the objects are equal in area.

If the minimum and maximum distance for a group were .1 and .3 respectively, the two cut-points and corresponding groups may look like this:

Cut-points:	.12	.14	
Group Numbers	1	2	3

That is, objects with values less than .12 would be allocated to the most-representative group (group 1). Those between .12 and .14 would be classed as of intermediate representativeness and allocated to group 2. Those greater than .14 would go to group 3 while those not in the target group are ignored.

GDEF option 9 requests two percentages to form the cut-points forming a set of three ER classes (1, 2 and 3). GDEF will automatically adjust the cut-points to attain, as close as possible, the percentage of objects in the first two groups. It may not be possible to place the exact percentage of objects in each class due to tied dissimilarity values or a small number of objects in a group.

Input values for the cut-points are assumed to be percentages if the values are greater than 1.0. If the values are less than or equal to 1.0, they are assumed to be the cut-points and the percentages in each class are derived. Using cut-values rather than percentages will not generally produce the desired effect because each group will have a somewhat different distribution of object-centroid distances. In other words, if the most representative 10% of the area is required, then use 10% as the first cut-point. The second cut-point is useful to identify areas that may assist in the construction of a contiguous reserve network, or to provide alternative regions where the most-representative class cannot be reserved.

The output *-.gav* file is unusual by PATN standards in that, by default, the format used is (2I4). The first column of figures represents the original group of the N objects. The second column represents the ER class, either 1 (=most representative), 2 (=intermediate) and 3 (=least representative). Thus, the two values taken together represent both group and representativeness class.

The *-.gav* file may be conveniently read by COLR and a distribution of the *representativeness* of the objects of a particular group or all groups examined spatially. In this case, the classification groups can be plotted by using a *-.gav* style of input and the format (I4). Plotting the ER zones uses the same file and the format (4X,I4); this skips the group information in the first four columns and reads the ER classes. Therefore, to view all information produced by ER, two runs of COLR are required. Alternatively, the *-.gav* file produced by GDEF option 9 can be appended to a set of coordinates (x/y or longitude/ latitude) and input to a plotting or GIS program (eg IDRISI, ARCVIEW, SPANS or even S+ or EXCEL).

If there is reasonable spatial contiguity of the groups, the result of this option on any one group often leads to a contouring pattern around the stereotypic location. An additional file with the extension *-.er* contains a summary of the cut-points used, the percentage allocations achieved on the data and number of objects in each ER class in each group.

Demo

What follows is an example run using GDEF option 9 on the results of a classification using ALOC to produce three groups from a 20 object by 10 uniform random attribute data set created by RAND. User input in **bold-**

```

GDEF
-----GROUPING OPTIONS:
  1 = Group definitions and comparisons from fuse
  2 = Terminal/file -> -.gdf
  3 = This option not available
  4 = -.gdf -> free format file
  5 = Expand groups of groups to members
  6 = Column -.gdf ordered by Kruskal-Wallis statistics
  7 = Row/Col -.gdf ordered by ordination
  8 = Create -.gav from a SUBSET -.gdf
  9 = Environmental Representativeness (I,D:1)    ? : 9
.....Data requirements =          644 bytes
.....Label requirements =          160 bytes
  Input file name (A43,D:
    <random.alo
    Cut point number 1 (<1=ass-->1=%)    ? : 15
    Cut point number 2 (<1=ass-->1=%)    > : 20

-----<random.er>
03/16/95 09:35:11.60 gdef Random-variate data set generated from RAND

-----GROUP      :    1
.....Cut Points :          0.135100E+00    0.196275E+00
.....Number     :          1          1          1
.....Percentage  :    33.33          33.33          33.33

-----GROUP      :    2
.....Cut Points :          0.173100E+00    0.191900E+00
.....Number     :          1          1          5
.....Percentage  :    14.29          14.29          71.43

-----GROUP      :    3
.....Cut Points :          0.165100E+00    0.198020E+00
.....Number     :          1          3          6
.....Percentage  :    10.00          30.00          60.00

  PRINT a copy of <random.er> (Y/N,D:N) ? : N
  Output file name (A43,D:random.gav) ? : random.gav
.....Current (default) format is : (2I4)
-----Enter output file FORTRAN format:      (2i4)

-----<random.gav>
  1  2
  2  3
  3  3
  2  3
  3  2
  3  3
  2  3
  3  2
  2  3
  3  3
  3  3
  3  3
  2  1
  1  1
  3  3
  1  3
  3  1
  2  2

  PRINT a copy of <random.gav> (Y/N,D:N) ? : N

```

The user first runs GDEF and selects option 9. The most representative class is selected as those 15% of objects closest to their group centroid. The intermediate group are those 20% of objects next closest to the centroid. Output is firstly to the file *random.er* and shows the percentages achieved at the cut values (in this instance Gower metric was used in ALOC so this is what these values refer to). Note that in some cases, the selected percentages could only be approximated. For example, the first group only contained three objects. It has been split into 1 object that was most representative (33%), one intermediate (33%) and one other (33%). This is acceptable.

The user is then asked if the file *random.er* is to be printed (no here), and the name of the output group-affinity-vector file (*random.gav* by default) and the FORTRAN format (2I4 by default). The class assignment of the original groups from ALOC forms the first column. The second column is the ER class. Class 1 is most representative, class 2 intermediate and class 3, horrible!

GROUP 'STATISTICS':

GSTA

OUTLINE

GSTA's task is to determine how well a set of attributes discriminate between a set of (pre-defined) groups. These groups *must* be summarised by a group definition file (-*gdf*). This file may have been created by GDEF either after FUSE or by any other partitioning of objects performed either in **PATN** (ALOC) or externally.

The statistical method of *discriminant analysis* is designed for determining group discrimination by a set of attributes. The results however, are based on a number of assumptions. Probably the most restrictive assumption is that a *linear* combination of attributes can be used to define group difference. This is rarely true. GSTA generally makes no assumptions about the distributions but consequently, can infer less. In other words GSTA can point to a number of significant attributes but it cannot say that ROW 23 belongs to group 15 with a probability of 85% as discriminant analysis can.

For presence/absence data, an attribute by group table is produced. Values in the table are related to what can be termed *constancy* and *fidelity* (see below). These terms are based on the work of Goodall (1969). For more continuous data, 'box and whisker'-type plots are produced that show the distribution of continuous attributes across the defined groups. In the latter case, constancy and fidelity are swapped for the use of the Kruskal-Wallis (non-parametric) statistic.

An alternative approach that could justify pursuit are the so called Classification and Regression Tree or Decision tree approaches typified by CART (Breiman *et al.*, 1984) and more recently Knowledgeseeker (Angoss Software, 1994). These techniques operate on a divisive monothetic (see PDIV) basis to split the objects into more homogeneous daughter groups on the basis of either intrinsic or extrinsic attributes. These techniques can be thought of as a type of regression where the variable to be predicted is the group to which each object has been assigned. A decision tree that accounts for most of the grouping is built by successive splits. These automated methods are in their infancy. In many cases, a more parsimonious explanation of the groups can be achieved by examining the output from GSTA.

Intrinsics & Extrinsics

GSTA is not limited to using attributes that were used in producing a set of groups. It is often the case that one set of attributes are *intrinsic* or used in the analysis procedure (eg.: ASO, FUSE, GDEF, SSH, MST etc.), while another set, termed *extrinsic*, are used to assist in the interpretation of the results. Intrinsic attributes may (eventually) be used to predict one or more extrinsic attributes, or vice-versa. Another possibility is that a set of objects are partitioned on different suites of attributes and the significance of components in each set is required. In any case, it is illuminating to subject any extrinsic attributes to GSTA for analysis.

As a hobby, I often request the user to partition objects into groups *before using PATN*. Usually, the person who collected the data has a good notion about what is causing the variation and how attributes are varying across objects. A group definition is quickly coded using GDEF option 2 and that, along with the data set is input to GSTA. This should determine how the groups were created; what attributes were significant in that persons classification. It is often useful, and interesting, to compare these results with the subsequent analysis.

Nominal & Ordinal-Ratio Attributes

The two options in GSTA relate to the type of attributes in the data matrix. Option 2 is designed primarily for use with presence/absence (binary) data. Option 1 is for all other data types ('continuous' if you like). Each option addresses different interests that are associated with the two different types of data. With continuous data, it is useful to have a display that shows the distribution of attribute values across groups. With binary data, the interest is in which attributes turn up faithfully in what groups.

NOTE: if a group by attribute means file is required, the first option (continuous attributes) must be used. If the input data is **mixed** in the sense that both types of attributes are present, MASK could be used to form separate data sets to be independently analysed by GSTA or both GSTA options could be used on the same data set (rename the first .gst file to something else). If the data being used by GSTA is not binary, the second option has a further sub-option to *interpret* the data as if it were binary (values greater than zero are given the value '1').

The Kruskal-Wallis statistic is based on average rank of each attribute within each of the NRG groups. It is a non-parametric equivalent to the f-ratio and should be relatively robust to anything that is thrown at it. The probability value should be treated with some caution as it is based on an asymptotic approach to the χ^2 distribution. The probability will increase in reliability in proportion to the number of objects in the smallest group.

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OPTIONS

Option 1 Continuous Attributes

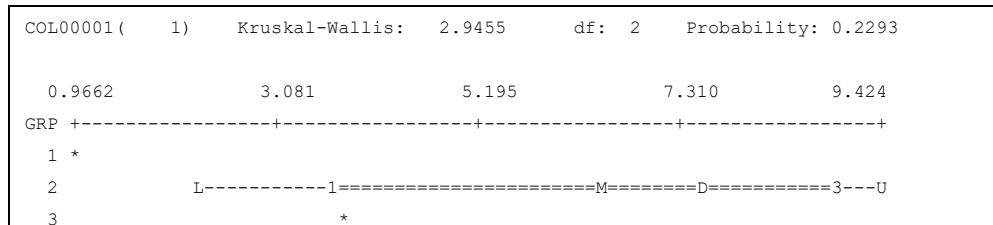
Four files are created with this option -

1. `-.gst`: a tabular form of the group by attribute distributions,
2. a numerical equivalent to (1) as `(-.sta)`,
3. a **PATN** group by attribute *means or medians* data file and
4. row and column labels and a parameter file for (2) `(-.clb, -.rlb)`.

The `-.gst` file contains a set of tables giving a visual estimation of the distribution of each attribute across all groups. Six parameters are determined for each attribute within each group:

1. Minimum (coded as 'L' on output)
2. 1st Quartile (" " '1' " ")
3. Median (" " 'D' " ")
4. Mean (" " 'M' " ")
5. 3rd Quartile (" " '3' " ")
6. Maximum (" " 'U' " ")

Output is in form of 'box and whisker plots'; one for each attribute across all groups. The values in the table above are coded onto a line that is used to represent the distribution of the attribute in a single group. The range between the 1st and third quartiles is coded using '=' while the values less than the 1st quartile and greater than the third quartile are coded using '-'. If more than one of the 6 parameters are located at the same character position on the graph, the location is coded as a '*'. For example a typical table for one attribute may appear as-



The numeric values used to generate these tables are written to the file `-.sta` in the form-

Group	Minimum	1st-Quartile	Median	Mean	3rd-Quartile	Maximum
----	-----	-----	-----	----	-----	-----
COL00001(1):						
1	0.9662	0.9662	0.9662	0.9662	0.9662	0.9662
2	2.291	3.689	7.429	6.463	8.875	9.424
3	3.780	3.780	3.780	3.780	3.780	3.780
COL00002(2):						
1	8.340	8.340	8.340	8.340	8.340	8.340
2	1.951	2.220	3.350	3.839	5.631	6.356
3	5.986	5.986	5.986	5.986	5.986	5.986
COL00003(3):						
1	9.477	9.477	9.477	9.477	9.477	9.477
2	2.426	2.752	5.996	5.972	9.017	9.822
3	3.254	3.254	3.254	3.254	3.254	3.254
.....						
.....						
COL00003(10):						

The second file (`-g.dat`), is a standard PATN data file (unformatted) containing NRG groups by M attribute medians or means. A corresponding set of label files (`-g.rlb`, `-g.clb`) and a parameter file (`-g.prm`) are also created. This data set may be activated as the default by using PRAM option 10.

Option 2 Nominal Data

For presence/absence data, distributional characteristics hold less interest. This option is oriented toward providing information on *constancy* and *fidelity*. Constancy is the proportion of objects that the attribute is present in. For example, if there are ten sites (objects) in a group and a species (attribute) occurs at eight sites, it's constancy is 80%.

Fidelity is the ability of an attribute to predict a group. If an attribute occurs only in a single group, it's presence indicates the particular group and it is said to have high fidelity. Low fidelity is where an attribute occurs in all groups; finding it gives no indication of which group it came from.

The table below is an example using ten attributes (labelled COL00001 to COL00010) and three groups. Group one has one member, group has four and group three has five. A value of 100 indicates that the attribute occurs in all objects within the group.

Entries are Constancy (% of objects in each group having the presence of each attribute).				
Group number :	1	2	3	
Number in group :	1	4	5	
COL00001(	1)	0	75	40
COL00002(	2)	100	0	20
COL00003(	3)	100	25	40
COL00004(	4)	0	0	80
COL00005(	5)	0	75	40
COL00006(	6)	0	50	20
COL00007(	7)	100	0	40
COL00008(	8)	0	50	20
COL00009(	9)	100	50	0
COL00010(	10)	100	0	20

The ordering of attributes in the table is the same as in the data. Dendrogram ordering of attributes may be useful. This may be achieved by using MASK to re-order the attributes with the *-.irm* file from DEND from a classification of the attributes. This would have to be done prior to GSTA. In other words, First, classify the objects as required. Then transpose the data and classify the attributes (consider using TWO-STEP in ASO for this). Run DEND to get the dendrogram as usual. This will also produce a file called *-.irm* which contains integers giving the *ordering* of items (in this case the attributes) down the dendrogram (See DEND). Using PRAM, return the objects by attributes data set and run MASK to re-order the attributes, producing a new data file. You will need to *copy* the object group definition file to the same name as this new data file. Then run GSTA option 2 and the order of listing the attributes will be as listed on the dendrogram.

Filtering Attributes

Not all attributes will be effective in discriminating between groups in terms of constancy and fidelity. This option can be used to eliminate attributes from the resulting table that do not have a constancy value for *any* group above a user-specified threshold.

For example, if a constancy value of '50' was supplied for this option, only those attributes which occurred in at least 50% of the objects in at least one group would be listed. All other attributes will not be tabulated. Legal value are 0-100. The larger the constancy, the fewer attributes will be tabulated.

Filtering Constancy Values

Once an attribute passes the threshold on constancy (above), it will be included in the tabulations. It is possible however to filter out all constancy values below a user-specified threshold by replacing numeric percentages with blanks. This option simply reduces the clutter in a table containing many attributes and highlights the significant associations between attributes and groups.

Permissible values here are between 0-100. If the attribute admission value (previous option) was 80 and this option were 50 (%), only constancy values of 50 or greater would be visible on those attributes that had at least one group with a constancy greater than 80. If these options were applied against the previous table, the following table would result:

Entries are Constancy (% of objects in each group having the presence of each attribute).				
Attributes with Constancy > 80 included				
Constancy values < 50 eliminated				
Group number :	1	2	3	
Number in group :	1	4	5	
COL00001(	1)		75	
COL00002(	2)	100		
COL00003(	3)	100		
COL00004(	4)		80	
COL00007(	7)	100		
COL00009(	9)	100	50	
COL00010(	10)	100		

Transforming data

In some circumstances, it may be useful to use this second option on continuous-type data that has been transformed to presence/absence. GSTA provides an option to TREAT the data as though it had been so transformed. *It does not alter any data values as maintained in your default data file.* The transformation is to set any value to '1' if it is greater than zero.

SORTED GSTA: SRGS

OUTLINE

This new module reads a standard GSTA-option 1 output file (*-.gst*) containing 'box and whisker'-type tables and sorts the file in order of decreasing Kruskal-Wallis statistic. The output file is named *-.gss*.

The purpose of SRGS is to enable fast re-arrangement of the GSTA output from the continuous GSTA option 1. The sorting of the tables in the file is determined by how well each attribute discriminates between the pre-defined NRG groups from some type of classification. The Kruskal-Wallice statistic was chosen for use with GSTA because it is a non-parametric equivalent to the f-ratio. This provides for a greater degree of robustness of the significance of attributes in group discrimination.

The output *-.gss* file is identical in format to the original *-.gst* file except that the order of the tables is by decreasing Kruskal-Wallice statistics. It is not however wise to rename *-.gss* to *-.gst* if the use of other options of PATN will *assume* the standard sequence-number ordered tables (for example, GDEF option 6).

REFERENCES

Kruskal, W.H. and Wallis, W.A (1952). Use of ranks in one-criterion variance analysis. Journal of the American Statistical Association, 47:583-621. Addendum (1953) 48: 907-911.

MERGING INFORMATION

TO DATA:

MERG

OUTLINE

This module performs a series of *left-right* and *up-down* merges of a variety of data items/structures. The object is generally to combine various types of data to be able to analyse results in a more efficient manner.

A familiar situation is the Left- Right merging of two data files. In the context of **PATN** the right-hand-side (RHS) file may correspond to additional data that had been collected, a set of extrinsic attributes (attributes that were not used in the analysis) or results from previous analysis. **PATN** can make good use of MERGED files. A few examples include -

1. GSTA option 1 (to a lesser extent 2)
2. SCAT all options but specially z-values,
3. TWAY with extrinsic attributes,
4. PCC with extrinsic data.

When merging data, wither left-right or up-down, the files need not contain a matched set of objects. More on this later. When merging ordination to data, objects must match. MERG can also be useful for reconstruction of files after using MASK to stratify or sub-set the data.

Default Data

MERG assumes that the LHS or TOP file is nominated by the current parameters. Information relevant to the RHS or bottom file is requested. Column labels for the RHS/bottom file may either be available in another file, entered from the terminal or automatically generated given a three letter base. A new file is always created, unless you decide to overwrite one of the input files. The latter is *not recommended*. Something may go wrong! A new parameter and label files are always created. The parameters are optionally updated to reflect the merge.

Under no circumstances should the output file have the same name as the input file when no updating is required. If this occurs -

1. *PATN.PRM* will point at the new file anyway (latest cycle is always accessed)
2. the latest cycle of the data file will not match the parameters or
3. the latest cycle of the row and column label files will not match the parameters.

REFERENCES

none

OPTIONS

Option 1 Merge Group Numbers

If the group number of each object is appended as an extra column, it may be used to superimpose classification groups onto a scatter plot (SCAT) based on geographic or ordination coordinates (PCA, SSH or SERE). In this way, the relationship between the classification groups and spatial relationship can be more clearly observed.

For this option, MERG will assume the existence of a group definition file and request its name. A single column containing the number of the group that each of the rows in is appended to the right-hand side of the file currently nominated by the parameter file.

This option is particularly useful when exporting results of PATN to statistical packages such as Splus (see DATN option 2). In this case, you may find it best to append group numbers (nominal attributes) to the right-hand side of an 'extrinsic' data file. For example, classify sites on the basis of biotic (intrinsic) data and then interpret the resulting groups in the terms of environmental (extrinsic) data. To do this, have both data sets known to **PATN** (as *-.dat/.prm.rlb/.clb* files), classify on intrinsics, then point (PRAM) to the extrinsic set and use this MERG option to append the intrinsic *.gdf/.gav* file. **Export for statistical processing could equally apply to MERG options 4,6, and 8.**

Option 2 Merge Classification Ordering

A group definition file can be used to provide an ordering of the rows of the data. Rows may then be sorted (see SAMP) to re-order the data so that it corresponds to the classification order (dendrogram in the case of FUSE). This could be useful to examine in more detail the differences between attributes within and between the defined groups.

A single column containing values corresponding to the order that each row was listed in the group definition file, is appended to the right-hand side of the file currently nominated by the parameter file.

Option 3 Merge a Data File

The merging of another data file may be the type of operation required during pre-processing. If a number of separate files exist corresponding to different samples on the same set of attributes, or one or more additional set of attributes to a common set of objects, MERG can fuse them together. To merge additional *rows*, use MERG option 9.

Mismatched ROWS

MERG will permit the left and right files to have a *different* number and order of columns. This is controlled by a sub-option asking if the number and order of the columns *perfectly matches* on the left and right files. If they do, then the operation is straight forward. If not, MERG needs to know the number of columns in the right file and the *name* of an *-.icm* file created by LABN option 6 on the columns. This LABN option is performed to create a set of indices of the right column labels with the current (top) column label file (as specified in *PATN.PRM*). The *-.icm* index file produced by LABN matches each right-column-label to the equivalent left-column-label. Indices of '0' mean that the right-column-label has no equivalent among the left-column-labels. MERG matches the position of the right-data using these indices; '0' indices become *extra* columns of the new output file.

Undefined data is filled with a user-supplied value. This value should be zero if the measurements were made and the values are really zero. A missing value should be used where the associated data was not measured at all.

MERG will request the name, format and number of columns of the right-hand-side file and will merge this to the current data file as nominated by the parameter file. MERG will also automatically *compare* all of the OLD column labels with the NEW column labels and if any matches are found, a set of options are displayed -

1. Skip the matching operation (leave duplicates),
2. leave the detected duplicate columns as is,
3. let column x = column x AND (+) column y
4. let column x = column x OR column y
5. delete column x
6. delete column y

Options 1,2,5 and 6 are straight forward. Options 3 and 4 are reasonable alternatives when the data file contains presence/ absence data (1/0) or quantitative data. The logical OR operation works like this-

If either column x OR column y contain a '1' then the new column x will contain a '1'.

The logical AND operation is a simple summation of column x and column y and is an appropriate operation where both columns contain correct and quantitative data.

Option 4 Merge an Ordination File

Ordination files are in ASCII format and contain headers. This option enables the ordination scores to be added, as additional columns, to the current data file. The LHS file may be the data as used in the analysis (intrinsic) or a file containing extrinsic data. The sequence of objects should of course match. The purpose for merging may be either to store the ordination scores in a more meaningful way, or to prepare a file for post-processing. If the latter, the ordination scores are available as base points when plotting z-values using the SCAT command. The z-values may correspond to group numbers or other intrinsic or extrinsic values.

For this option, MERG requests the name of the ordination file and the number of ordination vectors to be merged to the data file.

Option 5 Merge Row/Column Frequencies

The SCAN command produces a file containing the row and column frequencies. MERG can read this file and append an additional column representing the row frequencies and an additional row at the bottom of the file, representing the column frequencies. These frequencies are then available for post-processing in conjunction with some form of spatial information (x-y or ordination) and other intrinsic and extrinsic attributes. For example, site diversity could be plotted as a function of location.

This option will assume the existence of a SCAN file (*-.scn*) as produced by the current data file. MERG will request the name of the input SCAN file and append an additional column to the RHS representing the row frequencies and an additional row to the bottom of the file, representing the column frequencies.

Option 6 Merge Column Groups

This option adds one column to the default data file for each *column group* defined. Column group definitions are read from a standard group definition file (*-.gdf*). Each row is then scanned for the attributes in each of the column groups, one group at a time. For example if group 1 contains columns 2, 4, 6 and 8, these columns are scanned. The values placed in the new row 1, group 1 column depend on the values in row 1 columns 2, 4, 6 and 8 and the options -

1. if *any* values > 0, use '1',
2. add the *number* > 0,
3. add the *sum* of values or
4. use the *proportion* > 0.

An example: Consider a data matrix was 100 rows by 50 columns with 5 column groups, a non-zero entry in row 1, and column 5 (that belongs in column group 2) would produce (depending on the option) a non-zero value in column 52 of the output data matrix.

Why would you want to use this option? A typical analysis usually includes a classification of rows and columns. The rows may represent sites and the attributes the presence or absence of species. Classification of the species appears to show strong associations (say assemblages) and it is decided to attempt to predict the spatial distribution of these groups for all sites. What is required then is the distribution of richness for each species group.

Given the example above, five additional columns would be added to the data. The additional five columns could, depending on the sub-option, contain which species groups occurred at which sites (1), the number of species of each species group occurring at each site (2), the total number of species in each species group at each site (3) or the proportion of species in each group occurring at each site. For presence/ absence (1/0) data, sub-options 2 and 3 will produce identical results.

Optionally, the attribute-group data can be written (without the raw data) to a separate data file.

Option 7 Merge SUM of Selected Columns

This options was added to MERG as an interim method of creating a single new column as a function of existing column values. This option permits the selection, using the MASK style of coding, of which columns are to be used to form the sum. This is placed in the additional column on the RHS of the data file. For the example, a selection like columns

1 -5, 55 -100 200

could be used to sum the values in columns 1 through 5, 55 through 100 and 200 and place the result in a new column. If there were 200 columns in the example above, the column sum would be placed in column 201.

Option 8 Merge Group Number to Ordination

One of the most useful diagnostics is to plot the group numbers of the objects on their ordination base. There has been so many requests for this option, it has been implemented to save having to use a combination of option 4, then option 1 , then often masking the original data.

The new output file from this option is a **PATN data file**, not an ordination file! This facilitates the translation of ordination to data as well as opening up the possibility of merging other data to the RHS!

Option 9 Up/Down Data file

Older versions of **PATN** used only ASCII format data. Up-Down merges with this type of data could be performed simply by using an equivalent to the operating systems *append* command. With **PATN**'s new unformatted data format, this is no longer possible. Don't try it! Instead, use this option or option 3 as appropriate.

Mismatched COLUMNS

MERG will permit the up and down files to have a *different* number and order of columns. This is controlled by a sub-option asking if the number and order of the columns *perfectly matches* on the up and down files. If they do, then the operation is straight forward. If not, MERG needs to know the number of columns in the down file and the *name* of an *-.icm* file created by LABN option 6 on the columns. This LABN option is performed to create a set if indices of the down column labels with the current (top) column label file (as specified in *PATN.PRM*). The *-.icm* index file produced by LABN matches each down-column-label to the equivalent up-column-label. Indices of '0' mean that the down-column-label has no equivalent among the up-column-labels. MERG matches the position of the down-data using these indices; '0' indices become *extra* columns of the new output file.

Undefined data is filled with a user-supplied value. This value should be zero if the measurements were made and the values are really zero. A missing value should be used where the associated data was not measured at all.

TWO-WAY TABLES:

TWAY

OUTLINE

TWAY produces a two-way table using the default data file and two *-.gdf* files from separate classification of objects and attributes. The two-way table is a re-ordering and display of rows and columns in the data matrix in the same order as in the corresponding group definition files. These groups could have been produced by any of the classification techniques in PATN -

FUSE
MDIV
PDIV
BOND
GDEF (using Kruskal-Wallis statistic values or ordination scores)...

or even purposively. This type of table has been used historically as an aid to evaluating the classification of presence / absence data and appears useful for-

1. detecting data errors
2. finding mis-classifications
3. determining level of classification (how many groups)
4. producing a summary of analyses
5. detecting attributes responsible for groupings and
6. isolating noda (see below)

TWAY will however provide a two-way table of any data type, not only presence/absence. For the display of presence/absence data, only two symbols are needed, one to represent presence and one for absence. Standardisation of non-presence/absence data is available. This makes the symbols used in the table local to each column or the complete matrix. Any printable characters may be used to nominate the levels of data values.

If a group definition is based on FUSE, the order of objects or attributes in the resulting group definition file is in dendrogram order. TWAY writes the dendrogram row order, in a free format row mask file (*-.irm*) after receiving the name of the *-.gdf* file containing the **row** group definitions. This file may be used by MASK to produce a re-ordering of the rows within the data matrix itself. The re-ordered data may then be displayed using DATN.

You may abort TWAY, as any PATN program with a CTRL-z (UNIX: <CTRL>-d) sequence (hold down both the CTRL key and the z or d key) if the *-.irm* file or the two-way table itself is all you require.

In addition to the two-way table, TWAY produces a number of additional tables. The term **noda** is used to refer to a group of values in the two-way table defined by the intersection of a row and column group. For example, each one of the '.'s below represents a value in the data matrix. A group of the data has been highlighted by '*'. This corresponds to an intersection of a row group comprising ROWS 5,6 and 1 with a group of columns comprising 5, 4, 9 and 8:

	COLUMNS
ROWS	132654987.....
ROW 2	
ROW 4	
ROW 5	...****...
ROW 6	...****...
ROW 1	...****...
ROW 3	
ROW 4	
.	
.	
.	

The additional tables include, for each noda -

1. totals
2. proportions
3. average number of attributes/object group
4. average number of objects/attribute group
5. (observed-expected)/expected

The **totals** table is a summation for each noda, the number of presence's for presence/absence data or the sum for numerical data. The second table is produced by dividing the values in the first table by the number of values in the noda. The third and forth table are complementary. The third is the sum of the first table divided by the number of rows in each noda. The fourth is the same total divided by the number of columns in the noda. When data are sites and species, this corresponds to the average number of species per site and the average number of sites that a species group appears in.

The last table, is based on a *Chi-square* estimate of noda sums. For example, for row group i and column group j, the expected frequency for noda ij is-

$$\text{Sum row group } i * \text{Sum column group } j / \text{Grand sum}$$

Because the data may not be normally distributed, this table should only be used in a relative sense, not absolute.

REFERENCES

- Belbin, L., Faith, D.P. and Minchin, P.R. (1984). Some algorithms contained in the Numerical Taxonomy Package NTP. CSIRO Division of Water and Land Resources Technical Memorandum 84/23.
- Kruskal, W.H. and Wallis, W.A (1952). Use of ranks in one-criterion variance analysis. *Journal of the American Statistical Association*, 47:583-621.
Addendum (1953) 48: 907-911.

OPTIONS

Symbols

If a standard two-way table is being produced, the user must nominate a *set* of symbols for the table. In many cases, the data used for the table are of presence/absence type. In this case, only two symbols are required; one to represent the absences (zeros) and one for the presence's (ones). With other data, a display of other intermediate values, as separate symbols may be required.

If, for example, data values range from zero to four, five symbols will be used-

- 0 = first symbol
- 1 = second symbol
- 2 = third symbol
- 3 = fourth symbol
- 4 = fifth symbol.

For the above example you may like to use -

- 0 = (blank)
- 1 = .
- 2 = :
- 3 = *
- 4 = #

NOTE: TWAY always treats **ZEROS** as a special case. They are always uniquely allocated the *first* symbol. Consider an example where data values ranged from zero to ten with fractional values, and a five level partition was requested. Zero will be allocated to the first level and symbol. TWAY then splits all the remaining values into the remaining four levels. If the smallest positive value was 0.1, then TWAY would form equal categories with a range $(10-.1)/4=2.475$. The first class will then be 0.1 to 2.575.

Option 1 Row by Column Groups

This is the normal mode of operation of TWAY, where a re-ordering of rows and columns is achieved by way of row and column group definitions.

TWAY has two forms of standardisation -

1. by column or
2. by complete matrix.

With presence/absence data, all columns are in the same units, so it doesn't matter. If the data matrix is mixed however, and it is desired to equally weight each attribute (columns), then column standardisation should be selected. If the same symbols are used to represent the same values throughout the matrix, then matrix standardisation will be necessary for attributes on different scales or with different ranges.

Option 2 Row Groups by Row Groups

This option has been designed to perform a comparison of two different classifications (specifically partitions) of the same set of objects. In this situation, two group definition files are available and both refer to the *rows* in the current data matrix. With this option, the intersection or conjunction of the objects (rows) are delineated by the second user-defined character. For example, the position in the table where the object say 'site 01' as one of N rows corresponds to the column heading 'site 01', the second character will be found. This means that there will only be N such 'intersection' or 'coincidence' symbols in the table, one per row in the table. The first symbol entered will be used in all non-conjunction positions in the table.

The first *-.gdf* file will generate the row ordering of the objects in the two-way table while the second *-.gdf* file will provide the order of the columns in the table. The inter-group boundaries will be placed as specified by the two group-definition files. Two identical partitions of the same set of N objects should lead to a table with a diagonal of the user-specified 'coincidence-character'.

GROUPS BY ATTRIBUTES

CHI-SQUARE:

CHI2

OUTLINE

CHI2 is a program that generates a two-way contingency-table of the form NRG groups by M attributes where the table entries are the *sums* of the data values. CHI2 then calculates a standard χ^2 test on this table and presents the significance value and the associated degrees of freedom.

CHI2 assumes that the groups are defined by an additional column on the right-hand-side (RHS) of the current PATN data file. This is achieved by using MERG option 1 to append the group number (from a *-.gdf* file) to the RHS of the data file. This enables CHI2 to calculate a χ^2 value for multiple group columns. Groups must be contiguous and the right-most columns.

The multiple χ^2 approach was designed to evaluate the significance of various **levels** of grouping, for example, by cutting a dendrogram at different levels. CHI2 requires the sequence-number (from 1->M) of the *column* in the current PATN data file that contains the group number that each ROW belongs to. This column can be considered as a group-affinity-vector; values show the affinity of each object to the NRG groups (defined in the parameter file PATN.PRM). If more than one grouping attribute exists, these columns **must be** contiguous on the right-hand-side of the current PATN data file. That is, there can be NO data columns of the RHS of the grouping columns.

Use the negative of the first grouping column sequence number if there are more than one group columns on the RHS of the data. For example, a data set having 50 columns with the last 10 being various classifications, use -41 for the first parameter (1st column) and 50 for the second (last group column).

REFERENCES

none

OPTIONS

none

DISPLAYING GROUPS

BY COLOUR:

COLR

OUTLINE

COLR is available for PC: versions only. COLR is a simple routine for mapping classification results to a graphics monitor. If the classified objects have some spatial arrangement and extent, COLR may be used to graph a patch of colour for each object where the colour represents the group of each object. At this time PC-PATN cannot use the automatic colour algorithm as in Belbin *et al.* (1983). This is due to the current limitations of the graphics code being used and the colour palettes generally available. If a palette of 256 or more colours becomes a standard, automatic colouring would be feasible.

COLR picks up the title, number of rows and number of row groups from *PATN.PRM*. It then requires a definition of the groups, either the standard format (*.gdf*) file or a group affinity vector file (*-.gav*). Required also is an **ASCII** file that contains the coordinates of the each of the objects. The x and y *extent* of the objects is also requested. COLR assumes that all objects have the same geographic *shape* (some form of rectangle). Colour assignment, by group, is arbitrary. The colour assigned to each group of objects will depend on the graphics card selected for your system. If you exceed the number of colours permitted by your graphics card by having too many groups COLR will cycle the colours. This will generally be inconvenient as two groups will map to the same colour. Generally, around 5-10 colours should suffice.

COLR may be used for plotting ordination results. The only hassle is the format of the x-y coordinates. It is easier with headed files such as *.ssh* to copy them to another file and delete the header lines. The format for the x-y coordinates is then simply '(16x,2f8.4)'.

REFERENCES

Belbin, L., Marshall, C. and Faith, D.P. (1983). Representing relationships by the automatic assignment of colour. *Australian Computer J.* 15(4), 160-163.

OPTIONS

Graphics Card

There are currently ten different options for graphics cards-

4	= CGA	320 by 200 by	4 colours
5	= CGA	320 by 200 by	2 colours
6	= MONO	640 by 200 by	2 colours
13	= EGA	320 by 200 by	16 colours
14	= EGA	640 by 200 by	16 colours
15	= EGA	640 by 200 by	16 colours
16	= EGA	640 by 350 by	16 colours
17	= VGA	640 by 480 by	2 colours
18	= VGA	640 by 480 by	16 colours
19	= VGA	320 by 200 by	256 colours

The default is 18 (VGA 640 by 480 by 16 colours).

Group Definitions

COLR can pick up either a standard group definition file (*.gdf*) as option 1 or a group affinity vector-type file (*.gav*) as option 2. One of these files must be present, even if you desire to locate the objects and don't care about group colours. In this case, make up a simple group definition file using GDEF.

Box Sizes

This is the x and y spatial *extent* of the objects in the same units as the *location* of the objects. COLR assumes that all objects have the same shape; the same extent is applied to all objects. The measurement should be in decimal format. Convert degrees minutes and seconds for example, to decimal degrees.

If the geographic range of the objects is considerably greater than the extent of the objects, it won't matter if you scale the box-sizes up to produce a manageable display. For example, if sites range over two degrees of latitude and are only tenth hectare each, they may be represented by a single pixel on the graphics display. In this case, it is expedient to increase their size so as to cover say 5 by 5 pixels. A little experimentation may be required!

Coordinate Type

The file containing the coordinates of the objects may be any ASCII file. COLR will request the name and the FORTRAN format of the file. Note that free format by use of a '*' is permissible (see the chapter in the users guide on formatting).

COLR accepts three different coordinate formats and transforms all into an X-Y form-

1. latitude and longitude. This is considered to take the form '-Y' and 'X' (southing and easting in the southern hemisphere) so the latitude value is transformed on input to 90.0-latitude. Longitude already forms an 'X-coordinate'
2. -latitude and longitude. In this case, the data is assumed to be a 'Y' so the latitude values are not transformed. In this case, the input data is assumed to be of the form 'Y' & 'X'.
3. X and Y. The coordinates do not require transformation.

If your data is in some other form, it still may be able to be handled correctly by one of the three above and some judicious use of FORTRAN formats.

There is no facility for producing a hard-copy of the mapping at present. To get a copy of what is on the screen, I would suggest a HP-paintjet (colour inkjet) printer (around \$700) and a copy of the Pizazz Plus software (around \$200). Both are great value for money and can be used for a very wide range of applications.

Masking

COLR provides the ability, as many other **PATN** programs such as TRND, to select a sub-set of objects to process. The procedure used in COLR is the standard list (L) or MASK format. COLR gives the option of-

1. reading a list of objects to be used from the terminal (L-format),
2. reading a list of objects to be processed from a file (L-format) or
3. using all objects.

List format is summarised in the chapter on MASK but a small example should give you an idea of how to proceed-

```
1 -15 18 19 54 -88 e
```

says that the objects 1 through 15, 18, 19 and 54 through to 88 should be included in the processing.

Windowing

In addition to direct masking of particular objects to plot, COLR provides a simple rectangular masking window. If you select this option COLR will request-

1. the minimum x-coordinate to process,
2. the maximum x-coordinate to process,
3. the minimum y-coordinate to process and
4. the maximum y-coordinate to process.

Only objects having x-y coordinates that fall *within* this window are plotted at full scale on your monitor. This amounts to a basic zoom and pan operation. Supplying limits *OUTSIDE* the x-y limits of your data will simply be the same as not selecting the windowing option.

ANOSIM ANALYSIS:

ASIM

OUTLINE

The ANOSIM algorithm of Clark and Green (1988) was established as a means of determining if a series of replicate samples are more similar to each other than say to other samples. The method is essentially ANOVA in nature in that a comparison of 'within-group' and 'between-group' variation is determined. The differences from ANOVA are however twofold. First, a Monte-Carlo type randomisation is performed and second, the raw data is a symmetric association matrix between all pairs of what Clark and Green call 'replicates'.

In their paper, Clark and Green suggested that the test-statistic could use the comparison of average dissimilarity within a group (of replicates) as compared to average between group dissimilarity. This has been the approach used in PATN. The actual statistic that I have decided to use is on the F-ratio (which is unbounded):

$$\frac{\text{average between group dissimilarity}}{\text{average within group dissimilarity}}$$

An alternative test-statistic would be to use (between)/(between + within). This is equivalent to the Cramer value. The useful aspect of the Cramer statistic is that it is bounded between zero (no discrimination) to one (maximal group difference). I will stick to the F-type formulation until I have time to think more on the subject. Using the f-ratio, values >1 would suggest that 'groups' of some nature exist.

NOTE: RAND, unless forced to do otherwise generates 'random data'. This does not mean the data is 'structureless' by PATN's reckoning! PATN is very good at detecting very subtle patterns. If 'random' data is subjected to ASIM on groups generated by ASO and a good classification strategy (UPGMA or ALOC), it will probably tell you that no permutation will beat the original (classification). Of course, the poorer the pre-processing and classification strategy, the more likely is a random permutation to exceed the original.

The randomisation component is performed on the allocation of '*replicates*' to '*samples*'. In the generalised case used in ASIM, this means that objects are randomly allocated to a set of pre-defined groups. There are therefore two significant input files to ASIM, a standard association matrix (-.aso) and a standard group-definition-file (-.gdf). The latter is read into the equivalent of a group-affinity-vector and forms the basis of the randomisation. This would be equivalent in the paper of Clark and Green of creating a group-definition-file with each set of replicate (cores) defining a group.

In PATN, you may create the group-definition however you wish. It may be created from a 'standard' classification (eg.: FUSE, ALOC, PDIV...). In this case (if your choices for data coding, standardisation, association measure and classification option are 'wise'), it likely that the final test-statistic will << .05!

ASIM assumes that the input `-.aso` file is to be used 'as is'. Clark and Green suggested that the statistic could be made 'as non-parametric as possible' by ranking the association values. The basic test is already non-parametric. If you want to rank the association values, use TRNA option 12.

The output from ASIM is to the terminal. A running list contains:

1. the iteration (randomisation number),
2. the best test-statistic to-date,
3. the test-statistic for the 'real-grouping'
4. and the proportion of cases where (3) $\geq$ (4)

Ideally, the final value of (4) will be < 0.05 . That is, the proportion of cases that equalled or exceeded the statistic on the real data is less than 5%. This would imply that the 'actual' allocation of objects to groups is optimal in some sense.

In addition, ASIM provides some notion of the heterogeneity of each group. ASIM calculates and writes to the `-.Sam` file for the given (true or input) grouping, for each group:

1. the number of pairs of objects,
2. the maximum within-group association and
3. the average within-group association.

REFERENCES

Clark, K.R. & Green, R.H. (1988). Statistical design and analysis for a 'biological effects' study. Marine Ecology Progress Series, 46: 213-226.

OPTIONS

Number of Randomisation's

This is the number of times that ASIM is to permute the objects across the NRG pre-defined groups and calculate the test-statistic. The default is 100 but any value less than 99,999 will work...as long as you have time to spare.

Actually the algorithm used for the permuting of objects to groups and calculation of 'within-between' averages is very efficient. Try running 10 and then scale (linearly) up.

Random Number Seed

The usual initial seed for my random number generator in PATN. The same seed will always generate the same set of random numbers (machine independent). No particular values are more appropriate (than any other).

OUTPUT

A file *-.asm* containing a summary of the statistical results containing the best randomised statistic, the test-statistic for the original grouping and the proportion of randomisations where the base statistic was exceeded. In addition, a summary of group heterogeneity of the original input groups is listed by group.

PRINCIPAL AXIS

CORRELATION:

PCC

OUTLINE

PCC is essentially a multiple-linear regression program designed to see how well a set of attributes can be fitted to an ordination space. PCC assumes that the attributes to be used are in the columns of the current **PATN** data file while the ordination vectors are in any **PATN** ordination file (PCA, SSH, DCOR). The attributes may be intrinsic (used in the analysis) or extrinsic (used only in interpretation).

An example. A set of sites that have been measured for a range of species and a set of environmental attributes. A two-dimensional ordination from SSH of the sites based on the species data has been performed. How do a set of environmental attributes relate to the ordination? In ecological terms, the question is -

"is the species data responding in any systematic way to the environmental attributes?"

What PCC will do is to take each of the environmental attributes (or species for that matter) and find the location of the best fitted vector in the ordination space. You can eyeball a similar result in two-dimensions by using SCAT annotation option 4 where the environmental attributes are z-values. Two pieces of information result from PCC, the direction of best fit and the correlation with that direction. The correlation coefficient may be used as a rough indicator of the significance of each attribute. If PCC were limited to fitting attributes to single ordination axes, simple correlation using ASO option 3 would suffice. Trials of more general/non-linear methods appeared to be computationally demanding and somewhat problematical; producing degenerate solutions.

PATN uses a slightly unusual tabulation in the output file *-.pcc*. The file *-.pcc* is basically in **PATN** ordination format. It contains a copy of the original ordination that has been centred *plus*, as additional rows, each attribute that was fitted with PCC. In other words, the best linear fit of each attribute in the original supplied ordination space is given as the coordinates of the head of the vector in this space. The length of these vectors are arbitrarily standardised such that their lengths from the origin of the space are 1.0.

The correlation for each attribute is given as an additional column. In other words, if the original ordination was three-dimensional, then each attribute will be tabulated with four columns; the first three represent the coordinates of the best linear fit in the space and the fourth, the correlation in this direction between objects and the fitted attribute. All correlations will be positive as negative values represent a vector at 180 degrees from the those tabulated.

This form was selected because it enable the results to be neatly displayed by SCAT. All that is required is to alter the number of ROWS in the parameter file to be -

Number of Objects + number of attributes.

SCAT should be informed that there is one more dimension in the ordination. The correlations are then read as the additional dimension. SCAT may then be used to -

1. locate the objects in the ordination space
2. locate the best fit attributes in the ordination space AND
3. display the correlation of each attribute in the ordination space

(1) and (2) are displayed as usual, using SCAT annotation option 1 while (3) is displayed with annotation option 4. The objects will be read as having zero correlation while the attributes will always be greater than zero and hopefully close to 1.0. Note that because of the addition of the attributes (of unit length from the origin) into the ordination space, SCAT may not display the PCC file at the same scale as the original ordination, but it will still be drawn to scale if the 'dependent-scaling option' in SCAT is selected.

REFERENCES

Your reading it!

OPTIONS

none

MONTE-CARLO ATTRIBUTES AND ORDINATION: MCAO

OUTLINE

MCAO has been designed to automate the process of running Monte-Carlo randomisations in conjunction with PATN's PCC module. MCAO is a module that basically ties together and runs other PATN modules.

MCAO will assume that an ordination and PCC has already been run on the default data set. The module stores the correlations from this run and then reads the original data and permutes each attribute in the default set and re-runs PCC, one attribute at a time. Each time PCC is run, it is run on k-permutations of a single attribute. The process is repeated for each attribute in the default data set.

The user controls the number of permutations (default: 100), the name of the original *-pcc* and ordination files (and number of axes) and a random number seed. Output from MCAO is to a file *-.mca* that contains an ascending sort of the k-correlations (with the true correlation in the last position; one group for each attribute. In other words, for 10 attributes and 100 permutations, *-.mca* will contain 10 logical records each containing an independent ascending sort of 100 values with the 101st value the original PCC correlation.

REFERENCES

- Manly, B.F.J. (1991). *Randomization and Monte Carlo Methods in Biology*. Chapman & Hall, London, 281p.
- Mantel, N. (1967). The detection of disease clustering and a generalized regression approach. *Cancer Research*, **27**, 209-220.

OPTIONS

none

MONTE CARLO ORDINATION

DIMENSIONALITY:

MCSS

OUTLINE:

MCSS is an experimental program that uses permutations of the data matrix to examine differences between stress levels across different 'dimensionalities' in ordination. The *hypothesis* is that the difference between the true and permuted stress will be maximal for the 'true-dimensionality'. This is experimental and is released on the unsuspecting audience for evaluation and feedback.

MCSS is a front-end program that accepts from the user -

1. the type of association measure to be used,
2. the minimum number of dimensions for ordination using SSH
3. the maximum number of dimensions for SSH,
4. the number of permutations to use for each dimensionality,
5. a random number seed (for the permutations) and
6. all other parameters necessary for SSH.

MCSS generates input to SAMP to create a permutation of the original data to the output file called *mcss.clb*, *mcss.rlb*, *mcss.dat* and *mcss.prm*. This data set is then run through ASO and SSH using the user-defined parameters. MCSS then extracts the stress value and writes it to *-.mcs*. This process is repeated for the user-defined number of permutations.

Output from MCSS comprises a heading showing the dimension and the stress values for each permutation. MCSS does not currently produce the stress for the unpermuted data. This will need to be done either before or after the use of MCSS.

Warning: ordination is a particularly time-consuming algorithm in pattern analysis. SSH is no exception to this. So, I would advise that some caution is required if you are thinking of doing 1000 permutations of a 500 by 100 data set using 30 random starts in SSH between dimensionality 1 and 6! Suffice to say that this would be the type of exercise that is best left to running on your PC over the weekend!

PRINCIPAL AXIS ROTATION: PCR

OUTLINE

PCR implements a continua of rotation schemes for any multi-axis ordination configuration (SSH, PCA, or DCOR). The rotation methods provided are termed rotation to simple structure. A number of different criteria exist for the rotation. Those provided by PCR are-

Varimax rotation maximises the variance of the squared loadings on each axis or column by re-arranging them closer to -1, 0, or 1. This tends to be the most common form of rotation and greatly simplifies interpretation of the resulting vectors.

Quartimax rotation takes the alternate approach. It attempts to maximise the variance of each row vector. This method aims to load row (-1, 0 or 1) onto few PC vectors. This simplifies the interpretation of each row.

Equimax is a compromise between the Varimax and Quartimax approaches. By modifying the rotation parameter various effects can be obtained. At this time I have little experience with intermediate situations. I suggest varimax as a starting point.

NOTE: SSH does not rotate the final configuration to principal components (as KYST used to). This was to permit the use of PCR to provide a better lower dimensional solution. Principal components places the primary axis first. The position of all subsequent axes is restricted by the axes already positioned. In PCR, axes are positioned two at a time. This provides greater opportunity for squeezing more information into lower dimensions. Rotation of SSH using PCoA is an alternative.

Because PCR creates, in effect, a new set of attributes, these may readily be merged as additional columns to the DATA file (additional attributes). This is intended to be useful for subsequent analysis such as -

1. Scatter PLOTS of ordinations (SCAT)
2. Correlating ordinations with other attributes (PCC)
3. Determining row colour from ordinations (COLR)
4. Assessing the importance of ordination scores (GSTA)

The output file from PCR is in standard ordination form with the rows corresponding to the objects and the columns to the rotated vectors.

REFERENCES

Nie, N.H., Hull, C.H., Jenkins, J.G., Steinbrenner, K. and Brent, H. (1970). SPSS: statistical package for the social sciences. McGraw Hill.

OPTIONS

One floating point value is used to define the nature of the rotation criteria. Examples of the definition of some of the criteria are -

0.	=	quartimax (default)
1.	=	varimax
M/2	=	equimax

where M = Number of ordination vectors.

PROCRUSTES ROTATION:

PROC

OUTLINE

PROC is used to rotate one (orthogonal) ordination called here the *test* configuration or matrix to a *target* ordination using generalised Procrustean rotation. The name derives from a mythical Roman innkeeper who preferred to modify the clients (trim, legs...!) to fit the available beds rather than vice-versa! The ordination configurations may have been derived from any ordination method. PROC uses least squares to minimise the sum of the squared distances between the objects in the two configurations. Procedures used are -

1. rotation of the test matrix
2. translation of the test matrix and
3. dilation or contraction of the test matrix.

The procedure may be used to compare two ordinations of the same objects or to compare an ordination with an hypothesised or known configuration. An example of the latter would be to fit an ordination of a simulated data matrix to known sample positions on model gradients.

The analysis yields two measures of the overall fit. The first is the *root-mean-square residual*, which is the square root of the mean of the squared distances between corresponding points in the fitted and target configurations. This is expressed in the scale of the target matrix, which remains fixed throughout the analysis. In general, the RMS residual, when fitting A to B, will not be the same as the value obtained from fitting B to A. The second fit measure is the *root-mean-square symmetric error*, which has the same value regardless of the direction of the fit. This is a normalised average of the RMS residuals one would obtain in fitting A to B and B to A.

In addition to the overall fit measures, an additional column of what may be interpreted as *stress* values are appended as an additional column of the output matrix. These should be interpreted as standardised distances between the corresponding points in the two ordination, subsequent to rotation and optional scaling. A distance close to zero implies that the corresponding points were closely fitted while a value close to one suggests otherwise.

REFERENCES

- Gower, J.C. (1971) Statistical methods of comparing different multivariate analyses of the same data. In, F.R. Hodson, D.G. Jkendall & P. Tauto (eds.), *Mathematics in the Archaeological and Historical Sciences*. Edinburgh University Press, Edinburgh, pp 138-149.
- Gower, J.C. (1975) Generalised procrustes analysis. *Psychometrika* **40**, 33-51.
- Sibson, R. (1978). Studies in the robustness of multidimensional scaling: Procrustes statistics. *J. Roy. Stat. Soc*, **40**, 234-238.

OPTIONS

Standardisation

Both the test and target configurations are centred. If standardisation is selected, in addition to centring, *both* configurations are standardised such that the sums of squares of the loadings equals unity.

3-DIMENSIONAL ORDINATION

VIEWING:

TSPN

OUTLINE

The PATN module TSPN is an *interface* or pre-processor for the ACROSPIN(tm) program under MS/PC-DOS. ACROSPIN is a neat method of displaying and rotating objects and lines in colour in 3-dimensions. ACROSPIN accepts an ASCII file containing a list of points and lines and enables very (very!) fast rotation of the resulting configuration on any MS/PC-DOS machine. It is similar in aims to the Macintosh program MACKSPIN(tm).

ACROSPIN may be purchased for US\$27 (along with a number of other useful software for scientific use). The address is-

Pacific Ease
601 Pacific Street
Santa Monica CA
USA 90405
PH: (213) 396 5007.

TSPN reads from the following files-

PATN.PRM,
any *ordination* file (SSH, PCA, DCA, PCR),
the default group definition file (*-.gdf*),
optionally, the attributes section of the *-.pcr* file,
optionally, the *-.mst* (minimum spanning tree) file and,
optionally, the *-.sal* (travelling salesman circuit) file.

The ordination file provides the 3-d location of the objects in space and the group definition file provides the colours of the objects. Optionally, the location of the best fit linear vectors of a set of attributes from PCC may be plotted (as labels) in space. If a minimum spanning tree has been defined, it too will be plotted, in a separate colour, as lines, on the ordination base. The same is true of the travelling salesman circuit. If the *-.sal* file is detected, TSPN will prepare input for ACROSPIN for this circuit as a separate layer.

All layers generated for ACROSPIN can be toggled off and on. For example, if both *-.mst* and *-.sal* files are detected, they will be presented as two separate layers in ACROSPIN. If the MST layer is 3 and the SALE layer is 4, either or both can be switched off and on by pressing 'L' and the layer number (3 or 4). Each of the groups are also separate layers.

OPTIONS

Symbol-type

TSPN has three options for the TYPE of symbol that is used to define the location of the object in 3-d-

1. a dot,
2. an 'L' shape symbol and
3. the object label (1-8 characters).

When there are large number of objects (100+), it may be expedient to use a simple coloured dot to represent the objects. With ACROSPIN, these are one *pixel* each (small on a VGA or EGA system). The actual size of the pixels depends on the monitor being used. On CGA systems they will be larger than VGA systems. The colour of the dot is defined by the group definitions. An 'L-shape' is useful to achieve a better sense of 3-d shape and object location (front/back) during rotation. The size of the 'arms' on the 'L' are user defined. The third option is useful for locating a specific object, or when you are determined to know where all the objects are. Be warned, it gets cluttered when the number of objects increases (hence the other options)!

Symbol-size

The size of the 'L'-symbol or the characters is determined by a single floating point value. Size is related to the ordination coordinates, which are dependent on the algorithm that generated them. In the case of PCA, RA and DCA, coordinates are standardised in terms of eigenvalues. The sums of the squares of the coordinates on each axis are equal to the eigenvalue. In the case of SSH, values are generally normalised (subtract the mean and divide by the standard deviation). In any case, values generally range from ± 2 .

The maximum range of the coordinates is listed as a guide-line and the default value for this parameter is-

MAXIMUM (RANGE/50, .04) .

The default should be reasonable but feel free to experiment to suit your purposes.

COMPARING PARTITIONS:

RIND

OUTLINE

This module compares any two partitions defined by either

`-.gdf` or
`-.gav`

files. The method is based on a modification of the statistic of Rand (1971) by Hubert & Arabie (1985). Essentially, the two classifications/ partitions are tabulated in a contingency table. The first group definition entered forms the rows and the second, the columns. The columns of the table are ordered to maximise the diagonal entries. The extent of off-diagonal entries indicates the level of miss-match between the two partitions.

The number of groups in the two classifications does not have to be the same (but it helps!).

The Rand value ranges from zero implying chance levels of associations between the partitions to one indicating a perfect match. Output is to the file `-.rin` and contains the contingency table, the Hubert/Arabie/Rand statistic, and the extent of mismatches of each row, column and total.

REFERENCES

Hubert, L. & Arabie, P. (1985). Comparing Partitions. *Journal of Classification*. 2: 193-218.

Rand, W.M. (1971). Objective criteria for the evaluation of clustering methods, *J. Amer. Stat. Ass.*, **66**, 846-50.

OPTIONS

The first two parameters requested are the number of groups for each of the two classifications. This value is defaulted from NRG in the current PATN.PRM parameter file. Note: classifications do not need to have the same number of groups.

Group Format

The *form* or structure of the (ASCII) file nominating either of the two partitions may be

- 1 = standard *group-definition format* (-gdf) or
- 2 = *group-affinity-vector format* (-gav).

Either of the two classifications can be nominated in either form. See GDEF for details of these two formats. After the *form* of each classification is requested, the *name* of the file is entered from the terminal.

SENSITIVITY ANALYSIS:

SENS

OUTLINE

SENS aims to test the sensitivity of a classification to the omission of what it deems to be the attribute that shows least group discrimination. *Least significant* is defined in SENS as the lowest Kruskal-Wallis value from GSTA. The procedure iterates through the process of dropping an attribute (MASK), reclassifying the objects (using ASO, FUSE, GDEF or ALOC) and comparing the resulting groups with the original grouping (RIND). Iteration continues until the Rand-statistic from RIND drops below a nominated level.

This type of analysis could be done in a multitude of ways. This algorithm appealed to me at the time. Please let me know what you think.

The basic idea is to use GSTA (option 1) to generate a set of Kruskal-Wallis statistical values. This indicates the ranking of the M attributes across all NRG groups as defined by the current (default) `-.gdf` file (assumed to have been created).

The attribute with the *lowest* Kruskal-Wallis statistic (this is the least discriminating attribute among the NRG groups) is then MASKed out to create a data set with the root name *SENS* (**I assume that you have no files with this name!**).

This data set is then subjected to a classification. This can either be via ASO with a user defined association measure, and FUSE to create the new partitions, and finally to GDEF to define a new set of NRG partitions. The alternative is to use the non-hierarchical classification module ALOC (it does all three components). This new classification is then compared by way of RIND to the original and a single stress value obtained (Hubert & Arabie modified Rand statistic: see RIND). This value ranges from zero implying a chance association between the partitions to one, implying a perfect match of partitions between the two classifications. NOTE: The use of GDEF on FUSE output creates a partition rather than a hierarchy.

If the test value is greater than a user-supplied threshold, it is assumed that the omission of the attribute had little affect on the original classification. If so, iteration continues back to GSTA to locate the *now* lowest discriminating attribute of the new classification. If the Rand value drops below the supplied threshold, iteration stops and the history of iteration and dropped attributes is listed from the output file `-.sen`.

The procedure could operate on (at least) two alternative strategies of choosing which attribute to drop. It would be possible to use the original GSTA Kruskal-Wallis statistical information to determine the *complete* drop order. The alternative, as used here, re-runs GSTA on each sub-set data set and again sorts the attributes into a 'drop-order' on the basis of the Kruskal-Wallis values. My preference (obviously) is for the latter. Why? Firstly there is the problem of confounding and correlation between attributes; the sub-set of attributes cannot be influenced by what is not there. Consider it as the removal of 'noise' (not that I am for a minute suggesting that any species could be considered redundant! Heaven forbid). Comments welcomed.

Before implementing this procedure, I assumed that the drop in the relationship between the original data set and subsequent masked sets would be monotonic decreasing. This turned out not to be so in many situations. A common finding was a 'saw-tooth' curve that was certainly decreasing overall but had some intriguing 'high-points'. I am still trying alternative hypotheses as to why this occurs. Here is my favourite explanation (as of 3/93). A set of similarly structured 'noisy' attributes are removed with small decreases in the RAND value. When the last of the 'group' is removed, the next attribute is the 'noisiest in the next sub-structure'. When this attribute is removed, the classification suddenly 'bounces' back closer to the original. The 'noisiest' attributes with any 'substructure' of attributes are always removed first, so the removal of the 'most noise' will illicit a larger response as measured by the RAND statistic. The process repeats as more significant structural components are removed but usually to a lower 'high'.

REFERENCES

- Belbin, L., Faith, D.P. & Milligan (1992). A comparison of two approaches to β -flexible clustering. *Multivariate Behavioural Research*. 27:417-433.
- Belbin, L. & McDonald, C (1993). Comparing three classification strategies for use in ecology. *J. Vegetation Science*, 4:341-348.
- Hubert, L. & Arabie, P. (1985). Comparing Partitions. *Journal of Classification*. 2: 193-218.
- Kruskal, W.H. and Wallis, W.A (1952). Use of ranks in one-criterion variance analysis. *Journal of the American Statistical Association*, 47:583-621. Addendum (1953) 48: 907-911.
- Rand, W.M. (1971). Objective criteria for the evaluation of clustering methods, *J. Amer. Stat. Ass.*, **66**, 846-50.

OPTIONS

Classification strategy

There are two choices currently available on how you wish to re-partition the subsets of your objects, either via ASO-FUSE-GDEF or ALOC. The first alternative is appropriate for smaller sets (less than say 1,000 objects) that for one reason or another, you consider to be hierarchically related. While there is a choice of association measure from ASO, the default options in FUSE will be used (flexible UPGMA, $\beta=-0.1$).

If you are using largish data sets or you don't want to use an hierarchical clustering method, use ALOC. In this case, you will have a choice of association measure and defaults will be used elsewhere (to keep it simple).

Note, that default options for FUSE and ALOC are used. Other classification methods such as PDIV could be used but, as yet have not been implemented as options. Experience and testing (see Belbin, Faith & Milligan, Belbin & McDonald) suggests that FUSE and ALOC with default parameters work well in most circumstances.

Association Measure

The choice of association measure depends if ASO or ALOC is being used. If either case the options are as listed in those modules (17 for ASO and 3 for ALOC). Use of other methods such as GASO are not currently available.

RAND Threshold

This is the *minimum* value of the Hubert & Arabie modified Rand statistic permitted. As attributes are masked out of the SENS data set, the RIND module compares the new groups with the original, producing as a quantitative level of affinity, the Rand statistic.

If you want to see how many variables can be dropped before *any* change to the groups results, a value of .999 or so could be used. If you believe that the original classification is not necessarily 'optimal' due to the inclusion of 'noisy' attributes, lower Rand thresholds (say .5 or even .01) could be used. I find that I am always interested in the last attributes to remain so tend to use 0.01 here and get the complete picture.

An example. A threshold of .5 is used and results in a new classification that splits a major groups into two. Examination reveals that while the definition of the original group could be justified, a better explanation of the sub-groups was forthcoming.

BATCH OPERATIONS:

LOG2B

OUTLINE

LOG2B is a stand-alone utility that is distributed to enable batch processing of **PATN**. LOG2B reads *PATN.LOG* creating a batch file called *PATNBAT.BAT* and a set of additional files used for re-directed input to **PATN**. *Batch mode* is useful for re-running analysis on different data sets. In this mode input to **PATN** is from a file rather than a terminal. Output may be directed to a file or a screen. The operation of LOG2B is based on the UNIX/DOS concept of re-direction. For example, the command

```
SSH <test.inp >test.out
```

The '<' literally means to re-direct input from *test.inp* into SSH. Similarly, the '>' means to re-direct the normal output of SSH from the terminal to the file *test.out*. In theory, this should also work with MS/PC-DOS, but it doesn't! Ask Microsoft sometime. Redirection will work with individual **PATN** modules run stand-alone, but not with the front-end **PATN** itself ('**PATN**<test.inp' doesn't work). LOG2B gets around this by reading *PATN.LOG* looking for **PATN** commands (=PATN modules). It then creates a set of separate input files that are used for re-directed input into each requested **PATN** module (Eg. MASK).

LOG2B works the same on UNIX systems. The need for LOG2B on these operating systems is also in theory, not required. The problem I have encountered however on some systems (Pyramid for example), is that the system gets too smart for its own good, and gets fouled up. Consider this section of the log file *PATN.LOG*-

```
DATN          ! 06/19/90 ! 15:50:21.72 ! RANDOM DATA SET FROM RAND OPTION
              2                ! DATA I/O OPTION
Y              ! OUTPUT LABELS to ASCII FILE
              ! OUTPUT FILE NAME
              ! OUTPUT FORMAT
S              ! *STOP* TYPING FILE --> TERMINAL
N              ! PRINT FILE ?
HIST          ! 06/19/90 ! 15:53:31.54 ! RANDOM DATA SET FROM RAND OPTION
              2                ! ROWS(1) : COLS(2) : STOP(3)
              2                ! ROW/COL NUMBER_0=ALL
              18               ! NUMBER OF BARS
              3                ! ROWS(1) : COLS(2) : STOP(3)
              ! NEXT PAGE OF FILE --> TERMINAL
N              ! PRINT FILE ?
```

Running LOG2B on this will produce a three files. One called *DATN.INP* containing a copy of line 2 to the line before 'HIST'. The second file will be called *HIST.INP* and will contain all the lines after 'HIST'. Another file called *PATNBAT.BAT* contains the lines-

```
DATN <DATN.INP
HIST <HIST.INP
```

The reason for all this trouble is that the command-

PATN <*test.inp*

fails because once **PATN** activates the first option (say DATN), the DOS pointer to the next piece of re-directed input is *reset* to the top (first line) of the file *test.inp* which contains the command (DATN)! Once you have useful information in *PATN.LOG* for the creation of a batch file, run LOG2B and then simply type-

PATNBAT

to run the batch job. If any module is used more than once, LOG2B gives it a separate name. For example, the second use of MASK may result in the redirected input file *mask2.inp*. Commands to DOS (\$) as first character) are also handled correctly.

REFERENCES

none

OPTIONS

Output from the batch job created from LOG2B may be directed to the Terminal or a File. Enter a T or F depending on which you prefer. The File option is achieved simply by using re-directed output in the batch file *PATNBAT.BAT* to the file *PATN.OUT*. Using the example above with re-directed file output will result in-

```
DATN <DATN.INP >>PATN.OUT
HIST <HIST.INP >>PATN.OUT
```

The '>>' is a DOS re-directed *output* command with *append*. In this case output from DATN and HIST is appended to *PATN.OUT*. IF a single '>' was used, output from HIST would overwrite the previous output from DATN.

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